Applying Automotive EDR Data to Traffic Crash Reconstruction

Instructor: Richard R. Ruth

ID#: C1210



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Professional Development

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Learning Objectives: General

1. ***Describe the EDR system to others***
2. ***Explain the CDR report in the context of the case***
3. ***Locate data important to case within CDR report***
4. ***Understand the accuracy & reliability of EDR information***

Learning Objectives - Specific:

1. ***Determine if EDR recording is from your event***
2. ***Use Delta V of vehicle with EDR to get Delta V of vehicle without EDR***
3. ***Calculate Speed at Impact from Delta V and Post Crash Travel to Uncontrolled Rest***

Learning Objectives - Specific:

4. ***Calculate Closing Speed by combining 2 Delta V's and adjusting for restitution and offset***
5. ***Understand Principal Direction of Force (PDOF), what you can use it for and why it's important***
6. ***Calculate Speed at impact of Vehicle without EDR from Closing Speed***

Learning Objectives - Specific:

7. ***Recognize special circumstances when Delta V is understated or overstated in EDR***
8. ***Learn EDR speed is right if the speedometer is right – recognize 6 vehicle operational conditions and 3 equipment modifications affecting speed accuracy***
9. ***Calculate Min/Max range of speeds at impact from last EDR speed data point***

Learning Objectives - Specific:

10. Reconcile EDR speed at impact with other physical evidence
11. Do time distance analysis and place EDR data on google earth photo of scene
12. Use Accel Pedal, Brake, ABS and stability control data to better understand crash causation
13. Present EDR data reliability at trials and Frye/Daubert hearings

Instructor Intro

Rick Ruth



- Ford Motor Company Product Development Engineering - Airbag Design and Field Performance 1973-2006
- Ford's lead field user of EDR's, Ford EDR policy committee, Ford Rep to SAE & EDR Committee, Ford Rep to Bosch CDR
- 2007-present, Ruth Consulting LLC – Airbags and EDR in Recon
- Teach EDR tech, EDR Analyst 1 & 2 classes for IPTM and EDR for SAE. Presents regularly at national and regional conferences on reconstruction, on EDR's.
- Does research on EDR accuracy – 19 publications
- Helps prosecutors and cops nationwide in EDR cases – mostly for free (charge for written reports and testimony).
- Takes civil cases in airbags and EDR aspect of Recon

Student Background = tell us

- **Our typical student works in accident reconstruction and has been trained to do slide to stop and momentum calculations. But many SAE students are not typical.**
- **1. Are you typical? More or less trained in reconstruction? Specialties?**
- **2. Any prior EDR training (none expected)?**
- **3. If you are not “typical”, what do you hope to learn from this class?**

Student Roster 1

Ann	Grimes	Mecanica Scientific Svcs Corp
Ayoub-Gray	Alexandra	NHTSA
Carter	Chapman	Quest
Clare	Pierce	KIA America Inc
Colin	Truong	
David	Bizzak	Romualdi Davidson & Associates Inc
David	Plant	D P Plant & Associates
Elda Lilian	Garza	Explico
Greg	Wilcoxson	Wilcoxson Consulting
Justin	Smith	NHTSA

Student Roster 2

Kevin	**McGough**	**NHTSA**
Martin	**Landa**	**Auvion**
Matthew	**Lipscomb**	**Explico**
Sean	**Hays**	**NHTSA**
Tara	**Leipold**	**Explico**
Thomas	**Haugh**	**NHTSA**
Wesley	**Grimes**	**Mecanica Scientific Svcs Corp**
William	**Witt**	**Aperture Llc**



EDR Data Analysis Overview

- Introduction
- EDR Data is a game changer and gives new insight into crash reconstruction
- Minimum “basic knowledge” required to understand and apply EDR Data
- Lots of Details to check before declaring success/understanding
- This training module outlines the basic knowledge required.

Some Brief History

1. Pre-1994 Early airbag sensors were mechanical

2. Electronic sensors were first introduced by GM in 1994. Decision to deploy airbags is made in working (RAM) memory which only functions when power is on.

3. GM decided they needed to know is system worked properly, designed system to record basic inputs and outputs in EEPROM memory that can be retrieved after crash.

Early decisions in EDR's

1. GM measured acceleration 800 times per second.
2. The airbag usually deploys early in the crash, in the first 10-20 milliseconds
3. The typical frontal crash pulse lasts 100 to 150 milliseconds.

Early Decisions

1. There was not enough EEPROM memory to record all the acceleration data in those early days.
2. GM decided to calculate the change in velocity from the raw acceleration data and record that once every 10ms.
3. At that time the sensor measured only acceleration in the frontal (longitudinal) direction.
4. GM recorded in airbag deployments, but also decided to record in non-deployments

Early EDR History

1. In 1999 GM attached the airbag control module to the vehicle CAN bus communication system so it could receive information from other modules in the car.
2. GM phased in recording vehicle speed, throttle, brake on/off, and RPM.
3. The purpose of recording was still for GM to know if its airbag system worked correctly.
4. The tool to read this data was proprietary to GM.

Early History Continued

1. GM asked the supplier of its “Tech 2” dealer diagnostic tool, Vetronix, to make a tool to read crash data for sale to the public.
2. The Vetronix **C**rash **D**ata **R**etrieval system was born in 2000.
3. **Bosch Diagnostics** bought Vetronix in the mid 2000’s.

History Marches On

1. Ford began phasing in electronic sensors in 1997 and recorded limited info.
2. Ford contracted with Vetronix/Bosch to read its limited EDR's in the early 2000's.
3. Chrysler added electronic sensors but did not record anything useful until 2005MY.
4. In 2007 they also chose Bosch CDR to allow their EDR's to be read.
5. Toyota began recording in 2001, had proprietary read out tool, switched to Bosch CDR in 2010

US Government Involvement

1. The US government was reluctant to require EDR's – "big brother is watching".
2. Started regulatory process in 2002. In 2006 they adopted a regulation that would eventually ***standardize minimum content*** of **voluntarily** installed EDR's
3. It also CHANGED THE **PURPOSE** to be for ***effective crash reconstruction***.
4. The regulation required that data could be accessed by a **publicly available** tool (no monopoly given to Bosch).
5. Part 563 became effective September 2012.

Part 563 EDR Regulation



Conceived in 2002
Adopted in 2006
Effective 9-1-2012
11 model years of data now publicly available
THANKS, NHTSA!!!

(vs. China start 2021/2023, Europe 2022/2024)

Part 563 Min Data if EDR

Table I—Data Elements Required for All Vehicles Equipped With an EDR

	Recording interval/time[1] (relative to time zero)	Data sample rate (samples per second)
Delta-V, longitudinal	0 to 250 ms or 0 to End of Event Time plus 30 ms, whichever is shorter	100
Maximum delta-V, longitudinal	0–300 ms or 0 to End of Event Time plus 30 ms, whichever is shorter	N/A
Time, maximum delta-V	0–300 ms or 0 to End of Event Time plus 30 ms, whichever is shorter	N/A
Speed, vehicle indicated	–5.0 to 0 sec	2
Engine throttle, % full (or accelerator pedal, % full)	–5.0 to 0 sec	2
Service brake, on/off	–5.0 to 0 sec	2
Ignition cycle, crash	–1.0 sec	N/A
Ignition cycle, download	At time of download[3]	N/A
Safety belt status, driver	–1.0 sec	N/A
Frontal air bag warning lamp, on/off[2]	–1.0 sec	N/A
Frontal air bag deployment, time to deploy, in the case of a single stage air bag, or time to first stage deployment, in the case of a multi-stage air bag, driver	Event	N/A
Frontal air bag deployment, time to deploy, in the case of a single stage air bag, or time to first stage deployment, in the case of a multi-stage air bag, right front passenger	Event	N/A
Multi-event, number of event	Event	N/A
Time from event 1 to 2	As needed	N/A
Complete file recorded (yes, no)	Following other data	N/A

Mandatory in Europe July 2024

Table II—Data Elements Required for Vehicles Under Specified Minimum Conditions

Data element name	Condition for requirement	Recording interval/ time[1] (relative to time zero)	Data sample rate (per second)
Lateral acceleration	If recorded[2]	N/A	N/A
Longitudinal acceleration	If recorded	N/A	N/A
Normal acceleration	If recorded	N/A	N/A
Delta-V, lateral	If recorded	0–250 ms or 0 to End of Event Time plus 30 ms, whichever is shorter	100
Maximum delta-V, lateral	If recorded	0–300 ms or 0 to End of Event Time plus 30 ms, whichever is shorter	N/A
Time maximum delta-V, lateral	If recorded	0–300 ms or 0 to End of Event Time plus 30 ms, whichever is shorter	N/A
Time for maximum delta-V, resultant	If recorded	0–300 ms or 0 to End of Event Time plus 30 ms, whichever is shorter	N/A
Engine rpm	If recorded	–5.0 to 0 sec	2
Vehicle roll angle	If recorded	–1.0 up to 5.0 sec[3]	10
ABS activity (engaged, non-engaged)	If recorded	–5.0 to 0 sec	2
Stability control (on, off, or engaged)	If recorded	–5.0 to 0 sec	2
Steering input	If recorded	–5.0 to 0 sec	2
Safety belt status, right front passenger (buckled, not buckled)	If recorded	–1.0 sec	N/A

Mandatory in Europe July 2024

Table III—Reported Data Element Format

Details

Data element	Minimum range	Accuracy[1]	Resolution
Lateral acceleration	At option of manufacturer	At option of manufacturer	At option of manufacturer.
Longitudinal acceleration	At option of manufacturer	At option of manufacturer	At option of manufacturer.
Normal Acceleration	At option of manufacturer	At option of manufacturer	At option of manufacturer.
Longitudinal delta-V	−100 km/h to + 100 km/h	±10%	1 km/h.
Lateral delta-V	−100 km/h to + 100 km/h	±10%	1 km/h.
Maximum delta-V, longitudinal	−100 km/h to + 100 km/h	±10%	1 km/h.
Maximum delta-V, lateral	−100 km/h to + 100 km/h	±10%	1 km/h.
Time, maximum delta-V, longitudinal	0−300 ms, or 0−End of Event Time plus 30 ms, whichever is shorter	±3 ms	2.5 ms.
Time, maximum delta-V, lateral	0−300 ms, or 0−End of Event Time plus 30 ms, whichever is shorter	±3 ms	2.5 ms.
Time, maximum delta-V, resultant	0−300 ms, or 0−End of Event Time plus 30 ms, whichever is shorter	±3 ms	2.5 ms.
Vehicle Roll Angle	−1080 deg to + 1080 deg	±10%	10 deg.
Speed, vehicle indicated	0 km/h to 200 km/h	±1 km/h	1 km/h.
Engine throttle, percent full (accelerator pedal percent full)	0 to 100%	±5%	1%.
Engine rpm	0 to 10,000 rpm	±100 rpm	100 rpm.
Service brake	On or Off	N/A	On or Off.
ABS activity	On or Off	N/A	On or Off.
Stability control	On, Off, or Engaged	N/A	On, Off, or Engaged.
Steering input	−250 deg CW to + 250 deg CCW	±5%	±1%.

Are EDR's required in US? Proposed Yes, Implemented NO

1. NHTSA issued NPRM Dec 2012 to **require** 563-EDR's 9/14
2. NPRM proposed EDR as FMVSS 405, OEM's would have to self certify EDR's and recall defective EDR's. OEM's objected.
3. Officially withdrawn Feb 2019 https://www.federalregister.gov/documents/2019/02/08/2019-01651/federal-motor-vehicle-safety-standards-event-data-recorders
4. Mfrs were NOT expected to remove EDR's – But one Land Rover model may have – now has Bosch freeze frame recorder.
5. NHTSA estimates 99.5% of new US vehicles have an EDR

Require More Data in US EDR's?

- December 4, 2015 Section 24303 of the Fixing America's Surface Transportation Act (FAST Act), Pub. L. No. 119-14 required NHTSA to conduct a study "to determine the amount of time event data recorders installed in passenger motor vehicles should capture and record"
- May 2017 Event Data Recorders Duration Study completed by Virginia Tech (poorly)
- September 2018 Report to Congress summarizing pre-crash time interval results in determining crash causation
- NHTSA issued NPRM 6-22-2022 Docket NHTSA-2022-0021

Part 563 NPRM 6-22-2022

1. Proposed rule increases precrash recording time to 20 sec @10X/sec, effective 1-2 years after final reg. Did NOT address additional data elements.
2. Industry strongly disputed need for 20 seconds, noted significant effects on cost, package size, and development time.
3. Insurance, NTSB, others like more free data
4. No Final Rule yet as of 11/29/2023

EDR Propagation in US

1. There are about 300 million vehicles registered in the US. 65+% have an EDR with data accessible by the Bosch CDR system
2. 85+% of 2023's are supported by Bosch.
3. Another 15% have EDR accessible by other tools (10+% Kia/Hyundai, 3+% Tesla). **99+%** have EDR.

US Lt. Veh EDR & Tool Used

Manufacturer	2022 CY per Automotive News	Market Share	EDR First Known Use	Tool?
GM	2,258,283	16.3%	1994	Bosch CDR
Toyota	2,108,455	15.2%	2001	Bosch CDR
Ford	1,850,925	13.4%	1997	Bosch CDR 2001+
Stellantis	1,553,485	11.2%	2005	Bosch CDR
Hyundai/Kia	1,474,224	10.7%	2010	GIT tool co.
Honda	983,507	7.1%	2012	Bosch CDR 2012+
Nissan	729,350	5.3%	2006	Bosch CDR 2013+
VW/Audi	565,119	4.1%	2014	Bosch CDR
Subaru	556,581	4.0%	2012	Bosch CDR now
Tesla	491,000	3.5%	2012	Tesla Tool
BMW	363,612	2.6%	2013	Bosch CDR
Mercedes-Benz	350,949	2.5%	2014	Bosch CDR
Mazda	294,908	2.1%	2011	Bosch CDR 2011+
Geely (Volvo)	102,038	0.7%	2011	Bosch CDR 2011+
Mitsubishi	85,810	0.6%	2006	Bosch CDR now
Tata (Jag, LR)	65,602	0.5%	early	Send to Supplier
Other	7,944	0.1%		
				N/A
Industry Total	13,841,792	100.0%	########	Bosch CDR 85%
updated April 2023				Non CDR 15%

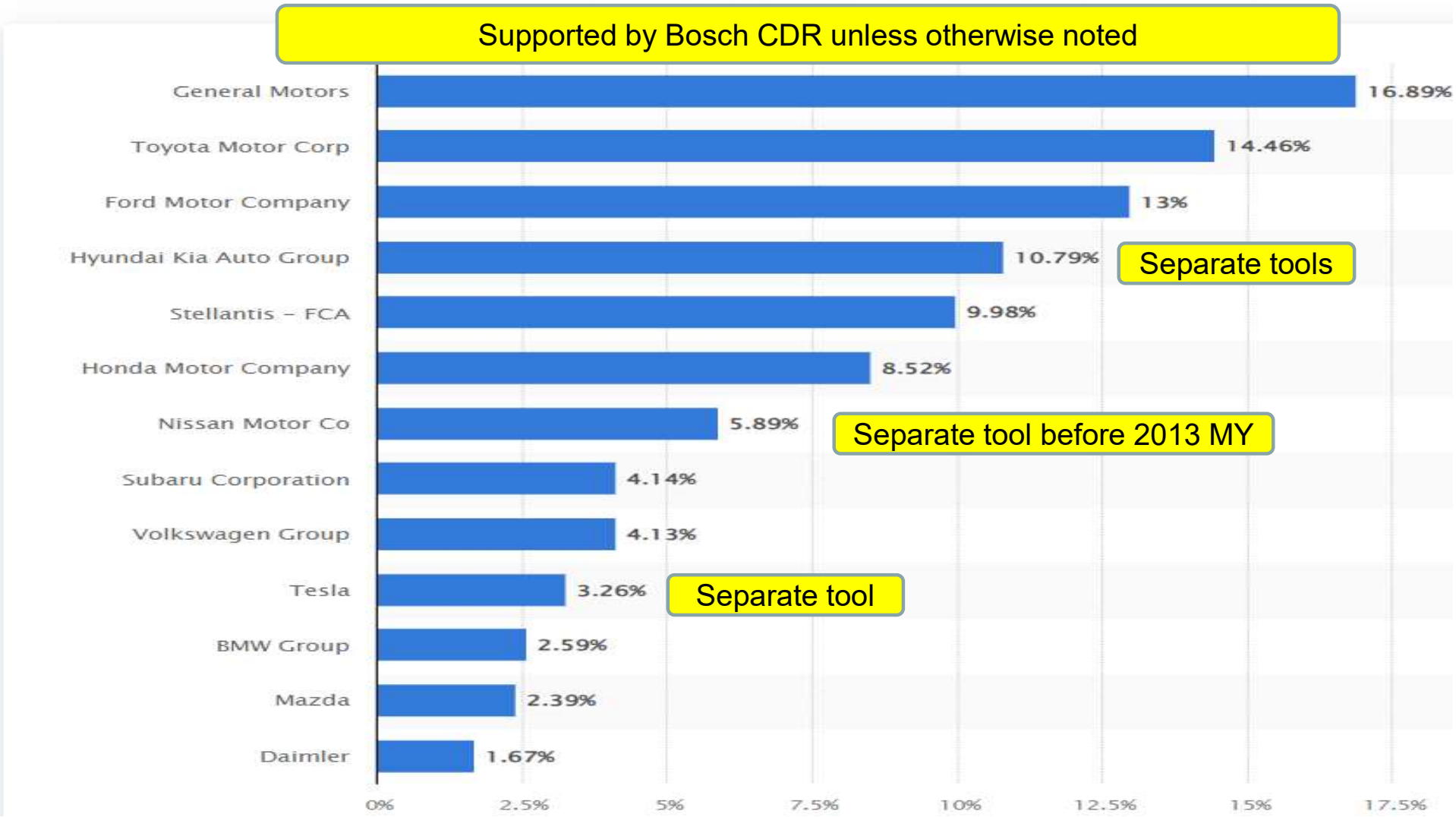
Includes ASCM/ FCM

Techstream for VCH

Porsche 2020 Taycan

CDR 2.0 ADS

Selected automakers' U.S. YTD market share



2023 US EDR Recording Capability

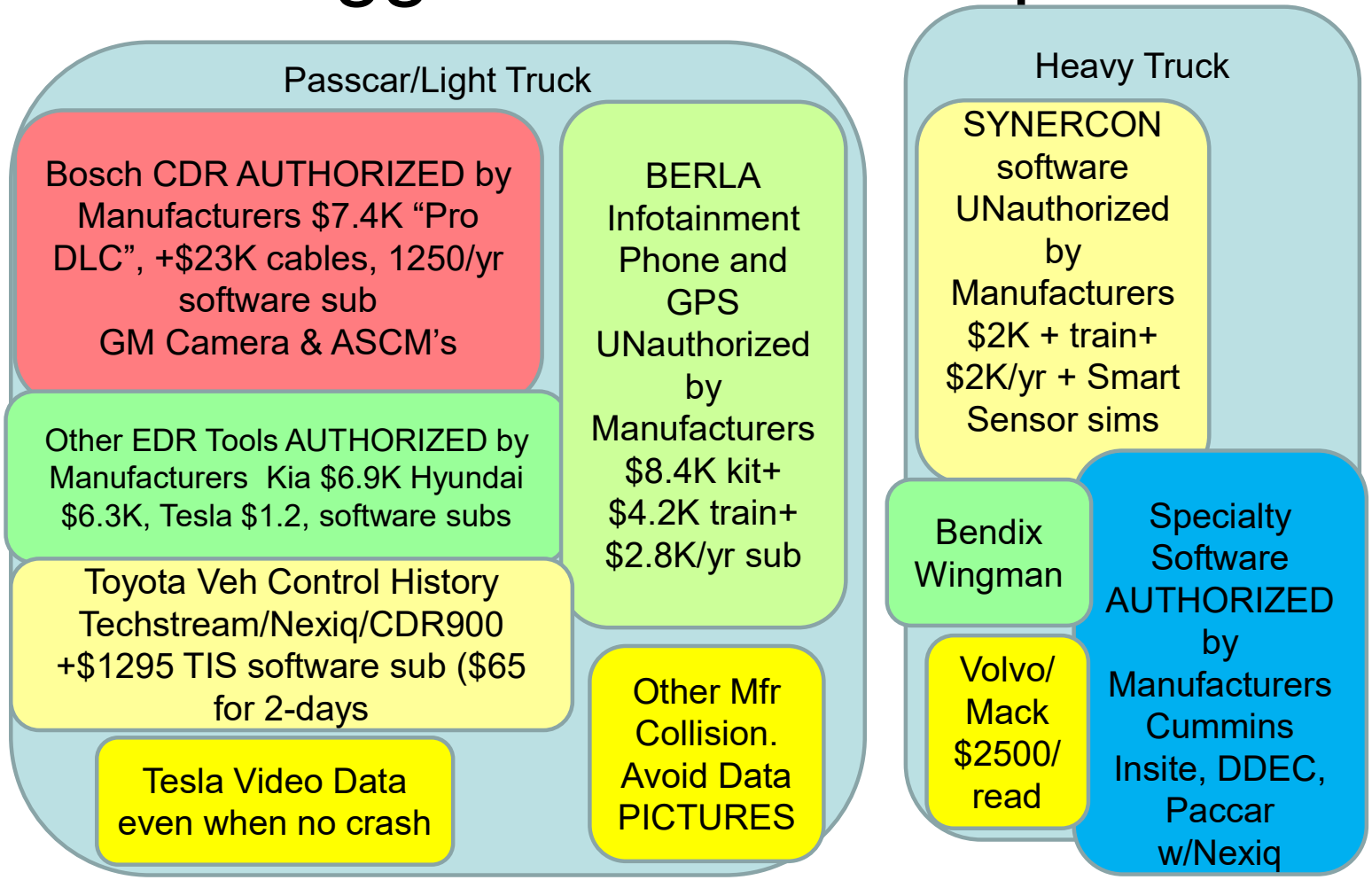
EDR Major Data Elements by Manufacturer (All Mfrs with EDR have the required 15 data elements, all are not listed)

Information is taken from Bosch CDR sample files, Ruth cases, and 2012 MY practice projected into 2019 model year

	Part 563 Table 1					Part 563 Table 2						Beyond 563									
Mfr	Event Complete	Key Cycles	Long. ΔV	Speed/Brake/ Throttle or Accel	Multi Event	Lateral ΔV	RPM	Steering Angle	ACM Acceleration	ABS on/off	Roll Angle	ESC Long Accel	ESC Lat Accel	Yaw Rate	Tire Pressure	Panic Brake Assist	Wheel Speeds	Roll Rate	Brake Pressure	OTHER
GM	X	X	X	5@0.5 TA	X	X	X	SDM50	, Z 300@2	X		50	50	50	LO		50	1 @ 10ms	50	50 GPS Location
Ford	X	X	X	5@0.5 A	X	X	X	5@0.1	-	X		5@0.1		5@0.1	LO		-	5@0.1	**	2017 AB12 precrash 0.2 sec
Chrysler	X	X	X	5@0.1 TA	X	X	X	5@0.1	-	X		-		5@0.1	PSI	X	X			2018 ACC set speed 2018, Avoidance
Toyota	X	X	X	5@0.5 TA	X	Side	X	5@0.5	-		Peak&atTRG	5	in roll	5@0.5	-	-	-	In Roll	5@0.5	Ped 2018, Veh history rpt
Honda	X	X	X	5@0.5 TA	X	X	X	5@0.5	X, Y, Z 250@10	X	-1+5@0.1	-		-	-	-	-			AEmBrk. Lanekeep
Nissan	X	X	X	5@0.5 A	X	X	X	5@0.5	X, Y 250@10	-	-	-		-	-	-	?			
Mazda	X	X	X	5@0.5 T	X	X	X		X, Y, Z 250@10	-							-			
Volvo	X	X	X	5@0.5T	X	2016 XC90	-	2016 XC90	X,Y,Z 250@10	2016 XC90	-1+0.3@0.1									
BMW	X	X	X	5@0.5A	X		X	5@0.5	X, Y 300@10ms	X	-									
Mercedes	X	X	X	5@0.5A	X	2017 E class adds														
VW	X	X	X	5@0.5A	X	X														Time of Crash
Hyundai/Kia	X	X	X	5@0.5	X	X	X	5@0.5	X,Y,Z 250@10	X	-1+5@0.1	-		-	-	-	-		19 Forte	
Subaru	X	X	X	5@0.5A	X	X	X	5@0.5		X								-1 to+1.5		Eyesight
Mitsubishi	X	X	X	5@0.5A	X	X	X	5@0.5	X, Y 250@10		-1+5@0.1									

CDR 2.0 ADS

The bigger vehicle data picture



Recent CDR Software Updates

- 21.2 Aug 3 GM Fwd Camera Module
- 21.3 Oct. 28 21 Volvo, 22 Ford, Audi. WIN11
- 21.4 Jan 5, 2022 - +GM ASCM, 22 FCA,
- 21.5 April 8– Ferrari, more ’22, mitsu
- 23.0 October 7– Many 2023’s
- 23.1 March 31, 2023 – Euro support, 2023/4’s
- 23.2 July 13
- 23.3 Oct. 18 “CDR2” Release
- 23.4 Dec. 18 Mercedes ADS
- 24.0 March 29, + patch immediately after

Which Interface Module by Manufacturer 2024

Manufacturer	CAN+ Only	CAN+ or CDR900	CDR900 Only
GM	All SDM pre-2016, Some 2016+		Some SDM 2016+, All ASCM, FCM
Ford	All pre-2021, Some 2021-24		Some 2021+
Chrysler	All Pre 2016, some 2017	Some 2017, All 2018+	
Toyota		All 2001-2020, some 2021+	Some 2021+
Honda		All 2012-2016, Some 2017-2023	Some 2017-2023, All 2024+
Nissan	All 2013-2020, Some 2021+		Some 2021+
Mazda	All 2012-22, Some 23+		Some 23+
Volvo		Most 2011-2024	Some 2022+
BMW		All 2013-2022	All 2023+
Mercedes		All 2014-2022	All 2023+
VW/Audi		All 2014-2023	All 2024
Subaru			All
Mitsubishi			All



HARDWARE CHECKLIST 24.0

New 3-digit Part ID

Part Number	Description	Released
1699200778	Cable, ID835, CDR Subaru ACM 1 — 2019+ Forester/Acsent	19.0
1699200779	Cable, ID836, CDR Subaru ACM 2 — 2015+ Legacy/Outback	19.0
1699200783	Cable, ID837, CDR Mitsubishi ACM 1 — Eclipse Cross	19.0
1699200784	Cable, ID838, CDR Mitsubishi ACM 2 — Outlander	19.0
1699200785	Cable, ID839, CDR Mitsubishi ACM 3 — Many Older models	19.0
1699200786	Cable, ID840, CDR Mitsubishi ACM 4 — Japan Only models	19.0
1699200776	Cable, ID833, CDR GM ASCM 1 — Advanced Safety	19.1
1699200777	Cable, ID834, CDR GM ASCM 2 — Advanced Safety	19.1
1699200797	Cable, ID842, CDR VW Group ACM 4 — 2020 Golf, Neo	19.1
1699200800	Cable, ID843, CDR Mazda ACM — 2019+ Mazda 3	19.1
1699200838	Cable, ID845, CDR GM ACM — SDM-50 2020 Corvette	19.1
1699200837	Cable, ID844, CDR Subaru ACM 3 — 2020 Legacy/Outback	19.4
1699200839	Cable, ID846, CDR Honda ACM — 2021+ Honda Gen 4	19.4
1699200787	Cable, ID841, CDR Daimler ACM (requires CDR 500 — 2021 S-class	19.6
1699200882	Cable, ID847, CDR PSA ACM 1 — 2021 Puegot in China	21.1
1699200883	Cable, ID848, CDR PSA ACM 2 — 2021 Puegot in China	21.1
1699200983	Cable, ID850, CDR VWG ACM 5 — 2021 Golf + China models	21.1
1699201032	Cable, ID853, CDR Mitsu ACM 5 — Outside US/Canada	21.1
1699503814	Cable, ID858, CDR Maserati / Ferrari ACM	21.4
1699503636	Cable, ID859, CDR Lucid ACM	21.4
1699200985	Cable, ID852, CDR GM ACM	23.0
1699503637	Cable, ID854, CDR GM ASCM3	23.0

Nothing New in 24.0 YET -

Coming soon: 855 GM ASCM and 856 GM FCM

New Bosch CDR Version Numbering

- First two digits used to be how many manufacturers were covered
- Industry consolidation kept changing the count
- New numbering system uses calendar year as first two digits
- Second digit is major release within calendar year
- Third digit is patch number
- We are at 24.0.055

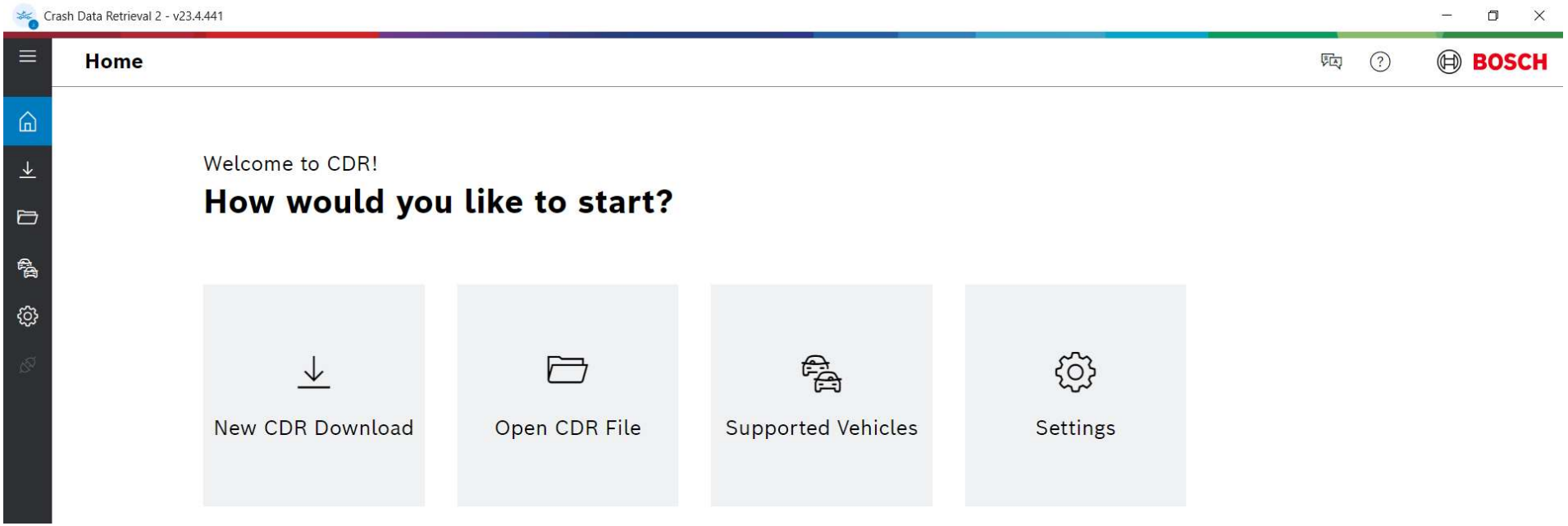
New Bosch “CDR2” Program 23.3

2024 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024

Mkt	Year/Make	Model	Module	New CDR2		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
1,10	2024 Mazda	CX-30	ACM	CDR 2	Data	CDR 900	CDR 900 - click here	Cable# 863 CDR 900 - click here (use listed cable above and no ACM adapter)	Center tunnel, forward of shifter
1	2024 Mazda	CX-5	ACM	N/A	Data	CANplus	F00K108287	F00K108387 & Cable# 812	Center tunnel, aft of shifter

- Next Generation of CDR software
- So far only a few models use it – Mazda CX30, Mazda 3, Mazda MX-5, Mazda CX-70, Mercedes EQS
- Click on help file link to open it

CDR2 slightly different look/feel, but otherwise similar



New Mercedes EQS ADS module

2 California and Nevada	2023 Mercedes-Benz	EQS (sedan)	ADS *(vehicles equipped with DrivePilot)*	CDR 2	Data	CDR 900	OBD Only CDR 900 - click here	does not support direct to module downloads	N/A

- Self driving car only offered in California and Nevada so far
- New separate module from ACM
- Click on “data” tab in help file to see contents
- GPS position, Nearby object locations, turn signal switch status,

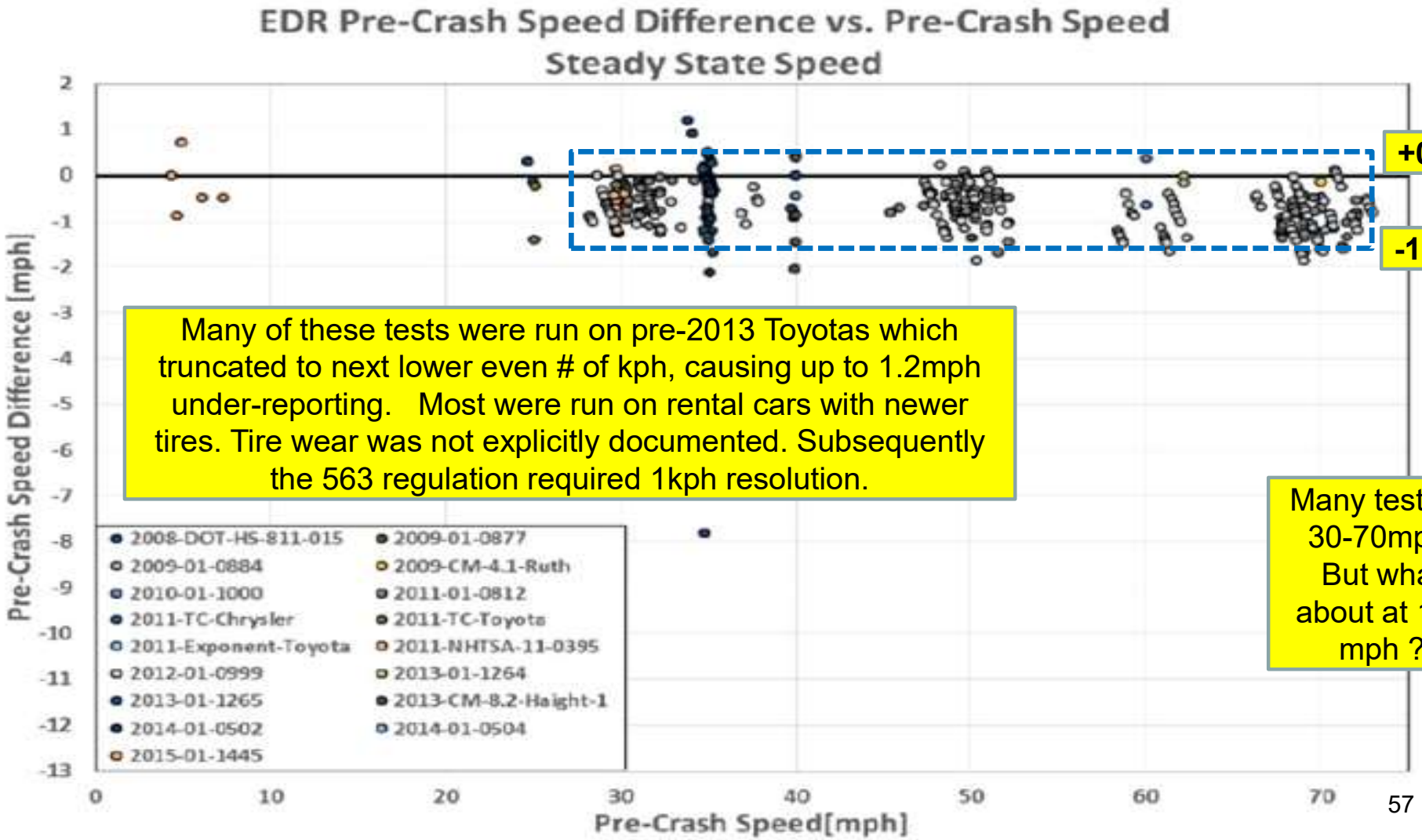
SAE INTRO ENDS HERE

Materials beyond this point are ”what’s new” and will be covered in other portions of the SAE class

Analysis Update: Speed Data

1. Accuracy: We taught +/-4% to speed data based on 1999 GM paper.
2. Part 563 speed data **+/-1 kph** (0.6 mph) as sold vehicles, only means EDR is within 1kph of speedo. Tire wear 2% not included.
3. SAE 2016-01-1497 "**Compendium** of Pass Vehicle EDR Literature and Analysis of Validation Studies" by Bortles et al, Kineticorp recaps accuracy as -1.5 mph to +0.5 mph (most testing @ 30 to 70 mph).
4. New WREX/SAE paper +/- 4% still good

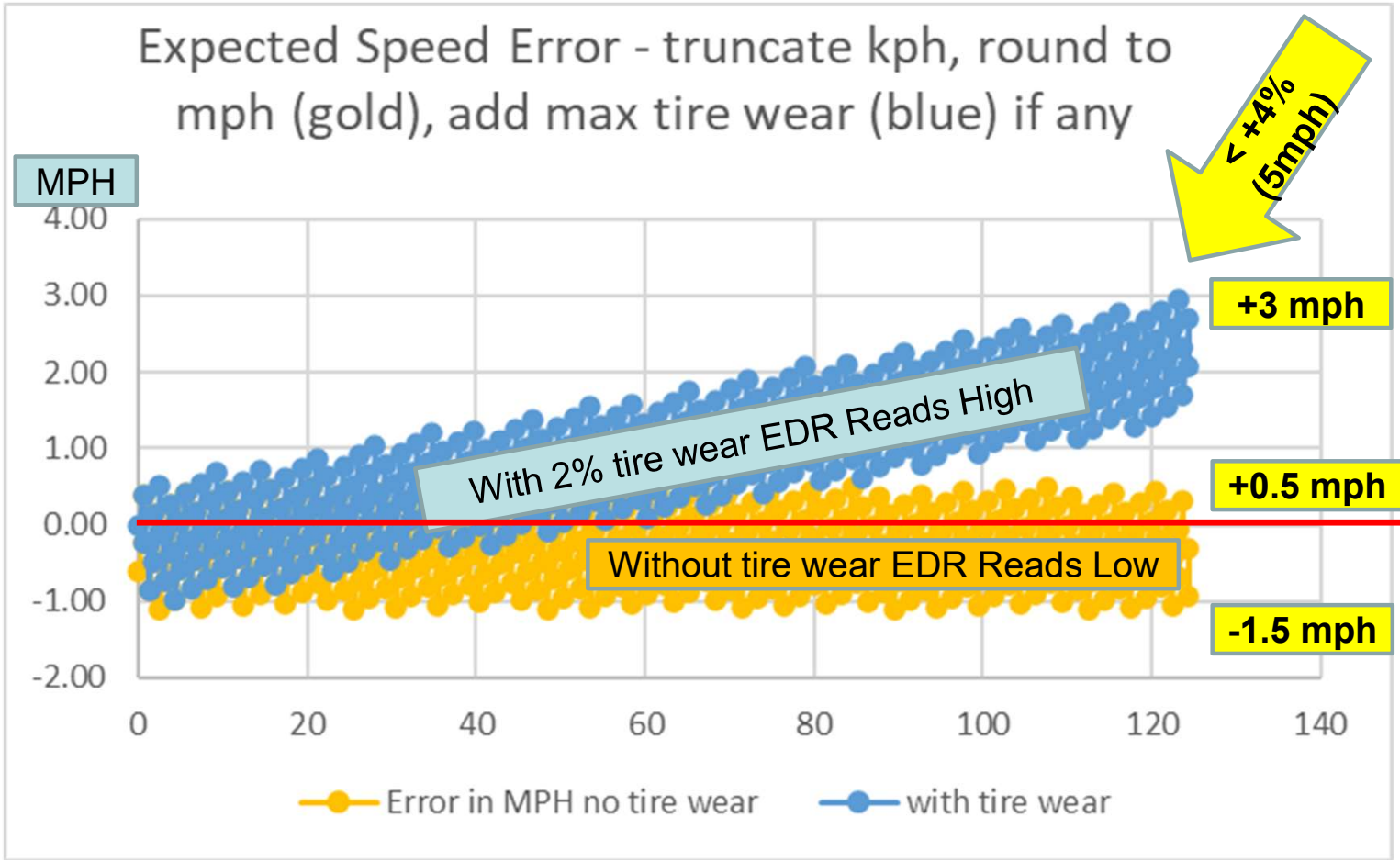
SAE 2016-01-1497 Data



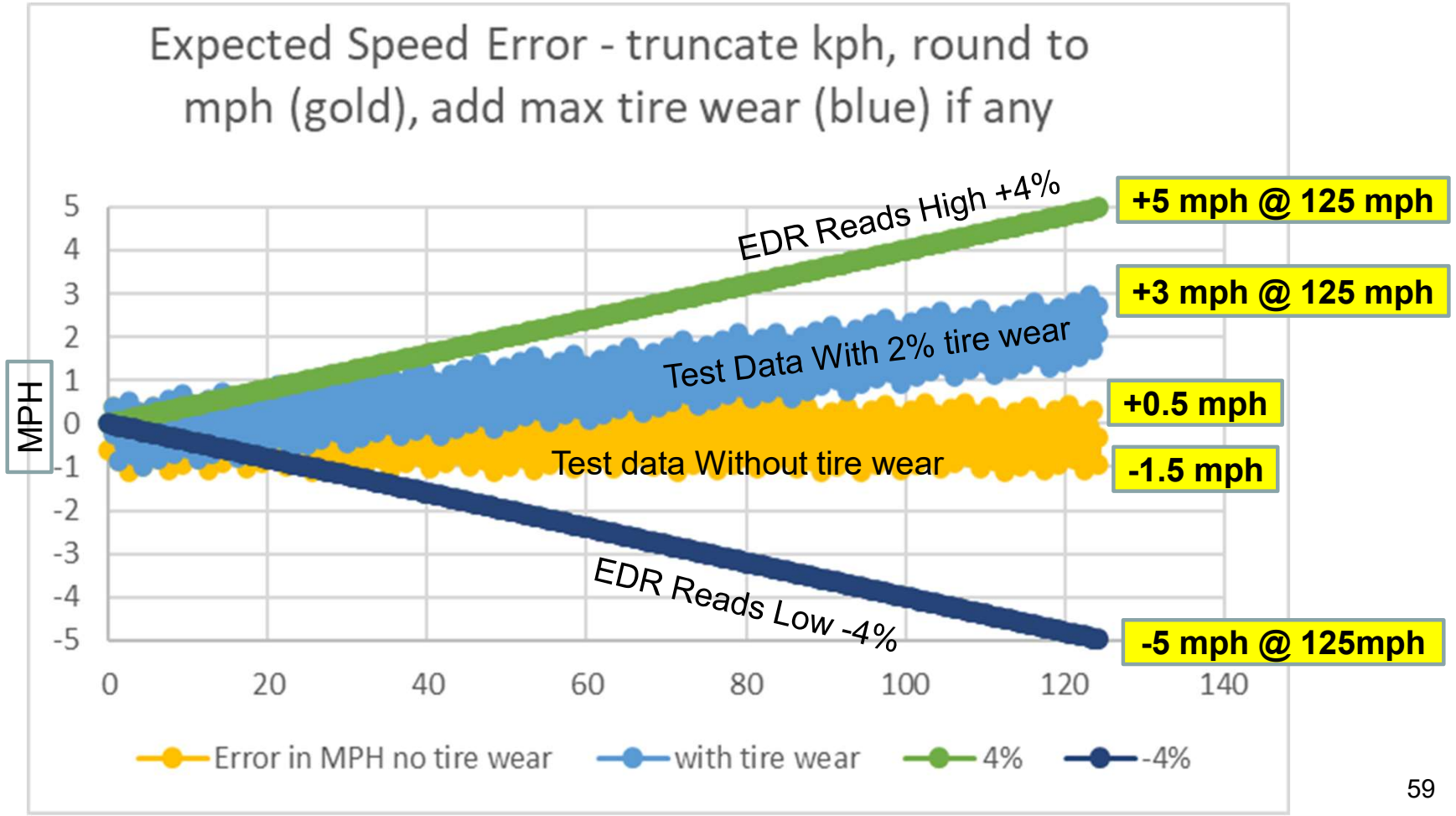
Many of these tests were run on pre-2013 Toyotas which truncated to next lower even # of kph, causing up to 1.2mph under-reporting. Most were run on rental cars with newer tires. Tire wear was not explicitly documented. Subsequently the 563 regulation required 1kph resolution.

Many tests at 30-70mph. But what about at 125 mph ?

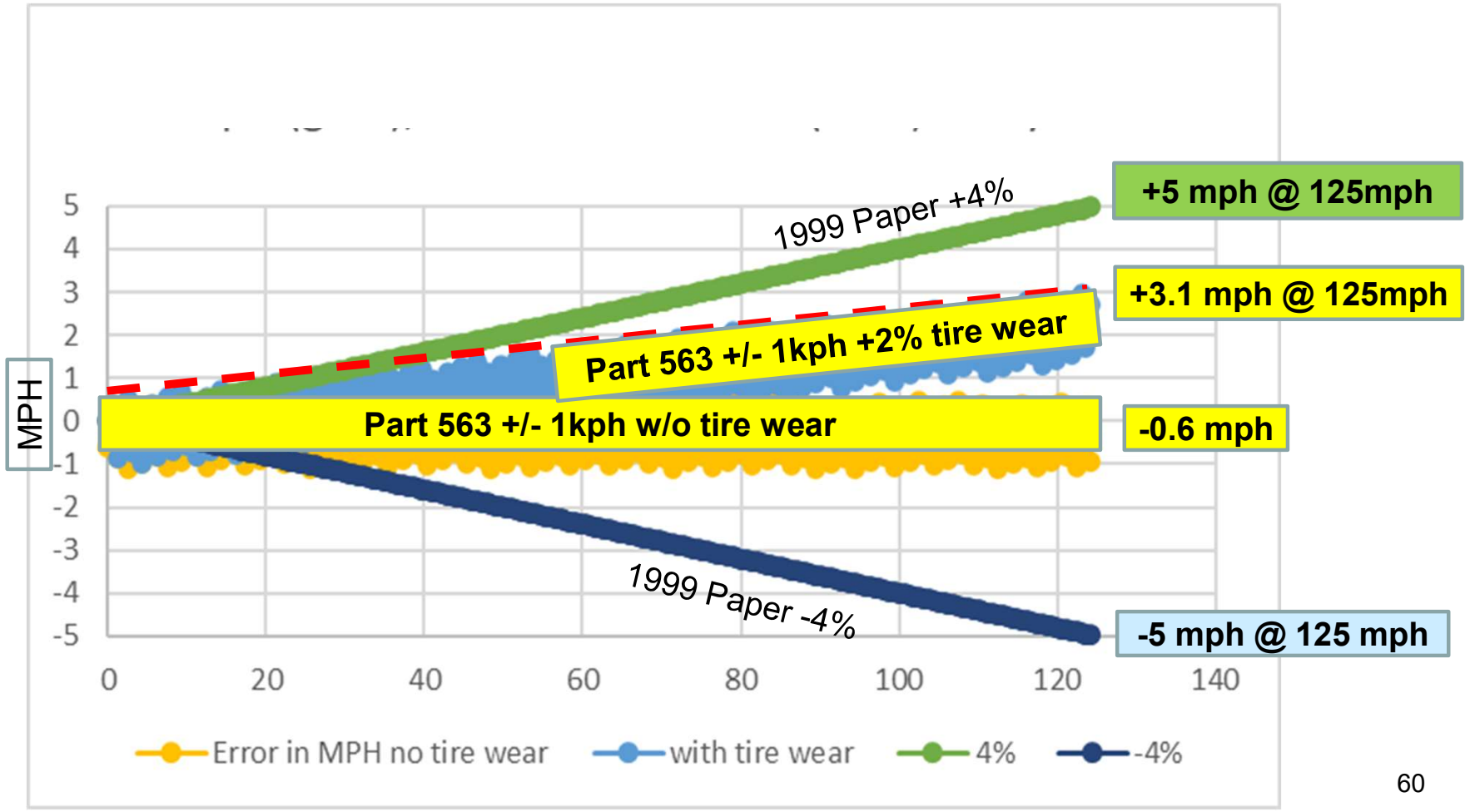
Add 2% allowance for tire wear to 2016 accumulated test data of -1.5 to +0.5 mph



Speed Error: Test Data+tire wear vs +/-4%



Speed Error: Part 563 vs +/-4%



Additional Testing Underway

- A sample of new model vehicle IIHS test EDR data vs reference instrumention was presented at WREX
- The sample size has been increased and will be published in SAE WCX in 2024
- The paper supports speed data is +/-4% in 40mph/64 kph tests, and longitudinal Delta V is within +/-10% in frontal impacts

Which Guidance to Use?

- You must evaluate the circumstances of your case and decide what is most relevant.
- Use AT LEAST +.5/-1.5 mph at low speed before 2013

  (At 5mph +/- 4% would be +/-0.2 mph, it is NOT that accurate)
- +/-4% at high speed will always cover any error and can be easily understood by juries – KISS?
- Reduce error by considering test data to -1.5 mph min and +0.5, +tire wear allowance (2% worst case, or less)
- If 2013 or later model, you can use +/- 1kph plus an allowance for tire wear (measure it & adjust or use +2% max)

Speed at Impact during braking - Timing of Last Speed Data Point

Red= Bad Green=Good

Time from last data point to impact		
(Time available for possible braking between last data point and impact)		
MANUFACTURER	**Model year**	
GM Gen 1	92-02	No precrash data
GM Gen 2	99-12	**0 to 1 interval between data points, 1 second**
GM Gen 3	2005-2012	0 to 1 interval between data points, some 0.5 **some 1.0 second**
GM Gen 4	2012-presen	0 to 1 interval between data points, 0.5 second
Ford ACM Gen 1	97-07	No precrash data
Ford PCM	2003-2011	0 to 1 interval between data points, 0.2 seconds
Ford ACM Gen 2	2005-2012	0 to 1 interval between data points, **1 second**
Ford Gen 3 RC6	2009-2010	**more than 1 interval between data points, use 1 second**
Ford Gen 3 RC6_2011 & Bosch AB10	2011+	0 to 1 interval between data points, 0.5 seconds
Ford Gen 3 RC7	2013+	0 to 1 interval between data points, 0.5 seconds
Ford Gen 3 AB12 & RC8	2017+	? Data point interval only 0.2 sec though, use that
Fiat.Chryslers	2006+	0 to 1 interval between data points, 0.1 sec (0.2 some minivan 2010-2012)
Toyota Gen 1 and 2	2001-2012	Last data point taken at impact **but 0 to 0.5 sec old**
Toyotoa Gen 3 & 4	2013+	**Last data point taken at imapct, within 24ms (use zero)**
Honda	2011+	Last data point taken at impact, within 0.1 sec of impact,but 0.3 sec delay for analog speedometer **0.9 sec for Digital speedo**
Nissan	2013+	Last data point taken at impact, within 0.1 sec?
Mazda	2011+	Last data point taken at impact, within 0.1 sec ?
Volvo	2011+	0 to 1 interval between data points, 0.5 second
BMW	2013+	0 to 1 interval between data points, 0.5 second
Mercedes	2014+	0 to 1 interval between data points, 0.5 second
Volkswagen	2015+	File gives time since last data point 0.0 to 0.4 sec - use that
Kia/Hyundai	2010+	Last data point taken at imapct, within 0.1 sec?
Subaru	2012+	Last data point taken at imapct, within 0.1 sec?
Mitsubishi (Chrysler clone)	2009+	? Assume 0 to 1 interval between data points, 0.1 sec
Mitsubishi (Part 563)	2013+	? Assume 0 to 1 interval between data points, 0.5 sec

Analysis Update: Delta V Usage - overview

VEHICLE WITH EDR

- Inline w/departure

  V1 = V3 - ΔVx

- Angled with Departure

  V1=V3cosβ - ΔVx

Note no assumptions about whether collision was central or vehicles reached a common velocity

VEHICLE WITHOUT EDR, ADD

- Inversely Proportional ΔV at centers of mass $\Delta V_1 = -\Delta V_2 \frac{W_2}{W_1}$
- Closing Speed Central Inline
- $\text{Closing Speed} = \left[\frac{1}{1+e}\right]\left[|\Delta V_1| + |\Delta V_2|\right]$ w/common vel.
- Closing Speed Offset Inline w/com vel.

  $\text{ClosingSpeed} = \left[\frac{1}{1+e}\right]\left[\frac{|\Delta V_1|}{\gamma_1} + \frac{|\Delta V_2|}{\gamma_2}\right]$
- Intersection inline approximation
- V1/V3/ΔV Triangle Vel Vectors
- ~~V1/V2/Closing Speed Triangle~~
- CS at intersection only works if one vehicle <20mph (left turning)



DeltaV – Example Application

$$\Delta V_1 = -\Delta V_2 \frac{W_2}{W_1}$$

Equation 12

- Purpose/Use of Equation: If you know one vehicle's DeltaV and the vehicle weights in a two car collision, you can calculate the other vehicle's DeltaV. This is ENORMOUSLY useful, it's an intermediate step to getting to speed at impact.
- This is just another application of Newton's 2nd and 3rd laws

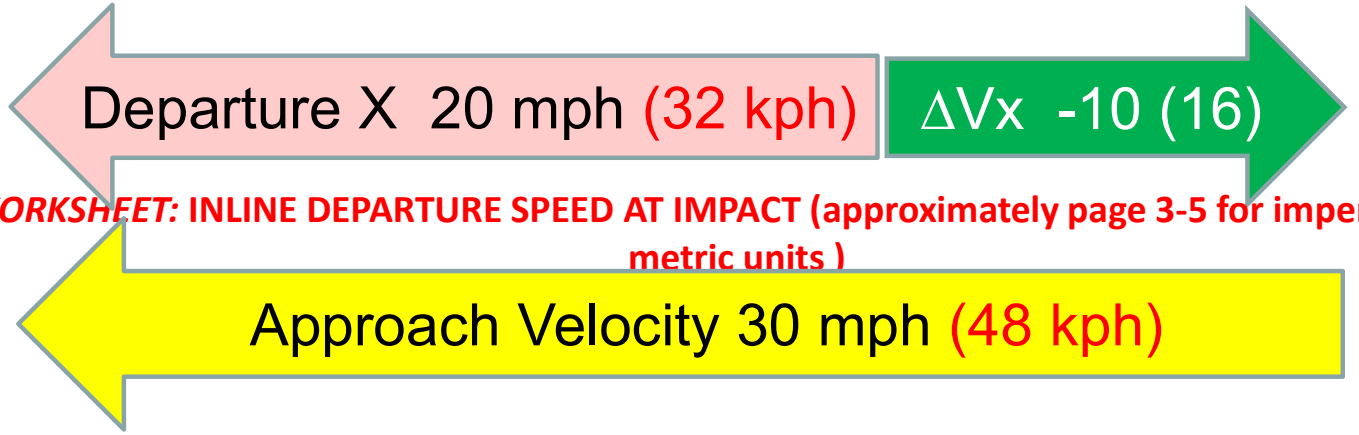
Inline *Speed at Impact* from ***Longitudinal* DeltaV** *and* ***Post Crash Speed***

Speed at Impact = Post Crash Speed – Delta V_x

V1 = V3 - ΔVx

For frontal crashes (negative Δv), minus a minus is like a plus

Equation 7

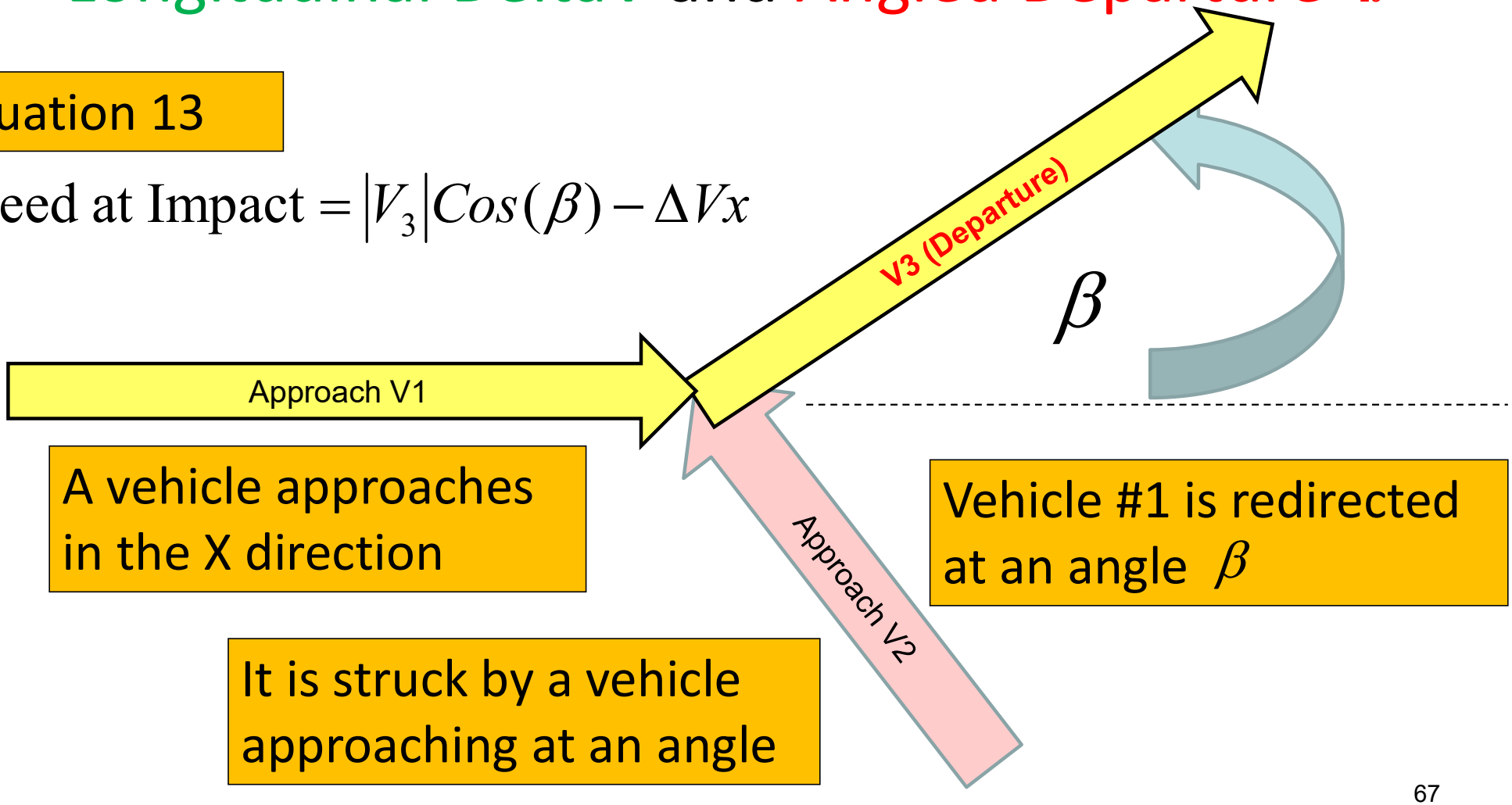


GO TO WORKSHEET: **INLINE DEPARTURE SPEED AT IMPACT (approximately page 3-5 for imperial units, 3-6 for metric units)**

Longitudinal DeltaV and Angled Departure 10

Equation 13

$$\text{Speed at Impact} = |V_3| Cos(\beta) - \Delta Vx$$



A vehicle approaches in the X direction

Vehicle #1 is redirected at an angle β

It is struck by a vehicle approaching at an angle

Drag factor if Post-crash Rotation

- In intersection crashes, if the crash is an offset collision (vs a central collision) there is often rotation post impact. To get the appropriate drag factor you need a “spin” analysis that considers the non-ABS roadway drag factor, whether any tires are locked and what percent of the weight is on the locked tires, and the degrees of rotation.
- A template has been developed to assist in calculating the appropriate drag factor. Please see the “Drag Factor with Vehicle Rotation post impact” template.

Drag factor vs spin angle template

Drag Factor with vehicle rotation post impact Calculation Template								
Assumptions made by this template:								
1. The vehicle is on level ground (ADD OR SUBSTRACT SLOPE IF NOT)								
2. The vehicle is rotating at a constrant rate (i.e., no roll out and the end)								
3. The change in vehicle heading to the departure angle happens instantly and is not part of the spin								
4. The rotation angle at rest is measured from the departure angle, not the approach angle.								
Yellow= User Entries								
Blue=Calculations								
0.7	Non-ABS Roadway Drag Factor (locked tires)							
0.05	rolling resistance no locked tires, 05 2WD, .06 AWD							
0	number of front tires locked 0 to 2							
0	number of rear tires locked 0 to 2							
55	percent of weight on front tires typically 50-60 (use 55 if not sure)							
0.050	Equivalent drag factor with locked tires straight ahead							
0.650	difference between roadway drag factor and straight ahead affected by sine							
Degrees Rotated	**sin of angle**	**drag factor this interval**		**avg drag factor**				
0	**0.000**	**0.050**		**0.050**				
1	0.017	0.061		**0.056**				
2	0.035	0.073		**0.061**				
3	0.052	0.084		**0.067**				
4	0.070	0.095		**0.073**				
5	0.087	0.107		**0.078**				

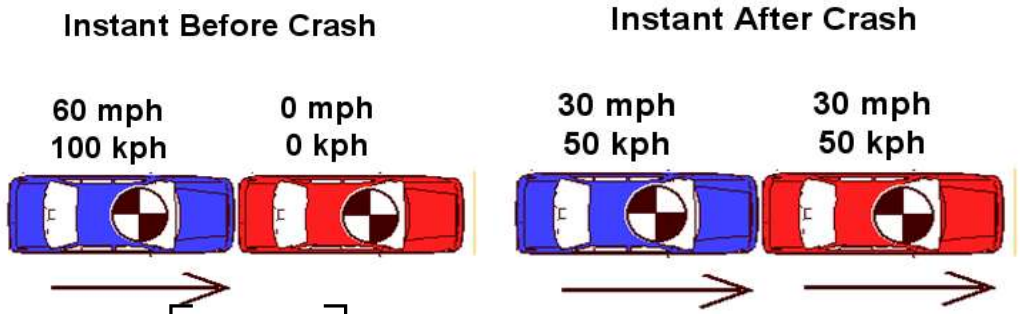
Restitution ADDS to Delta V; it must be ADJUSTED OUT of Closing Speed (imperial)

No Restitution

DV_{bullet} = -30 mph

DV_{target} = +30 mph

0 sep/60 close = 0% e



$$\text{Inline Closing Speed} = \left[\frac{1}{1+0}\right]\left[\left|-30\right| + \left|30\right|\right] = 60mph$$

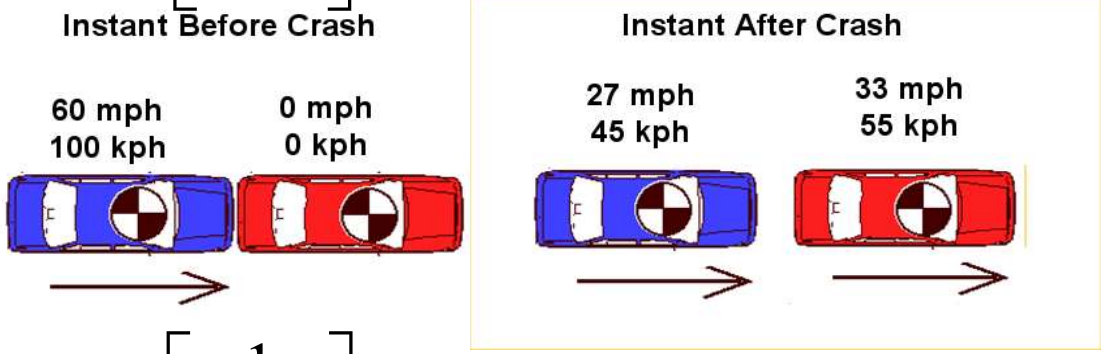
Equation 9

W/Restitution

Dv_{bullet} = -33 mph

DV_{target} = +33 mph

6 sep/60 close = 10% e



$$\text{Inline Closing Speed} = \left[\frac{1}{1+.1}\right]\left[\left|-33\right| + \left|33\right|\right] = 60mph$$

6. CLOSING SPEED central v offset

- In a central collision, the Effective Mass Ratio "gamma" is 1.0, and the term "drops out" leaving us with the central collision closing speed formula 9A presented earlier

$$\text{ClosingSpeed} = \left[\frac{1}{1+e}\right]\left[\frac{|\Delta V_1|}{\gamma_1} + \frac{|\Delta V_2|}{\gamma_2}\right]$$

Equation 9C –for offset (non-central) inline collision where the damage centroids reach a common velocity

$$\text{Closing Speed} = \left[\frac{1}{1+e}\right]\left[|\Delta V_1| + |\Delta V_2|\right]$$

Equation 9A – for inline CENTRAL collisions only where the damage centroids reach a common velocity

- When using Delta V to calculate closing speed, if the offset is not considered, the closing speed will be ***under***-estimated.
- These equations good for inline collisions, give CS in direction of PDOF. May understate total CS in intersection collisions

TRIANGULAR VELOCITY VECTOR METHOD

APPOACH VELOCITY = V1

DEPARTURE VELOCITY = V3

CHANGE IN VELOCITY = ΔV

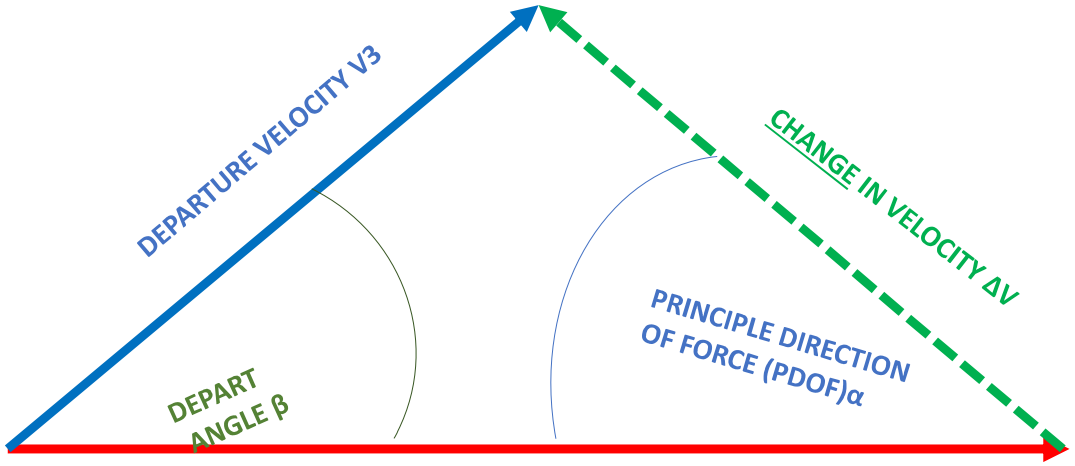
DEPARTURE ANGLE = β

PDOF = α

SIDES AND OPPOSITE ANGLES (PAIRS) ARE SHOWN IN THE SAME COLOR

IF WE KNOW ANY 3 SIDES OR ANGLES WE CAN FIGURE OUT THE REST

.

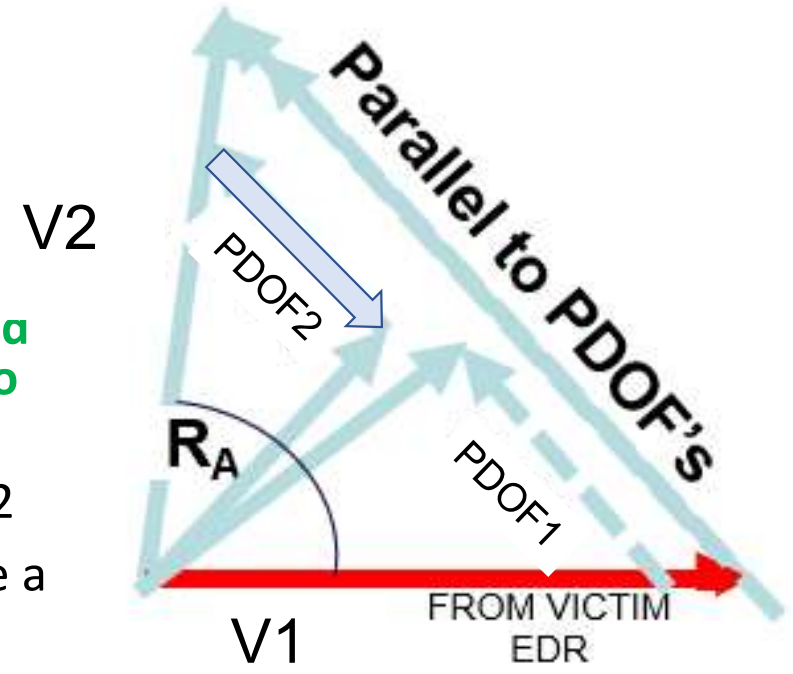


PDOF Triangle -2nd Triangle

1. **Must know relative approach R_A**
2. **Get PDOF1 from V1 EDR**
3. **PDOF2=180- R_A -PDOF1**
4. **ΔV2 = -ΔV1(W1/W2)**
5. **Must have departure speed or angle (3rd piece) of V2 to solve**
6. Use law of sines to solve V2

Note small inner triangles have a parallel edge (parallel PDOF's)

Outer triangle is the PDOF triangle.



ACM Not at CG

Thanks to Andy Rich for developing most of this section, and Bob Scurlock, Ph.D. for researching it





ACM Not at CG
Two possible conditions

Force applied closer to CG



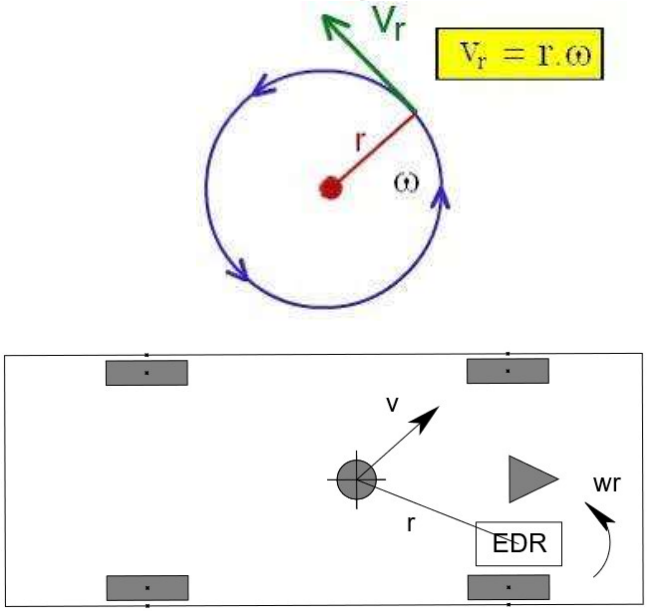
EDR under-reports delta-V

Force applied closer to EDR



EDR over-reports delta-V

ACM Not at CG – Theory
Over/Under reporting is caused by rotation



$$\Delta v^{EDR} = \Delta v^{CG} + \omega r$$

ACM Not at CG

- The following equation assumes that the EDR is mounted at the centerline (CL) of the car and that the CG is also at the CL.
- If the EDR is not at the CL see

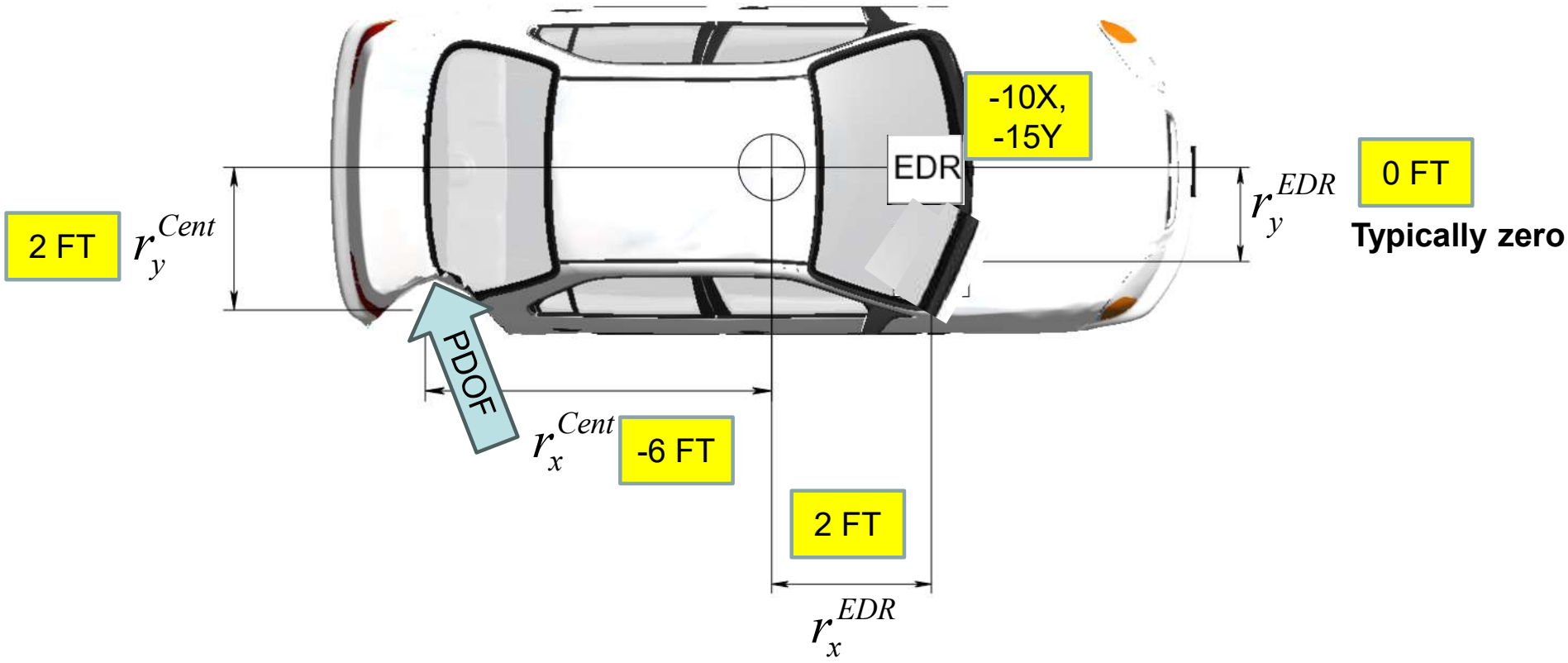
Corrections to Off-Axis Δv Measurements from Event Data Recorders

Bob Scurlock, Ph.D., ACTAR, Andrew Rich, BSME, ACTAR, and Kyle Poe

- June 2021 Collision Magazine
- Available at https://arxiv.org/abs/2001.03165
- Heavy math needed to handle all EDR locations and CG's– will strain most brains – but with EDR on CL in most typical cases it simplifies to:

$$\Delta v_y^{CG} = \frac{\Delta v_y^{EDR} \cdot k^2 + \Delta v_y^{EDR} \cdot r_y^{cent} \cdot r_x^{EDR}}{k^2 + r_x^{cent} \cdot r_x^{EDR}}$$

ACM Not at CG Inputs



Correction for Delta V at EDR to CG

Pass rear side impact example

$$\Delta v_y^{CG} = \frac{\Delta v_y^{EDR} \cdot k^2 + \Delta v_x^{EDR} \cdot r_y^{cent} \cdot r_x^{EDR}}{k^2 + r_x^{cent} \cdot r_x^{EDR}}$$

Solve for
- Δv_y^{CG} is the lateral delta-V at the CG

-15 mph
- Δv_y^{EDR} is the lateral delta-V reported by the EDR

21.4
- k^2 is the radius of gyration squared

10 mph
- Δv_x^{EDR} is the longitudinal delta-V reported by the EDR

2 FT
- r_y^{cent} is the lateral displacement of the damage centroid from the CG (Polarity positive if on pass side)

2 FT
- r_x^{EDR} is the longitudinal displacement of the EDR from the CG (RHR) (typically 0 to +2 feet)

6 FT
- r_x^{cent} is the longitudinal displacement of the damage centroid from the CG (negative if centroid rear of CG, positive if fwd)

Correction for EDR Delta V to CG

Pass rear side impact example

-15 mph | 21.4 | -10 mph | 2 FT | +2 FT

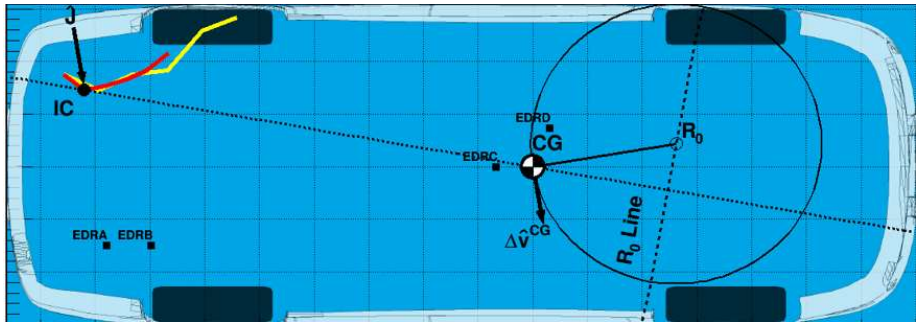
$$\Delta v_y^{CG} = \frac{\Delta v_y^{EDR} \cdot k^2 + \Delta v_x^{EDR} \cdot r_y^{cent} \cdot r_x^{EDR}}{k^2 + r_x^{cent} \cdot r_x^{EDR}}$$

21.4 | -6 FT | +2 FT

-321-40 = 38.4 mph
21.4-12 (61.8 kph)

- unknown • Δv_y^{CG} is the lateral delta-V at the CG
- -15 mph • Δv_y^{EDR} is the lateral delta-V reported by the EDR
- 21.4 • k is the radius of gyration
- -10 mph • Δv_x^{EDR} is the longitudinal delta-V reported by the EDR
- 2 FT • r_y^{cent} is the lateral displacement of the damage centroid from the CG (Polarity positive if pass side)
- 2 FT • r_x^{EDR} is the longitudinal displacement of the EDR from the CG (RHR) (typically 0 to +2 feet)
- -6 FT • r_x^{cent} is the longitudinal displacement of the damage centroid from the CG (Negative if centroid rear of CG, positive if fwd)

Beware the R_0 Line & the R_0 Point!



The R_0 line is a function of r_x^{EDR}

$$r_y^{EDR} = -\frac{k^2}{r_y^{cent}} - \left(\frac{r_x^{cent}}{r_y^{cent}}\right) r_x^{EDR}$$

- There is an imaginary line (the R0 line) where $\Delta v_y^{EDR} = 0mph$ so we cannot solve for a unique Δv^{cg}.
- When an accelerometer (EDR) is close to the R_O line, your analysis may be very sensitive to uncertainties in accelerometer location.
- You should do a sensitivity analysis considering actual accelerometer position in EDR may be +/-3" of the module center you measured. If results diverge excessively, use caution.

Beware the R_0 Line!

- Sensitivity Analysis:
  - Measure or look up the location of the EDR with respect to the CG.
  - Note that some BMWs have EDR forward of CG under dash, but the sensors are at CG!
  - Measure or determine the dimensions of the EDR. Include those measurements in your uncertainty analysis (except if you know exactly where the accelerometer is for your ACM).
  - Don't forget delta-V accuracy. See the Scurlock/Rich/Poe paper and other EDR technical papers for detailed discussions.

Analysis Update: Delta V Data

1. **Ground Forces were typically ignored in reconstruction**
2. **In small ΔV's (ped/mc) ground forces become significant**
3. **If car locks up brakes before hitting ped, Long. ΔV of ped hit is overstated**
4. **If MC t-bones car, Lat ΔV understated**
5. **ΔV is multiplied by big weight ratio, Closing speed error gets bigger**

INSPECTION PROCEDURE CHANGES

- Measure tread depth of tires in cases where speed is critical
- Look up ACM location. If **NOT** on CG, ***and*** there was significant rotation during crash, then
  - Document exact location of ACM vs CG at inspection
  - Document exact location of damage centroid vs CG

    (may be done later if vehicle was scanned)

Draw vehicles at max engagement, draw PDOF line thru damage centroid, understand which side of CG PDOF line passes by and approximate distance from CG

NHTSA CISS DATABASE UPDATE

1. https://crashviewer.nhtsa.dot.gov/CISS/SearchIndex
2. 14,689 cases available from 2016-2021 calendar years – last updated Dec 2022 for 2021 CY
3. Searchable by manufacturer, model year, model, EDR data available, driver assist technology available, etc
4. Find similar cases, check polarity, see typical EDR report in 7710 of those cases
5. Link to VIN decoder, Canadian Specs, Vin check digit calc'tr Welcome to VIN Decoding :: provided by vPIC (dot.gov)

Chevrolet SDM Phase In 21_5 87%

SDM50

X DV

+ SPEED

+Y DV

563

3 events

	GREEN = 300 ms Longitudinal Delta V					Red = Precrash Data & 150ms Longitudinal Delta V						Blue = Precrash Data & 300ms XY Delta V					3 Events		Part 563 Intent								SDM50	GM		
Model	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Astro			2888	2888	2888	2888	2829	2829	2829	2829	2829	2829																		
Avalanche									2829	3003	3003	3003	3003	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	*									
Aveo																3320	3320	3320		454	*				454					
Beat (India)																					454	454								
Blazer					2888	2888	2829	2829	2829	2829	2829	2829*	*													454	454	454	454	454
Bolt EV																									454	454	454	454	454	454
Camaro			2888	2888	2888	2829	2829	2829	2829								454	454	454	454	454	454	454	454	454	454	454	454	454	454
Caprice	3002	3002	3002															452	**454**	police	police	police	police	police	police					
Captiva																	454	454	454	454	*	Sport	Sport							
City Express																						780	780	780	780	780	454	454	454	852
Colorado											3003	3003	3003	3003	3003	3003	3003	3003	3003			454	454	454	454	454	454	454	454	454
Corvette				2888	2888	2829	2829	2829	2829	2829	2829	3003	3003	3003	3003	3003	3003	3003	3003	3003	454	454	454	454	454	454	845	845	845	845
Cruze																	454	454	454	454	454	454	454	454	454	454	454			
Equinox												3004	3004	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454	454	454	454	454	454
Express			2888	2888	2888	2888	2888	**2829**	**2829**	3003	3003	3003	3003	3003	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454
Impala		3002	3002				2829	2829	2829	2829	2829	2829	3320	3320	3320	3320	3320	3320	**454**	454	454	454	454	454	454	454	454			
Kodiak										2829	2829	2829	2829	2829	2829	2829			**2012**	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Low Cab Fwd																										454				
Malibu				2888	2888	2888	2829	2829	2829	2829	3321***	3321 mix	3321	3321	3321	3321	3321	3321	3321	3321 (2012	454	454	454	454	454	454	454	454	454	454
Matiz																			454											
Orlando																			454	454	454									
Silverado				2888	2888	2888	MIX	2829	2829	3003	3003	3003	3003	3320 ROS*mix	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454	454	454	454	454	454	852
Sonic																			454	454	454	454	454	454	454	454	454			
Spark	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011		454	454	454	454	454	454	454	454	454	454	
SSR										3003	3003	3003	3003									SS	**454**	**454**	**454**					
Suburban				2888	2888	2888	MIX	2829	2829	3003	3003	3003	3003	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454	454	454	454	845	845	852
Tahoe				2888	2888	2888	MIX	2829	2829	3003	3003	3003	3003	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454	454	454	454	845	845	852
Tracker			DLC	DLC	DLC	DLC	DLC	DLC	DLC	DLC	DLC														454	454	454	454	454	
TrailBlazer									2829	2829	2829	3293	3293	3293	3293	3293												454	454	454
Traverse																3320 ROS	454	454	454	454	454	**454**	**454**	**454**	**454**	454	454	454	454	454
Trax																				454	454	454	454	454	454	454	454	454	454	
Volt																		454	454	454	454	454	454	454	454	454				
	14%	48%	76%	94%	100%	20%	74%	86%	96%	98%	100%	14%	43%	71%	79%	82%	91%	92%	22%	96%	100%		100%							
PHASE IN	Delta V	Delta V	Delta V	Delta V	Delta V	Pre-Crash	Pre-crash	Pre-crash	Pre-Crash	Pre-Crash	Pre-Crash	Pre + XY	Pre + XY	Pre + XY	Pre + XY	Pre + XY	Pre + XY	Pre + XY	563	563	563	563								

Freeze Frame Data

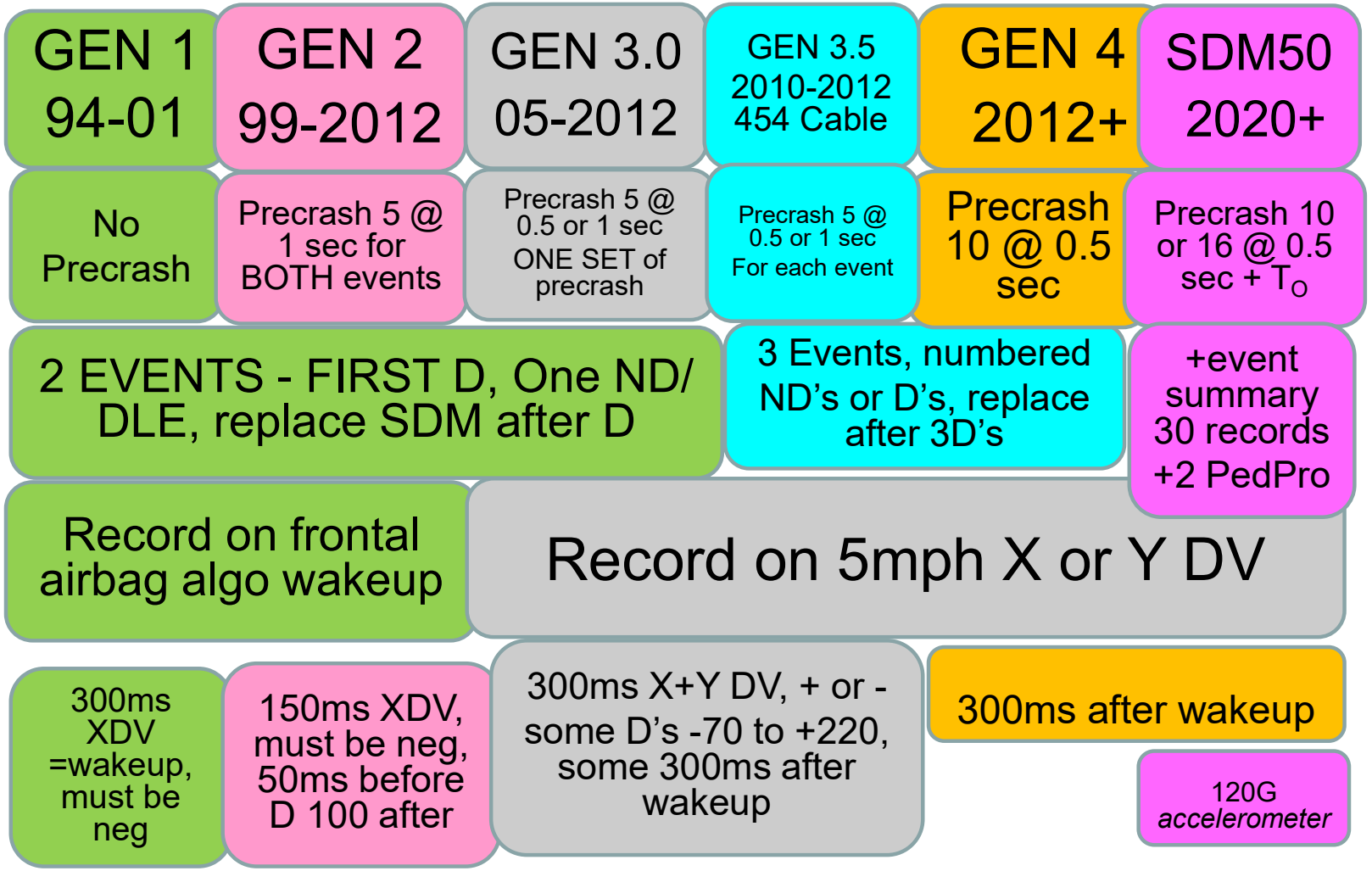
** = 150ms of Longitudinal Delta V instead of 300ms

Chevrolet ACM Recent Changes 87% 563

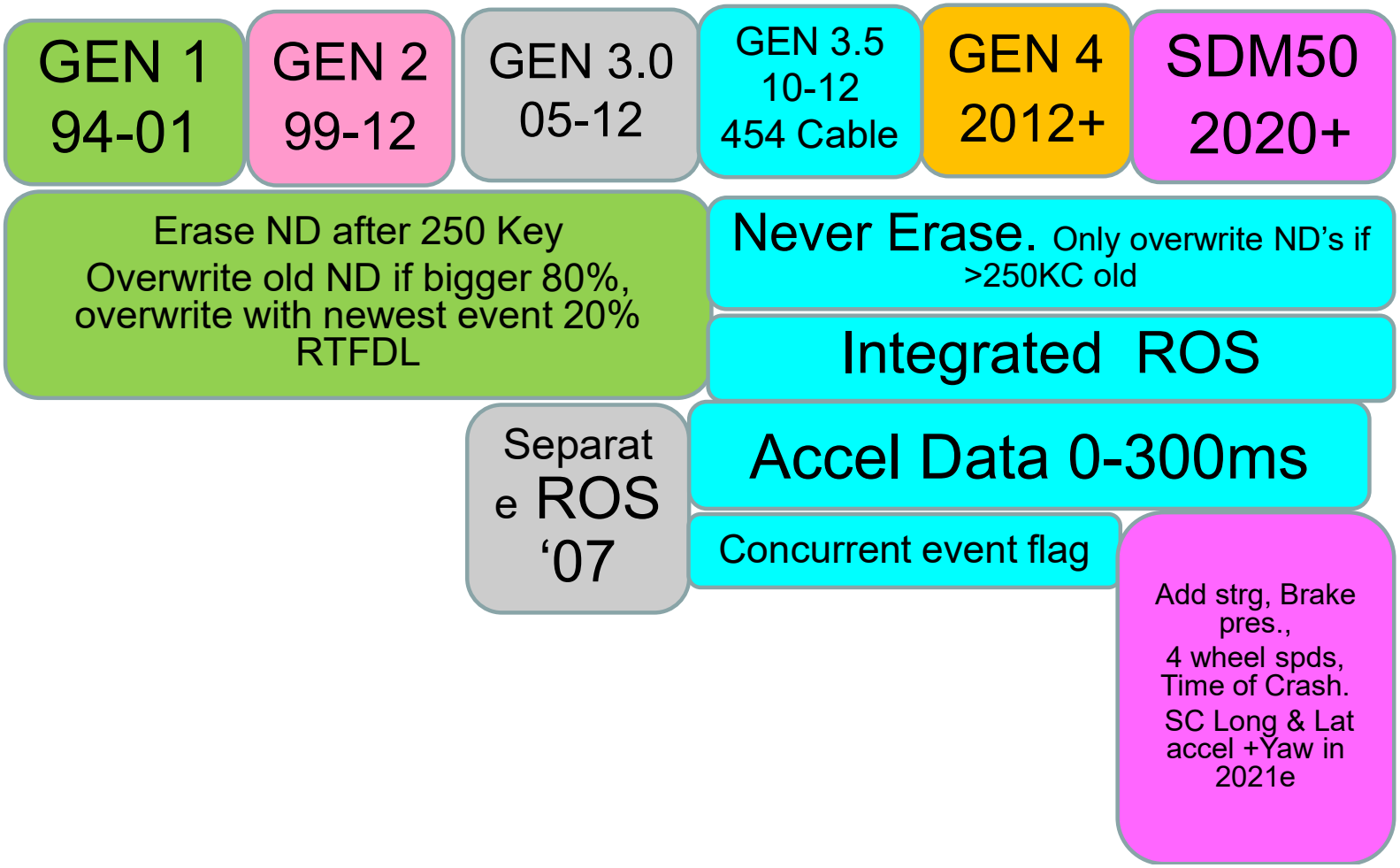
	Part 563 Intent								SDM50	GM			
Model	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	2024
Astro													
Avalanche	454	454	*										
Aveo		454	*				454						
Beat (India)			454	454									
Blazer								454	454	454	454	454	
Bolt EV							454	454	454	454	454	454	
Camaro	454	454	454	454	454	454	454	454	454	454	454	454	
Caprice	**454**	police	police	police	police	police	police						
Captiva	454	454	*	**Sport**	**Sport**								
City Express				**780**	**780**	**780**	**780**	**780**					
Colorado	3003			**454**	**454**	**454**	**454**	**454**	454	454	454	**852**	**852**
Corvette	3003	3003	454	454	454	454	454	**454**	**845**	**845**	**845**	**845**	845
Cruze	454	454	454	454	454	454	454	**454**	**454**				
Equinox	454	454	454	454	454	454	454	**454**	454	454	454	454	
Express	3320 ROS	3320 ROS	3320 ROS	3320 ROS	454	454	454	**454**	**454**	454	454	454	
Impala	**454**	454	454	454	454	454	454	**454**	**454**				
Kodiak	**2012**	**2013**	**2014**	**2015**	**2016**	2017	2018	2019	2020	2021	2022	2023	
Low Cab Fwd								454					
Malibu	3321	3321 (20	2 454	454	454	454	454	454	**454**	454	454	454	
Matiz	454												
Orlando	454	454	454										
Silverado	454	454	454	454	454	454	454	454	**454**	**454**	**454**	**852**	**852**
Sonic	454	454	454	454	454	454	454	454	**454**				
Spark		454	454	454	454	454	454	454	**454**	**454**	**454**		
SSR				**SS**	**454**	**454**	**454**						
Suburban	454	454	454	454	454	454	454	454	**454**	**845**	**845**	**852**	**852**
Tahoe	454	454	454	454	454	454	454	454	**454**	**845**	**845**	**852**	**852**
Tracker							454	454	**454**	**454**	**454**		
TrailBlazer										**454**	**454**	**454**	
Traverse	454	454	454	**454**	**454**	**454**	**454**	454	**454**	**454**	**454**	**454**	**852**
Trax		454	454	454	454	454	454	454	**454**	**454**	**454**		
Volt	454	454	454	454	454	454	454	454					

No Precrash in 2013-5 Express/Savanna

GM EDR Evolution



GM EDR Evolution #2



2020+ Corvette SDM50 – More Data!

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 1 of 6

Time (sec)	Accelerator Pedal, % Full (%)	ABS Activity	Brake Pedal Position (%)	Driver Applied Brake Pedal Pressure (kPa)	Driver Applied Brake Pedal Pressure Detected	Brake Pedal Moderate Travel Achieved
-8.0	0	Off	0	0	False	False

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 2 of 6

Time (sec)	Cruise and Speed Limiter Switch Status	Cruise Secondary Switch Status	Cruise Control Active	Adaptive Cruise Control Active Authenticated	Transmission Estimated Gear	Transmission Shift Lever Position	Reduced Engine Power Mode Indicator On
-8.0	No Activation	No Activation	Inactive	Inactive	Ninth	Forward Range A	Off

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 3 of 6

Time (sec)	Engine RPM (RPM)	Engine Torque (foot-pounds)	Engine Throttle, % Full (%)	Speed, Vehicle Indicated (MPH [km/h])	Right Front Wheel Angular Velocity (RPM)	Left Front Wheel Angular Velocity (RPM)
-8.0	1,301	50.5	16	44.1 [71]	595	598

SC accel + Yaw 2021

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 4 of 6

Time (sec)	Right Rear Wheel Angular Velocity (RPM)	Left Rear Wheel Angular Velocity (RPM)	Steering Wheel Angle (deg)	Propulsion System Active	Run/Start Input	Ignition Prolongation Time (sec)	Secondary Collision Prolongation Timer (sec)
-8.0	599	598	0	True	High	5.0	0.0

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 5 of 6

Time (sec)	Antilock Brake System Failed	Brake Pedal Override Flag	Traction Control System Present	Traction Control System Failed	Traction Control System Enabled	Traction Control System Active	Traction Control System Driver Intent
-8.0	False	False	True	False	True	Traction Control Off	Enabled

More 2020 Corvette SDM50

- GPS Lat/Long at crash
- Turn Signal Switch Status
- Second and third row (if equipped) seat belt status
- Up to 30 Event Summary records give clear order of events
- Outside Air Temp, Brake Overrides
- Traction Control activate
- Auxilliary Crash Sensor Data (recon does not use)

EVERYTHING GM COULD IMAGINE!!!!!

Uses of SDM50 Data similar to Ford & Stellantis

- Use individual wheel rpm's as additional independent measurements of speed to verify speed
- Use Brake Pedal Position and Brake Pressure to determine how hard driver is pushing on the brake pedal (classify panic braking vs normal braking)
- Use Longitudinal Accel from stability to more precisely get speed at impact if braking from a stable speed
- Use lat accel and yaw to get change in heading and approach angle from original, for momentum & triangle calcs

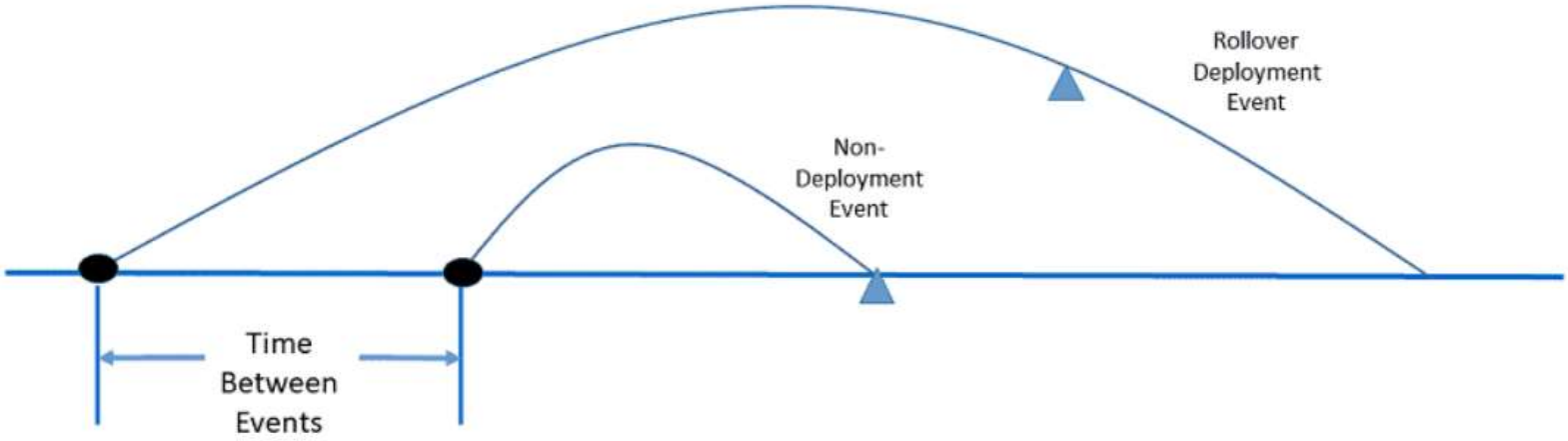
Pre SDM50: GM Gen 3.5 & 4 Features

1. 2010 454 cable SDM's do not require replacement after a D (until 3 D's).
2. Events & deployments all numbered vs. labeled D/ND.
3. Events may **not** be in order algorithms started, they are in order they **qualified** for recording (see graphic next page)
4. **Align precrash data for all events to see relationship**
5. Black plastic SDM housings
6. Accelerometer max values listed in DL's

95

SDM 10-40 Concurrent Event Flag

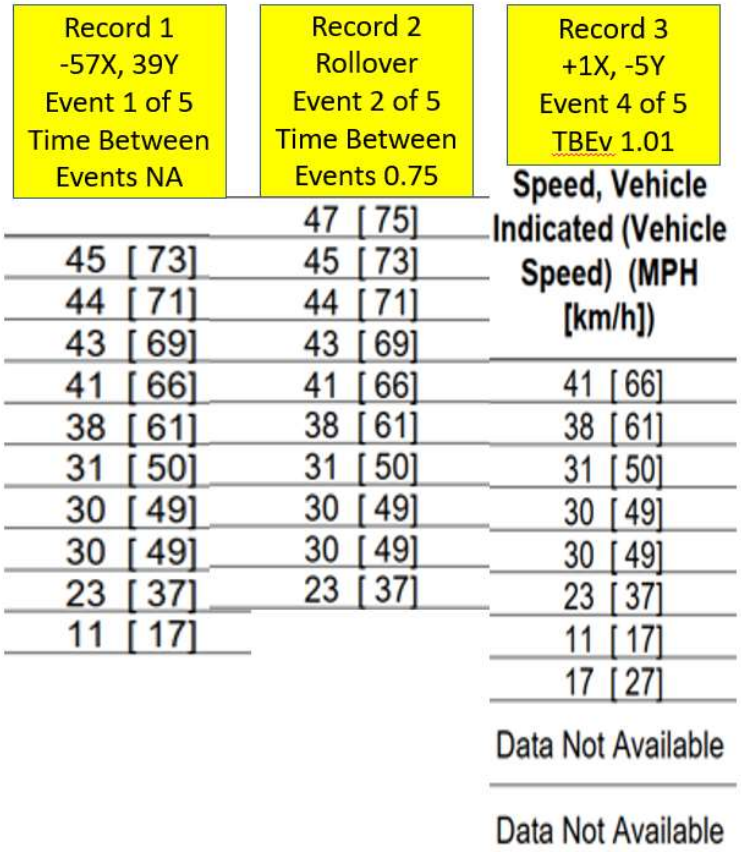
-A Concurrent Event is when two events are happening nearly simultaneously. The "Concurrent Event Flag Set" parameter will indicate "Yes" if one event begins, but before that event is qualified, another event begins and is qualified.
A Non-Deployment event typically becomes qualified if that event exceeds the 5 MPH (8 km/h) delta V recording threshold and the event has concluded. A deployment event (FSR or Rollover) becomes qualified when a deployment has been commanded for that event.
Example of a Concurrent Event:
A Rollover event begins. Before the Rollover event is qualified, a Non-Deployment event begins and is qualified. Sometime after the Non-Deployment event is qualified, the Rollover event is qualified. The Non-Deployment event will be recorded in the first open record even though the Rollover event enabled before the Non-Deployment event. The Rollover event will be recorded in the next open record. The "Concurrent Event Flag Set" parameter will indicate "Yes" for the Rollover event. The "Time Between Events" parameter will indicate the time from the start of the Rollover event to the start of the Non-Deployment event.



Event Recor #1	Event Record #2
Event Record Type = Non – Deployment	Event Record Type = Rollover
Concurrent Event Flag = No	Concurrent Event Flag = Yes
Time Between Events = NA	Time Between Events = XX seconds

When Concurrent Event Flag is Yes, Overlay Precrash Data from all events

- The event number gives when the event QUALIFIED for recording, not when I started
- Time between events is time between consecutive wake-ups, NOT always time between qualifying events
- Recap Delta V or, if rollover, event type over precrash data to assist understanding



Noteworthy GM news

1. Conti SDM-30 data incomplete file not accurately interpreted by CDR – not fixable. Must send to Conti for readout and interp. (data lims detail how)
2. GM note 6 (no key needed) vehicles initially had 2 fuses, SDM bat and SDM ignition. Newer GM vehicles may have only ONE SDM fuse, normally bat, but police packages may only power it with ignition. SDM 50 ***requires key to be on*** to read SDM.

GM Active Safety Control Module

1. First installed in 2013 Cadillac ATS
2. Only with GM option UGN or UKL
3. Initially released in CDR 17.9 to get raw data file only for future interpretation.
4. Interpretation capability added in 19.1
5. REQUIRES CDR 900 interface!
6. Some vehicles have TWO ASCM modules.
7. Typically located in luggage compartment on the side, behind trim

GM ASCM FCM CDR 23.1 Coverage

Model	Location	Model Year											
3 digits Bosch CDR cable ID shown		2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
BUICK													
ENCLAVE	cargo left front of wheel well						ASCM						
ENCORE									FCM				
ENVISION	rear seat left trim by wheel wl				ASCM			FCM					
LACROSSE	trunk left					ASCM							
REGAL	trunk left						ASCM						
CADILLAC													
ATS								FCM					
CELESTIQUE													ASCM+FCM
CT4									ASCM				
CT5									ASCM				
CT6 w/o UKL	Trunk right					ASCM	without supercruise						
CT6 w/UKL	Primary right, secondary le				834		Supercruise Option Code UKL						
CTS SEDAN	trunk left		ASCM										
CTS	trunk left				ASCM			FCM					
ESCALADE	under left qtr window trim			ASCM									
LYRIC												ASCM	ASCM
SRX	under left rear seat	ASCM											
XT4	cargo left front of wheel well							ASCM					
XT5	cargo left front of wheel well					ASCM							
XT6										ASCM			
XTS								FCM					
CHEVROLET													
BLAZER	trunk left							ASCM					
BOLT												ASCM/F	ASCM/FCM
CAMARO												FCM	
CRUZE								FCM					
EQUINOX								FCM				FCM	
MALIBU	trunk left				ASCM								
SILVERADO								FCM					
SPARK								FCM					
SUBURBAN												ASCM	
TAHOE												ASCM	
TRAVERSE	trunk right						ASCM						
VOLT	trunk left					ASCM		FCM					
GMC													
ACADIA	cargo left front of wheel well					ASCM							
HUMMER EV												ASCM	
SIERRA								FCM				ASCM/FCM	
SIERRA HD												FCM	
TERRAIN								FCM					

GM Active Safety Control Module

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
4	2018 Cadillac	ATS (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, left side, near rear, on outboard side of brace, behind trim
6	2018 Cadillac	ATS-L (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel
4,6	2018 Cadillac	CT6 (note 6)	ACM	Data	CANplus or CDR 900	CANplus - F00K108287 CDR 900 - click here	CANplus - F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
			ASCM (Note2) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 834	In the luggage compartment, right center, mounted to the right side in the battery well. The secondary ASCM is on the left.
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, right center, mounted to the right side in the battery well

GM Active Safety Module



1. Must be downloaded separately from ACM
2. Icon will appear after VIN input for any vehicle that MIGHT have Super Cruise
3. Module /data will only be present if option “UKL” (Super Cruise) or “UGN” (Adv Driver Assist) has been purchased for the subject vehicle (most DO NOT have it).

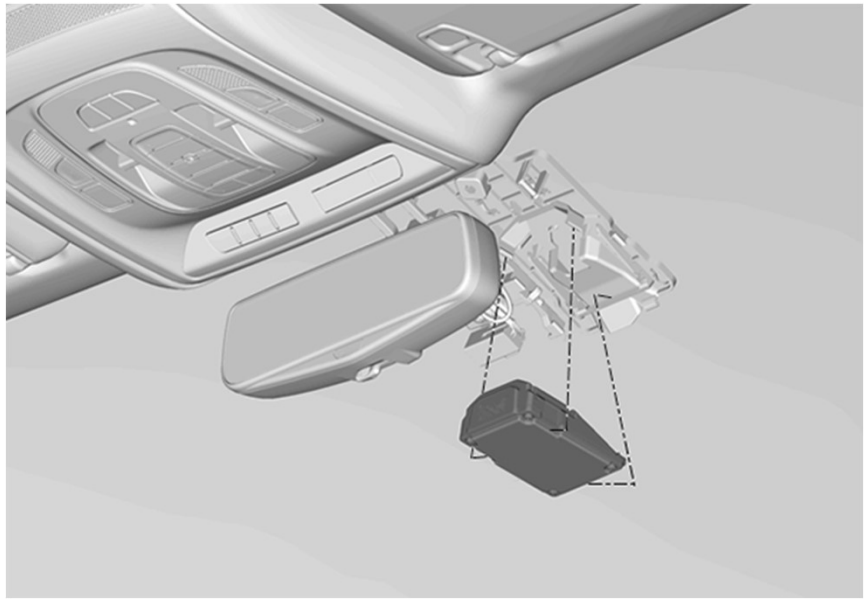
GM Active Safety Module

1. Data can be captured every 80 milliseconds and can record up to 50 records for a total of 4 seconds of data. The data recording window can be either 4 seconds before the end of an auto braking event or 4 seconds before an airbag deployment. ONE EVENT ONLY
2. Data is recorded in non-volatile memory at the next **proper power-down** of the module. Data may **not** be stored if the battery power is lost before the power-down.

GM Auto Emergency Braking

1. AEB will prevent many crashes and reduce the severity of crashes still happening
2. If there is still a crash, the AEB system may store up to 3 camera images in FCM*:
   a) 4 sec before event (*Fwd Camera Module)
   b) At event
   c) 4 sec after event
3. Some accessible with CDR 21.2 – others must be requested with proper legal authority from GM

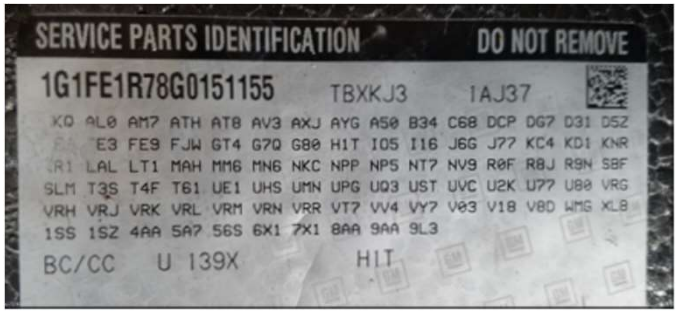
Front Camera Module (FCM)



FCMs are mounted on the windshield near the rear view mirror.

GM FCM EDR

- GM released the ability to read data from Front Camera Modules(FCM) with V 21.2 of the CDR Software (Date: 8/2021)
- The data consists of photos taken before, during, and after the crash.
- Not all GM cars are equipped with FCMs
- Some cars have FCMs but are not supported at this time
- We can identify which cars have an FCM by
  - Looking for presence of the camera module near the rear view mirror
  - Verifying the (RPO) UEU and UGN in the vehicles Service Parts Identification sticker. New QR code on the FMVSS Sticker does NOT have enough detail – get list of options by giving dealer a VIN. (see photos)





Current FCM access released in GREEN

Model	Location	Model Year											
3 digits Bosch CDR cable ID shown		2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
BUICK													
ENCLAVE	cargo left front of wheel well						ASCM						
ENCORE									FCM				
ENVISION	rear seat left trim by wheel wl				ASCM			FCM					
LACROSSE	trunk left					ASCM							
REGAL	trunk left						ASCM						
CADILLAC													
ATS								FCM					
CELESTIQUE													ASCM+FCM
CT4									ASCM				
CT5									ASCM				
CT6 w/o UKL	Trunk right					ASCM	without supercruise						
CT6 w/UKL	Primary right, secondary le				834		Supercruise Option Code UKL						
CTS SEDAN	trunk left		ASCM										
CTS	trunk left				ASCM			FCM					
ESCALADE	under left qtr window trim			ASCM									
LYRIC												ASCM	ASCM
SRX	under left rear seat	ASCM											
XT4	cargo left front of wheel well							ASCM					
XT5	cargo left front of wheel well					ASCM							
XT6										ASCM			
XTS								FCM					
CHEVROLET													
BLAZER	trunk left							ASCM					
BOLT												ASCM/F	ASCM/FCM
CAMARO												FCM	
CRUZE								FCM					
EQUINOX								FCM				FCM	
MALIBU	trunk left				ASCM								
SILVERADO								FCM					
SPARK								FCM					
SUBURBAN												ASCM	
TAHOE												ASCM	
TRAVERSE	trunk right						ASCM						
VOLT	trunk left					ASCM		FCM					
GMC													
ACADIA	cargo left front of wheel well					ASCM							
HUMMER EV												ASCM	
SIERRA								FCM				ASCM/FCM	
SIERRA HD												FCM	
TERRAIN								FCM					

GM FCM EDR

- The Lookup file will identify which GM vehicles may have an

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
4,6	2019 Cadillac	XTS (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel, forward
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, under the left side trim panel
			FCM (Note1) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 856	Center - top interior of windshield

ACM Data may be present

ASCM Data may be present

FCM Data may be present

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GM FCM EDR

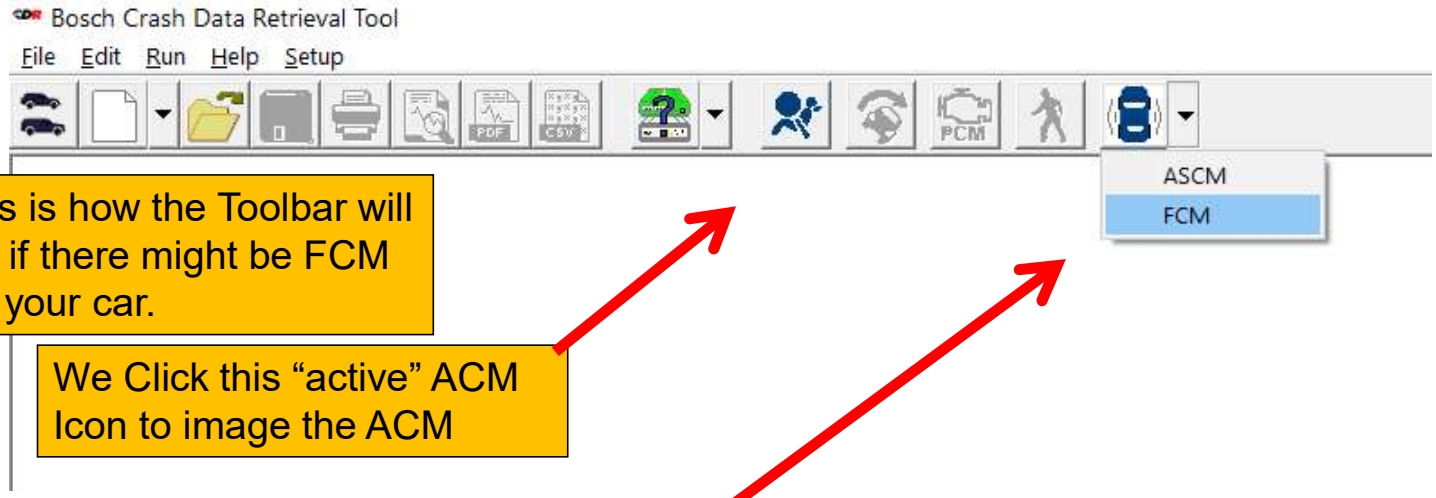
- The Lookup file will identify which GM vehicles may have an

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
4,6	2019 Cadillac	XTS (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel, forward
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, under the left side trim panel
			FCM (Note1) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 856	Center - top interior of windshield

Note we get hints on how to DL the FCM – How to image

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GM ASCM EDR – Bosch CDR 900



And this is how the Toolbar will appear if there might be FCM data in your car.

We Click this “active” ACM Icon to image the ACM

Then once the ACM Download is complete and saved, we click this “active” Dropdown Icon to image the FCM and then save that file as well. Now we have 3 things to image!!

GM ASCM EDR – Bosch CDR 900



We repeat the idea that we discussed in the DLC training: If you fail to download and save the data from all available modules you are driving away and leaving unread data in the car.

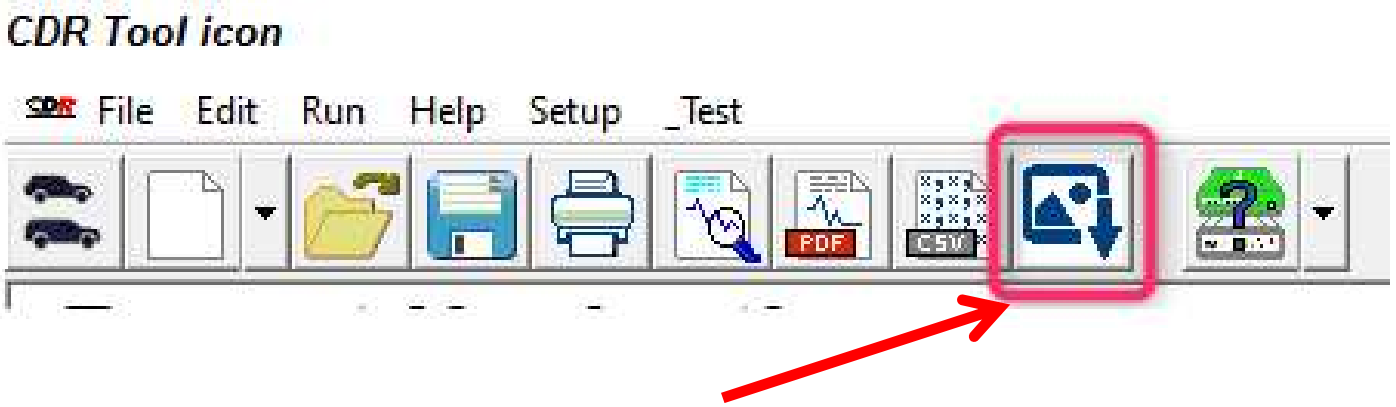
And, just because you can read the ACM via the DLC does not guarantee that you have communication with the FCM, ASCM, FCM, ROS, etc .

GM FCM EDR

- There are several network pathways in modern vehicles
- Damage may allow DLC communication to one module and prevent comm to another module.
- Do the research to see if your car is equipped with all indicated modules (ACM, ASCM, FCM) and read as necessary (DLC or DTM or maybe one of each)
- At this time FCMs are only readable on 2019MY and later GM's
- If you previously read the SDM and ASCM on an FCM now-supported vehicle, you may have to return to read the data from this new module type
- The FCM raw data was NOT saved in any ASCM files you may have read.

GM FCM EDR

- When a CDR file contains image data, the user can export the data using options from the CDR software:



GM FCM EDR

- The Front Camera Module (FCM) can record two events. Recording is triggered by “events of interest”. The following are possible “events of interest” that trigger data recording listed in the priority order:
- Airbag Deployment Event,
- Pretensioner Deployment Event,
- Non-Deployment Event,
- Pedestrian Braking,
- Collision Imminent Breaking (CIB)
- Pedestrian Warning,
- Front Collision Alert,
- Lane Departure Warning.

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GM FCM EDR

- The new event is stored only if it is a higher priority than the priority of one of the currently stored events. The new event overwrites the lowest priority event that is already stored.
- If all previously stored events are of equal priority (and are lower priority than the new event), the oldest stored event is the one being overwritten.
- The image data is not rendered directly in the CDR program .

- It must be exported to a file in the images' native format.

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GM FCM EDR

- Photos are black and white *.pgm files (**P**ortable **G**ray **M**ap)
- You will need a *.pgm file viewer – many free *.pgm apps are available.

  (Try www.coolutils.com/online/PGM-to-JPG)
- Users can regenerate the images collected by the FCM as many times as they wish recreating the file as it existed when the data was collected from the vehicle.
- Yet again we see that having the Raw file IS the evidence – As an analyst, *if sharing with another,* you want the raw FCM cdrx file, not someone else's downloaded *.pgm images.
- The data in the FCM CDRx file remains untouched and cannot be changed.
- The images can be downloaded and generated from the Raw file as many times as necessary.



GM FCM EDR

- If the exported image file appears to be changed by another user, you can re-export the image directly from the CDR file it was created; therefore, removing any doubt that the data source was changed after the file was generated.
- One more issue… The FCM image will be upside down.



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GM FCM EDR

- The solution is to “Invert” or Flip the photo. DO NOT “rotate” it.



- Original FCM “upside down” photo



- A successfully “inverted” or “flipped” photo

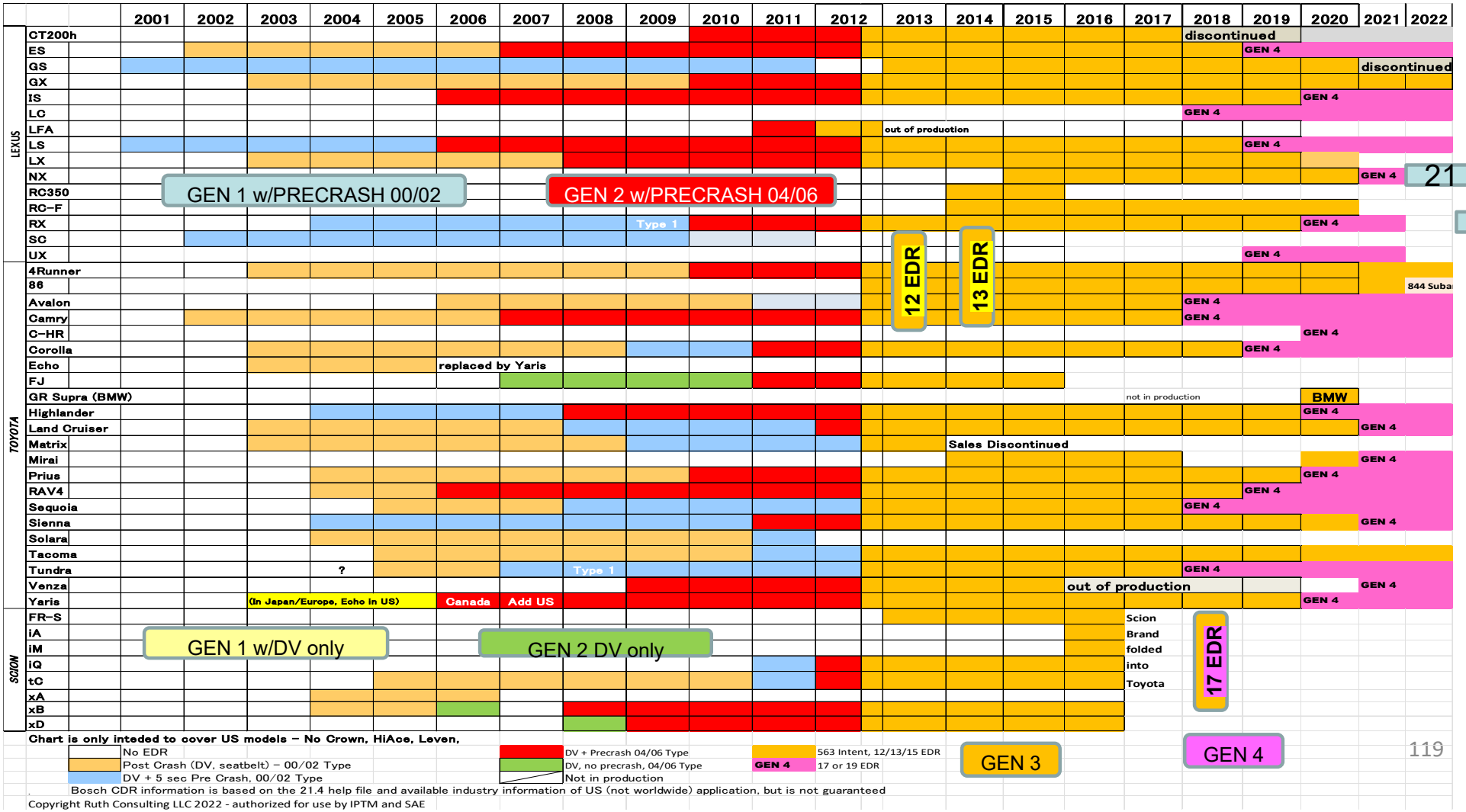
(if you convert pgm to jpg, grab the photo at the top and drag the top thru the bottom of the photo to invert it)

- A confusing, inaccurate “rotated” photo.

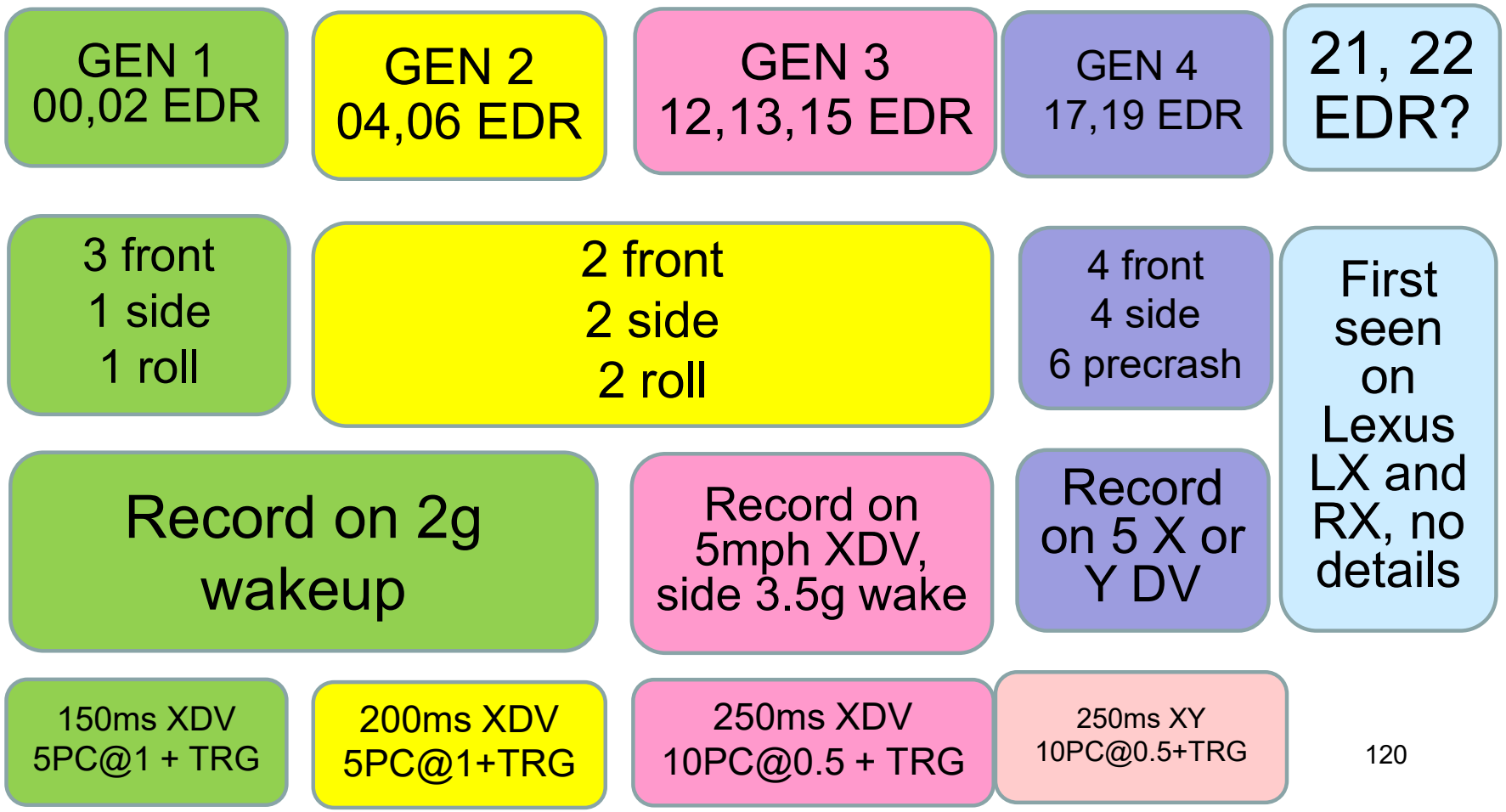
(Note car appears to be driving on *wrong side of road*)



Toyota US CDR 21.4 Coverage 84%



Toyota EDR Evolution



Toyota EDR Evolution cont'd

GEN 1 00,02 EDR	GEN 2 04,06 EDR	GEN 3 12,13,15 EDR	GEN 4 17,19 EDR	21, 22 EDR?
70ms YDV@ACM+sats		12 YDV 70ms sats only 13 250ms ACM	250ms YDV	
REFRESH 500 MS SPEED		REFRESH 24 MS SPEED		
78 MPH TOP, 2kph res	75 MPH TOP. 2kph res	124 MPH TOP 1 kph res		
Accel Pedal Volts		Accel Pedal % & Throttle %		

Toyota EDR Evolution cont'd

GEN 1
00,02 EDR

GEN 2
04,06 EDR

GEN 3
12,13,15 EDR

GEN 4
17,19 EDR

21, 22
EDR?

RPM to next lower 400

RPM to 100

No Key Cycles

Key Cycles

Time Between Events < 5 sec

Time Between Events <16 sec

Brk Oil Pres, Stab. Cont. Long Accel

Strg & Yaw

New “17 EDR & “19 EDR”

- “Front/Rear” events also triggered by 5mph Y DV or side deployment
- Y Delta V from ACM added to frontal events for recon
- More precrash registers (6 vs 2 prior)
- More front/rear & side registers (4 vs 2)
- Separate side events still reported, but do NOT have precrash associated with them. Side events have detail of side satellite sensors (generally ***not used*** by Recons).
- Odometer data element added
- 19 EDR adds date/time

17 EDR Event Record Summary
more events, detail on recording status
analyze 1st event associated with crash 1st

Event Record Summary at Retrieval

Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash Recording Status	Diagnostic Data Recording Status	Occupant Data Recording Status	Crash Info Recording Status	Time Series Recording Status
Most Recent Event	10	Side Crash	0	N/A	Complete	N/A		plete
1st Prior Event	9	Frontal/Rear/Si	-898	Complete	plete	Complete	Complete	Complete
2nd Prior Event	8	ide Crash	-925	N/A			N/A	Complete
3rd Prior Event	6	Side Crash	N/A		plete	N/A	N/A	Complete
4th Prior Event	5	Rollover	N/A	omplete	Complete	Complete	Incomplete	Incomplete
5th Prior Event	4	Frontal/Rear/Side Crash	N/A	Complete	Complete	Complete	Complete	Complete
6th Prior Event	3	Side Crash	N/A	N/A	plete	N/A	N/A	Complete
7th Prior Event	2	Rollover	N/A	Complete			Complete	Complete

Pages NOT numbered like Gen3

#7 missing

Lost count due to 7 missing

The big event – start here

No precrash for sides

Same Data, New Look to Precrash

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-5.00	29.8 [48]	49.5	24.5	Invalid	2,400	Invalid	OFF
-4.50	32.3 [52]	47.0	24.5	Invalid	2,500	Invalid	OFF
-4.00	34.2 [55]	38.0	20.0	Invalid	2,700	Invalid	OFF
-3.50	35.4 [57]	32.5	16.0	Invalid	2,800	Invalid	OFF
-3.00	36.7 [59]	26.5	13.5	Invalid	2,400	Invalid	OFF
-2.50	37.3 [60]	22.5	11.0	Invalid	2,400	Invalid	OFF
-2.00	37.3 [60]	19.0	8.5	Invalid	2,200	Invalid	OFF
-1.50	38.5 [62]	19.0	6.0	Invalid	1,900	Invalid	OFF
-1.00	37.9 [61]	18.0	5.5	Invalid	1,900	Invalid	OFF
-0.50	37.9 [61]	18.0	5.0	Invalid	1,800	Invalid	OFF
TRG(0)	36.0 [58]	100.0	26.0	Invalid	1,700	Invalid	OFF

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-5.00	OFF	OFF	0.00	1.579	-0.98	-4.5	D
-4.50	OFF	OFF	0.00	1.795	0.00	-1.5	D
-4.00	OFF	OFF	0.00	1.292	0.49	0.0	D
-3.50	OFF	OFF	0.00	0.861	0.00	-1.5	D
-3.00	OFF	OFF	0.00	0.718	0.00	-1.5	D
-2.50	OFF	OFF	0.00	-0.072	0.00	-1.5	D
-2.00	OFF	OFF	0.00	0.000	0.00	-1.5	D
-1.50	OFF	OFF	0.00	0.000	-0.49	-1.5	D
-1.00	OFF	OFF	0.00	0.287	0.00	-1.5	D
-0.50	OFF	OFF	0.00	-0.144	0.00	-1.5	D
TRG(0)	OFF	OFF	0.48	-1.507	61.00	-19.5	D

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 3 of 4

Time (sec)	Sequential Shift Range	Cruise Control Status	VSC Control Status	READY Signal	Drive Mode, Power Train	Drive Mode, Snow	Drive Mode, EV
-5.00	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid

Pedestrian Protection Module

System Status at Pedestrian Event (1st Prior Event, TRG 1)

TRG Count (times)	1
TRG Count not for Pedestrian (times)	SNA
Ignition Cycle, Crash (times)	353
Odometer signal (miles [km])	1,195 [1,923]
Trip count (times)	392
Time count (msec)	1,283,900
Time count input system	Normal
Airbag Warning Lamp, On/Off	OFF
Pedestrian Warning Lamp, On/Off	OFF
Time from Previous TRG (msec)	32767 or greater
Time from Pre-Crash to TRG (msec)	300
Time between TRG not for Pedestrian and TRG for Pedestrian (msec)	SNA
TRG Establishment not for Pedestrian, within 5 sec.	ON
Freeze Signal	ON
Recording Status, Pre-Crash	Complete
Recording Status, Crash Info.	Complete
Recording Status, EDR Correlation Info.	Complete

Deployment Command Data (1st Prior Event, TRG 1)

Front Pop-Up Hood, Time to Deploy (msec)	3.0
Rear Pop-Up Hood, Time to Deploy (msec)	3.0
Airbag for Pedestrian Deployment, Time to Deploy (msec)	SNA

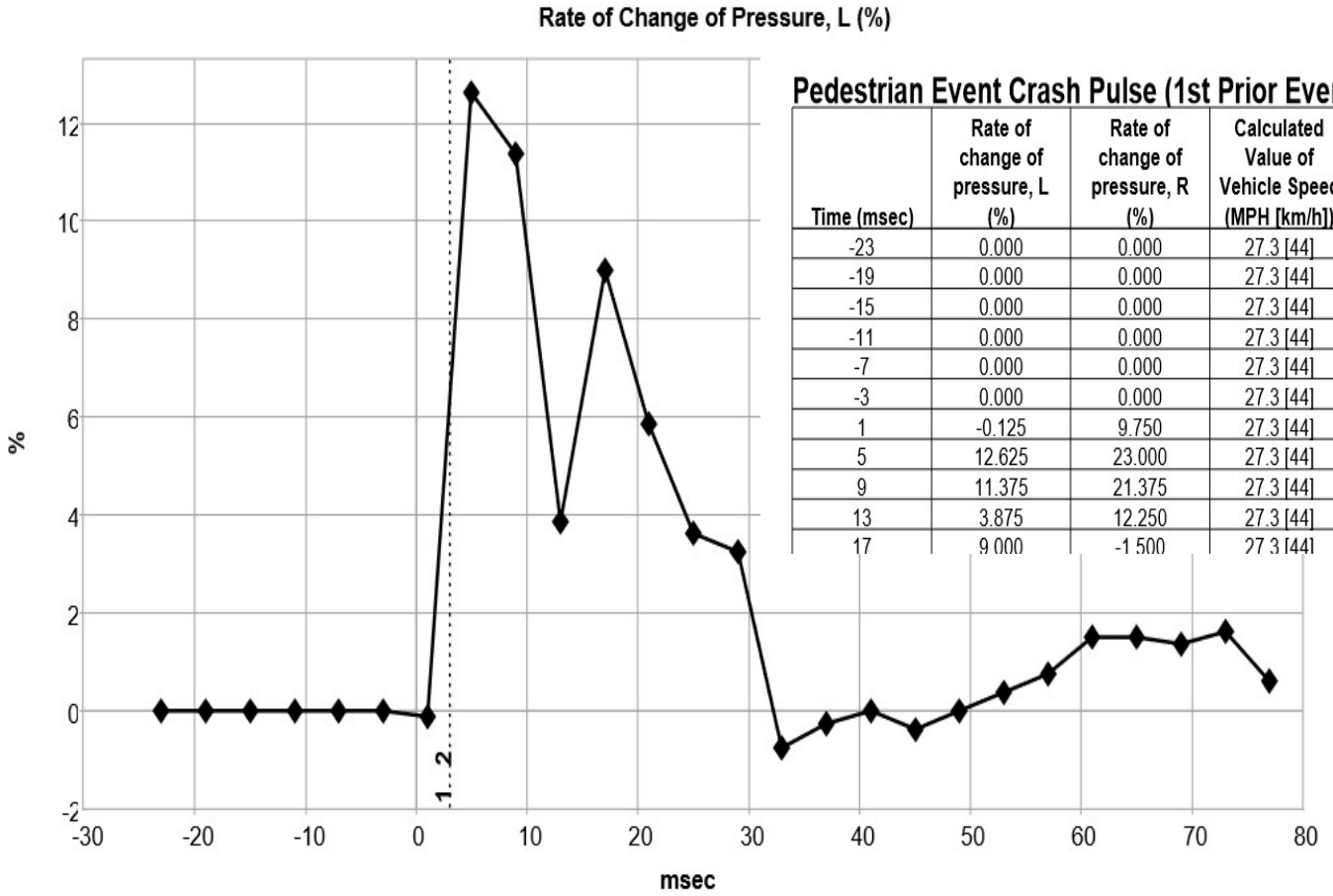
DTCs Present at Time of Event (1st Prior Event, TRG 1)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Warning ON Time Since DTC was Set, for Pedestrian (min)	0
Diagnostic Trouble Codes	None

Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	1

Data from L/R bumper pressure sensors on 15 & 17 EDR



Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

Time (msec)	Rate of change of pressure, L (%)	Rate of change of pressure, R (%)	Calculated Value of Vehicle Speed (MPH [km/h])
-23	0.000	0.000	27.3 [44]
-19	0.000	0.000	27.3 [44]
-15	0.000	0.000	27.3 [44]
-11	0.000	0.000	27.3 [44]
-7	0.000	0.000	27.3 [44]
-3	0.000	0.000	27.3 [44]
1	-0.125	9.750	27.3 [44]
5	12.625	23.000	27.3 [44]
9	11.375	21.375	27.3 [44]
13	3.875	12.250	27.3 [44]
17	9.000	-1.500	27.3 [44]

Toyota Help File Organization

Supported Toyota Vehicles

Use this section of the help file to determine if a vehicle is supported by the CDR tool and to look up the correct cable (and in some cases, adapter) to use for EDR data directly from the ACM. It is important to note that even if the vehicle is listed in this section, CDR must also support the part number on the ACM. Cli list of supported ACM part numbers.

Click here for tips on how to use this *Supported Vehicles* help topic

This Started in Nov 2015

IMPORTANT: To prevent damage to Toyota ECUs when imaging EDR data directly from the ECU, you must always use the appropriate cable called out in this the help file.

General Toyota Application Notes:

The CDR Tool software will accept most Toyota VINs entered by the user and the *Collect Airbag Control Module Data* icon will no longer be grayed out. **This is n indicator that the vehicle is supported.** Only the vehicles listed in the help file below are supported as indicated by the list and any associated coverage not

All supported vehicles listed can be imaged through the vehicle's DLC connector using the CDR OBDII DLC Cable (F00K108287) unless otherwise stated.

All supported vehicles require the vehicle's ignition key to be switched on prior to and during EDR record imaging through the vehicle's DLC connector.

Vehicle Specific Application Notes:

Click here to view all vehicle specific Toyota coverage notes

Vehicles Are Listed Alphabetically
Not by Model Year

Toyota Models begin with: 4 8 9 A B C D E F H I K L M N P R S T V W X Y Z

4Runner (Toyota) 2003 and later Market: 7

Year	EDR Generation	CDR Cable & Adapter	Module	Module Location
2002 - 2008	Gen 1 (02EDR)	F00K108617 (no adapter)	ACM	Under Center Stack
2009 - 2011	Gen 2 (06EDR)	F00K108616 & F00K108387	ACM	Under Center Stack
2012 and later	Gen 3 (12EDR/13EDR)	F00K108616 & F00K108387	ACM	Under Center Stack

Help file may be CALENDAR year not model year,
EDR may not yet be present in Model Year of first year listed

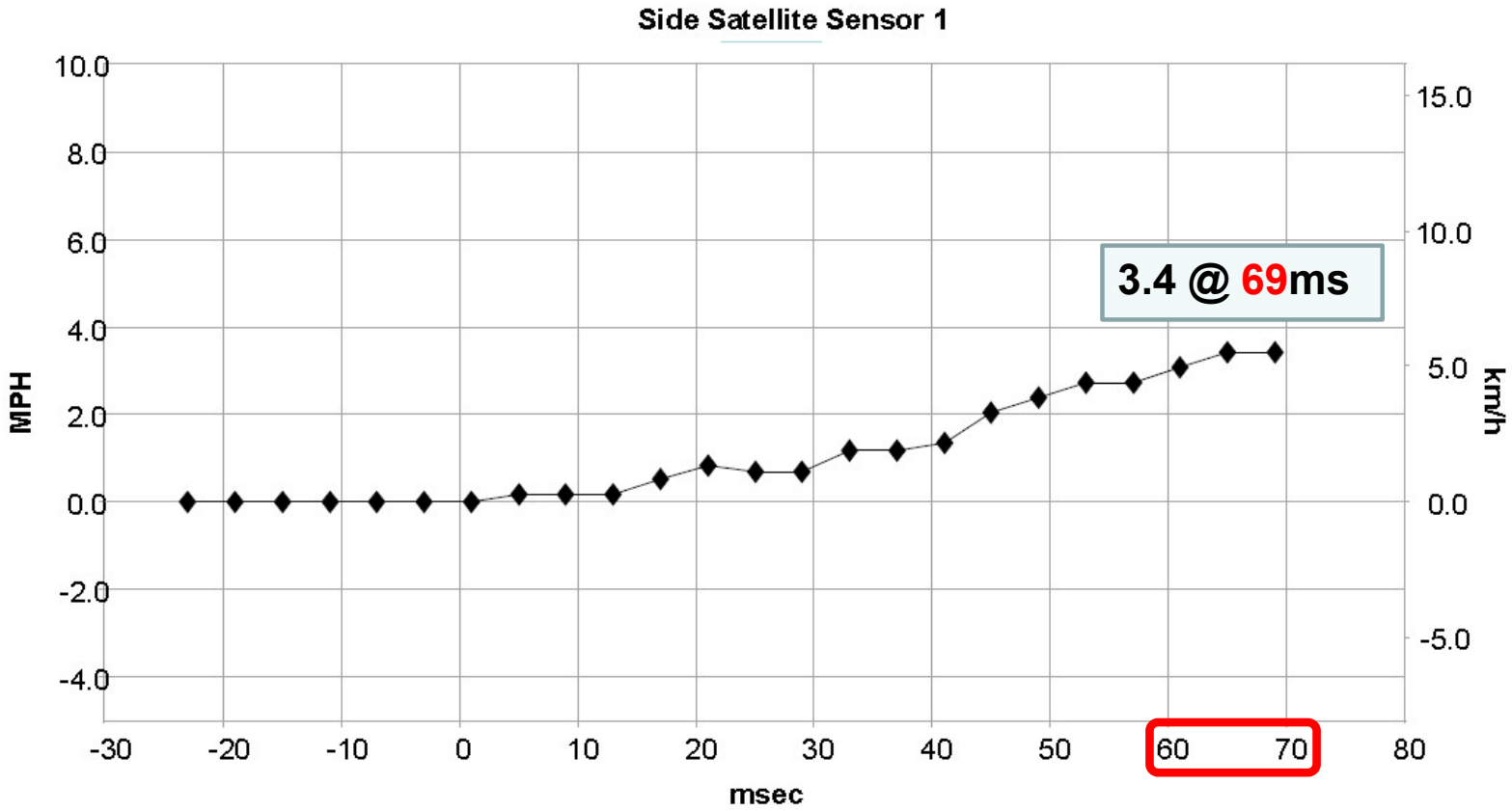
Toyota Vehicle Lookup



Toyota 12 EDR Drawback

1. Lateral acceleration is only recorded in a side algorithm wakeup.
2. When it is recorded, it is only for 68ms
3. It is only recorded for the side satellites, not for the central control module
4. If trying to calculate a PDOF you must be very aware of the different recording timing and sensor location
5. Fixed in “13EDR” – get 250ms of Y accel

12 EDR Lateral Delta V -



TOYOTA/LEXUS Vehicles with Vehicle Control History Data

		2013	2014	2015
LEXUS	ES			
	GS			
	GX			
	IS		250 and 350	
	LC			
	LS			
	LX			
	NX			
	RC			
	RX			
	UX			
TOYOTA	4Runner			
	Avalon			
	C-HR			
	Camry			
	Corolla			
	Highlander			
	Land Cruiser			
	Mirai			
	Prius			C and V
	RAV4			
	Sequoia			
	Sienna			
	Tacoma			
	Tundra			
	Venza			
	Yaris			
SCION	iM			

Tabular Data
2013-2015
No photos yet

Toyota/Lexus with Potential Photos stored

Gen 1 TSS/LSS pictures only with auto braking, Gen 2 pictures in more circumsta

		2015	2016	2017	2018	2019	2020	2021	2022	2023
LEXUS	CT200h				discontinued					
	ES		LSS	LSS	LSS	?	?	?	2.5	
	GS		LSS	LSS	LSS	?	?	discontinued		
	GX						LSS	LSS	LSS	
	IS			LSS	LSS	?	?	?	2.5	
	LC				LSS	LSS	LSS	LSS	LSS	
	LS				LSS+	2.0	2.0	2.0	2.5	
	LX		LSS	LSS	LSS	?	?	?	2.5	
	NX				LSS	?	?	?	3.0	
	RC-F				LSS	?	?	?	LSS	
	RX		LSS	LSS	LSS	?	?	?	2.0	
	UX								2.0	
TOYOTA	4Runner						P	P	P	
	86								Subaru	
	Avalon		P	P	P	P	P	P	2.5+	
	Camry				P	P	P	2.5+	2.5+	
	C-HR				P	P	P	2.5	2.5	
	Corolla			P/C	P	2.0	2.0	2.0	2.0	
	Echo									
	FJ									
	Highlander			P	P	P	2.0	2.5+	2.5+	
	Land Cruiser		P	P	P	P	P	P		
	Mirai					2.0	2.0	2.5+	2.5+	
	Prius		P/C	P/C	P	P/C	P	2.0	2.0	
	RAV4		P/C	P/C	P	2.0	2.0	2.0	2.0	
	Sequoia				P	P/C	P	P	P	
	Sienna				P	P/C	P	2.0	2.0	
	Supra (BMW)						(made by BMW)			
	Tacoma				P	P	P	P	P	
	Tundra				P	P	P	P	2.5	
	Venza	out of production						2.0	2.0	
	Yaris			C	C					

Chart is only inteded to cover US models. Content MAY depend on trim level or model.

TSS 1.0 (TSS-P, TSS-C, LSS) Photo Storage

- For Images to be recorded and linked the

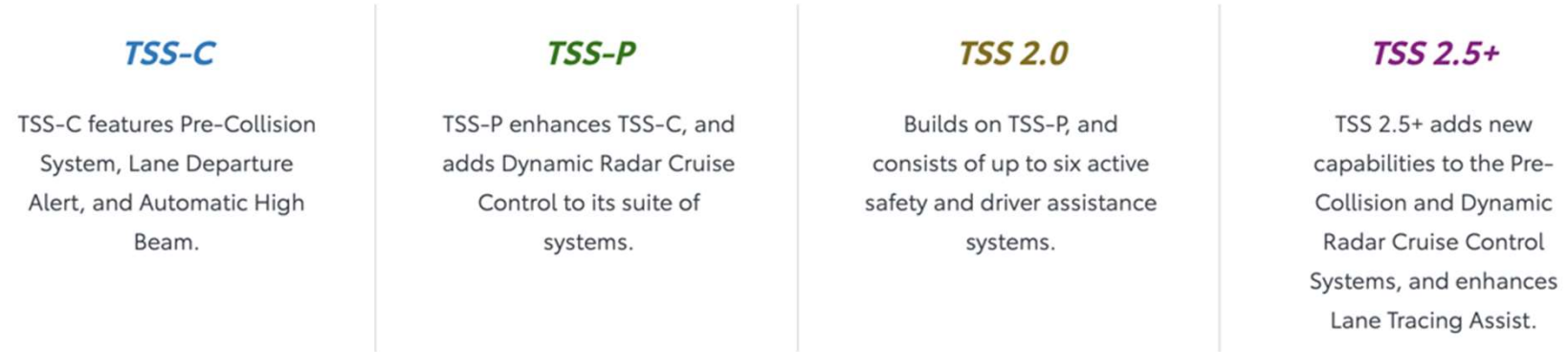
  VEHICLE must brake
- A PCS warning buzzer is not sufficient for images to be linked with VCH and recorded
- If vehicle begins to brake and driver responds, then related images should be stored
- CHECK: perform CDR download FIRST. Look for deceleration with no driver input: i.e. no brake switch = on, no brake oil pressure, but speed drops more than idling down

TSS 2.0 Photo Storage

- Adds new Image capture categories:
  - Vehicle behavior related image data: 1 Set of images
  - PCS related image data: 3 Sets of images (Pre-Crash, AEB)
  - LKA related image data: 1 Set of images (Lane Keep)
  - ICS related image data : 3 Sets of images
    (Intelligent Clearance Sonar – short range)
  - Collision related image data : 2 Sets of images

TSS 2.5 / 2.5+

- 2021 Camry and Highlander and Lexus IS to start
- Camera / Millimetre Wave Radar optimized improve response range and enhanced low light pedestrian and daytime bicycle recognition.
- Pre-Collision System adds new Features



TSS-C

TSS-C features Pre-Collision System, Lane Departure Alert, and Automatic High Beam.

TSS-P

TSS-P enhances TSS-C, and adds Dynamic Radar Cruise Control to its suite of systems.

TSS 2.0

Builds on TSS-P, and consists of up to six active safety and driver assistance systems.

TSS 2.5+

TSS 2.5+ adds new capabilities to the Pre-Collision and Dynamic Radar Cruise Control Systems, and enhances Lane Tracing Assist.

20 Images from the Toyota FCM @ 0.6 sec



Sample VCH Summary

Vehicle Diagnostic Report

2020 Rav4 [redacted]
004320 mile

[redacted]

[redacted]

Printed By: Default User(1)

[redacted]

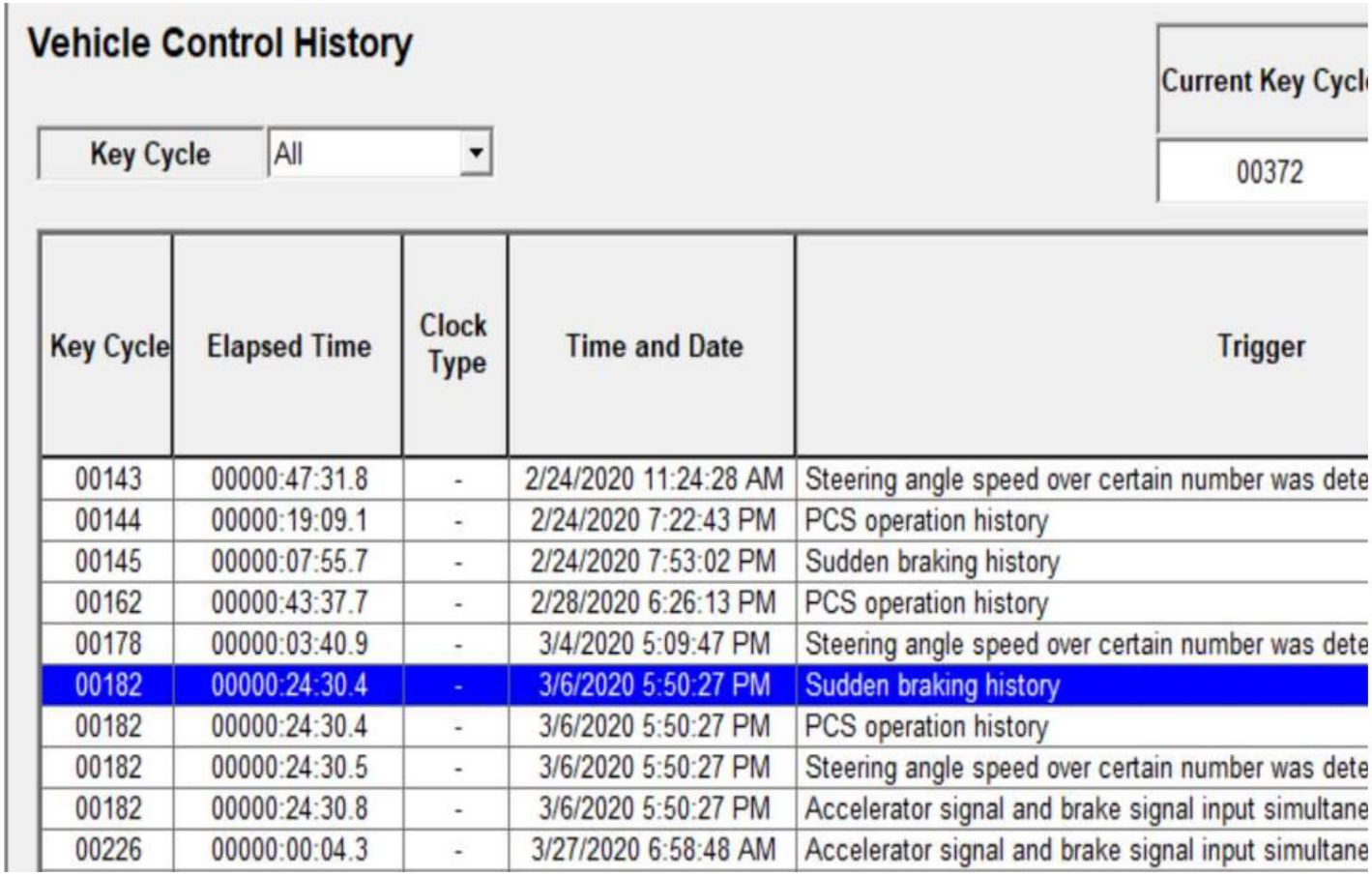
Vehicle Control History Report
(1 of 2)

Current Key Cycle : 00372
Current Key Cycle Elapsed Time : 00000:17:23
Maximum Number of Overwrite of the Recording Area : 30000

Key Cycle : All

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area
00030	00000:53:56.4	-	2/1/2020 4:24:56 PM	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	Icon	51	1
00044	00000:19:06.4	-	2/4/2020 11:00:03 AM	PCS operation history		264	1
00051	00000:01:15.3	-	2/6/2020 9:30:21 AM	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	Icon	315	1
00055	00000:24:15.8	-	2/7/2020 8:50:59 AM	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	Icon	340	1

A way to get ped hits??



Vehicle Control History

Current Key Cycl

00372

Key Cycle: All

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger
00143	00000:47:31.8	-	2/24/2020 11:24:28 AM	Steering angle speed over certain number was dete
00144	00000:19:09.1	-	2/24/2020 7:22:43 PM	PCS operation history
00145	00000:07:55.7	-	2/24/2020 7:53:02 PM	Sudden braking history
00162	00000:43:37.7	-	2/28/2020 6:26:13 PM	PCS operation history
00178	00000:03:40.9	-	3/4/2020 5:09:47 PM	Steering angle speed over certain number was dete
00182	00000:24:30.4	-	3/6/2020 5:50:27 PM	Sudden braking history
00182	00000:24:30.4	-	3/6/2020 5:50:27 PM	PCS operation history
00182	00000:24:30.5	-	3/6/2020 5:50:27 PM	Steering angle speed over certain number was dete
00182	00000:24:30.8	-	3/6/2020 5:50:27 PM	Accelerator signal and brake signal input simultane
00226	00000:00:04.3	-	3/27/2020 6:58:48 AM	Accelerator signal and brake signal input simultane

Toyota “13 EDR” & “15 EDR”

1. **13 EDR** Adds Lateral ***Accel*** data from ACM in frontal events for 200ms (Fixes biggest drawback of **12 EDR**)

Longitudinal/Lateral Crash Pulse (Most Recent Event,

Time (msec)	Longitudinal Delta-V (MPH [km/h])	Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2)	F
0	0.0 [0.0]	0.0	
10	0.3 [0.6]	-1.9	
20	0.5 [0.8]	0.0	

2. ***Calculate*** Lat **Delta V** for each 10ms interval
3. $\mathbf{DV_{10ms}}$ = ***Accel***/9.81*(32.2/1.466)*0.01s
4. Add up intervals to get ***total*** **Delta V**
5. **Template available to calculate it**

2018+ Toyota “17EDR”

1. “Front/rear” 5mph X events replaced by “Front/Rear/Side”, events triggered by 5mph X ***or Y*** DV or side deploys. Y DV graph added.
2. 6 precrash memory spaces, 4 DV spaces for each crash type (vs 2 each prior)
3. Separate side events still reported with 70ms side accel in m/sec^2, but do NOT have precrash associated with them. NOT normally used in recon.

17EDR Precrash (new look, same data)

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-5.00	29.8 [48]	49.5	24.5	Invalid	2,400	Invalid	OFF
-4.50	32.3 [52]	47.0	24.5	Invalid	2,500	Invalid	OFF
-4.00	34.2 [55]	38.0	20.0	Invalid	2,700	Invalid	OFF
-3.50	35.4 [57]	32.5	16.0	Invalid	2,800	Invalid	OFF
-3.00	36.7 [59]	26.5	13.5	Invalid	2,400	Invalid	OFF
-2.50	37.3 [60]	22.5	11.0	Invalid	2,400	Invalid	OFF
-2.00	37.3 [60]	19.0	8.5	Invalid	2,200	Invalid	OFF
-1.50	38.5 [62]	19.0	6.0	Invalid	1,900	Invalid	OFF
-1.00	37.9 [61]	18.0	5.5	Invalid	1,900	Invalid	OFF
-0.50	37.9 [61]	18.0	5.0	Invalid	1,800	Invalid	OFF
TRG(0)	36.0 [58]	100.0	26.0	Invalid	1,700	Invalid	OFF

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-5.00	OFF	OFF	0.00	1.579	-0.98	-4.5	D
-4.50	OFF	OFF	0.00	1.795	0.00	-1.5	D
-4.00	OFF	OFF	0.00	1.292	0.49	0.0	D
-3.50	OFF	OFF	0.00	0.861	0.00	-1.5	D
-3.00	OFF	OFF	0.00	0.718	0.00	-1.5	D
-2.50	OFF	OFF	0.00	-0.072	0.00	-1.5	D
-2.00	OFF	OFF	0.00	0.000	0.00	-1.5	D
-1.50	OFF	OFF	0.00	0.000	-0.49	-1.5	D
-1.00	OFF	OFF	0.00	0.287	0.00	-1.5	D
-0.50	OFF	OFF	0.00	-0.144	0.00	-1.5	D
TRG(0)	OFF	OFF	0.48	-1.507	61.00	-19.5	D

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 3 of 4

Time (sec)	Sequential Shift Range	Cruise Control Status	VSC Control Status	READY Signal	Drive Mode, Power Train	Drive Mode, Snow	Drive Mode, EV
-5.00	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-4.50	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid

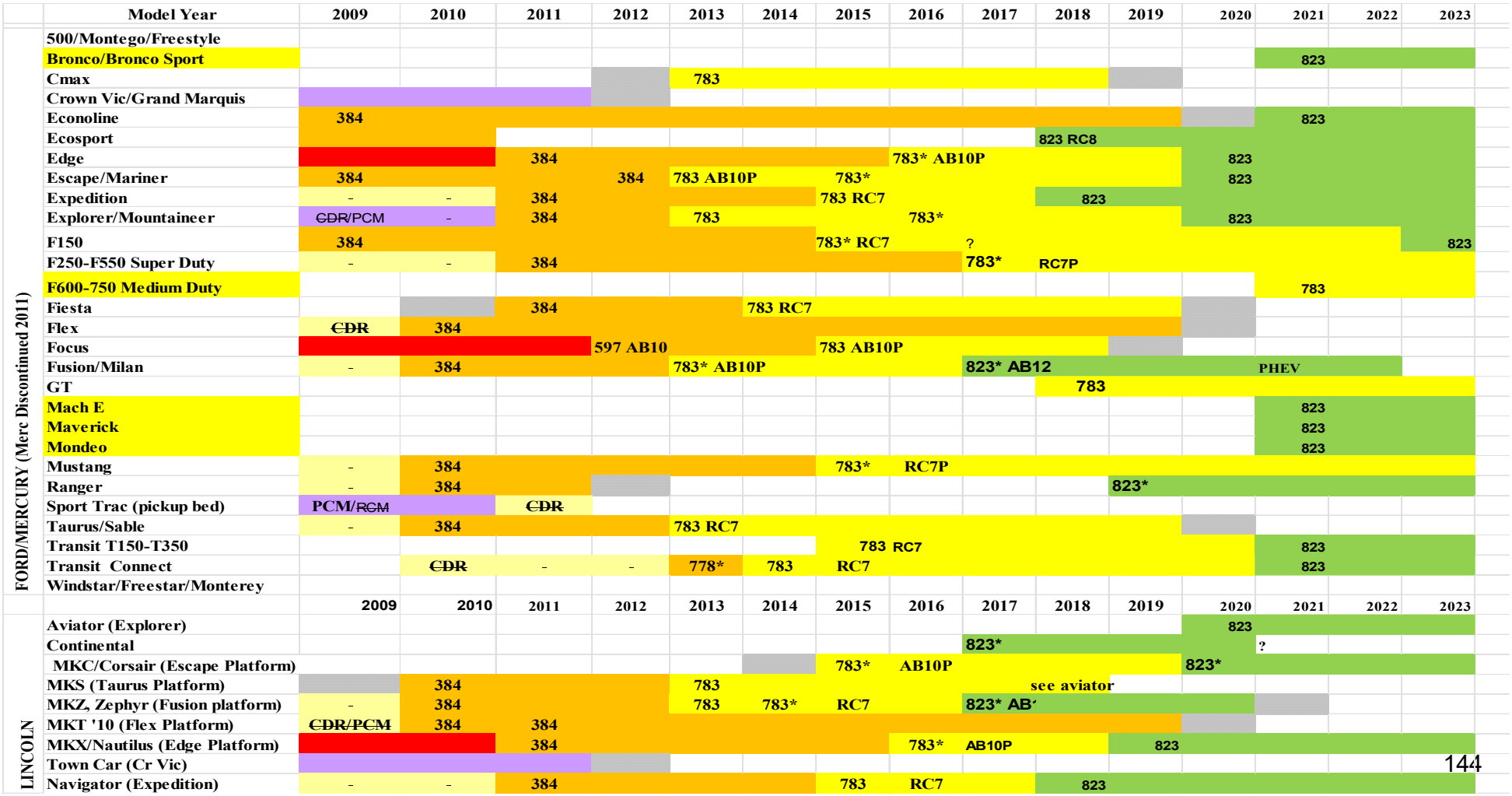
17 EDR time data & ped events

System Status at Pedestrian Event (1st Prior Event, TRG 1)

TRG Count (times)	1
TRG Count not for Pedestrian (times)	SNA
Ignition Cycle, Crash (times)	353
Odometer signal (miles [km])	1,195 [1,923]
Trip count (times)	392
Time count (msec)	1,283,900
Time count input system	Normal
Airbag Warning Lamp, On/Off	OFF
Pedestrian Warning Lamp, On/Off	OFF
Time from Previous TRG (msec)	32767 or greater

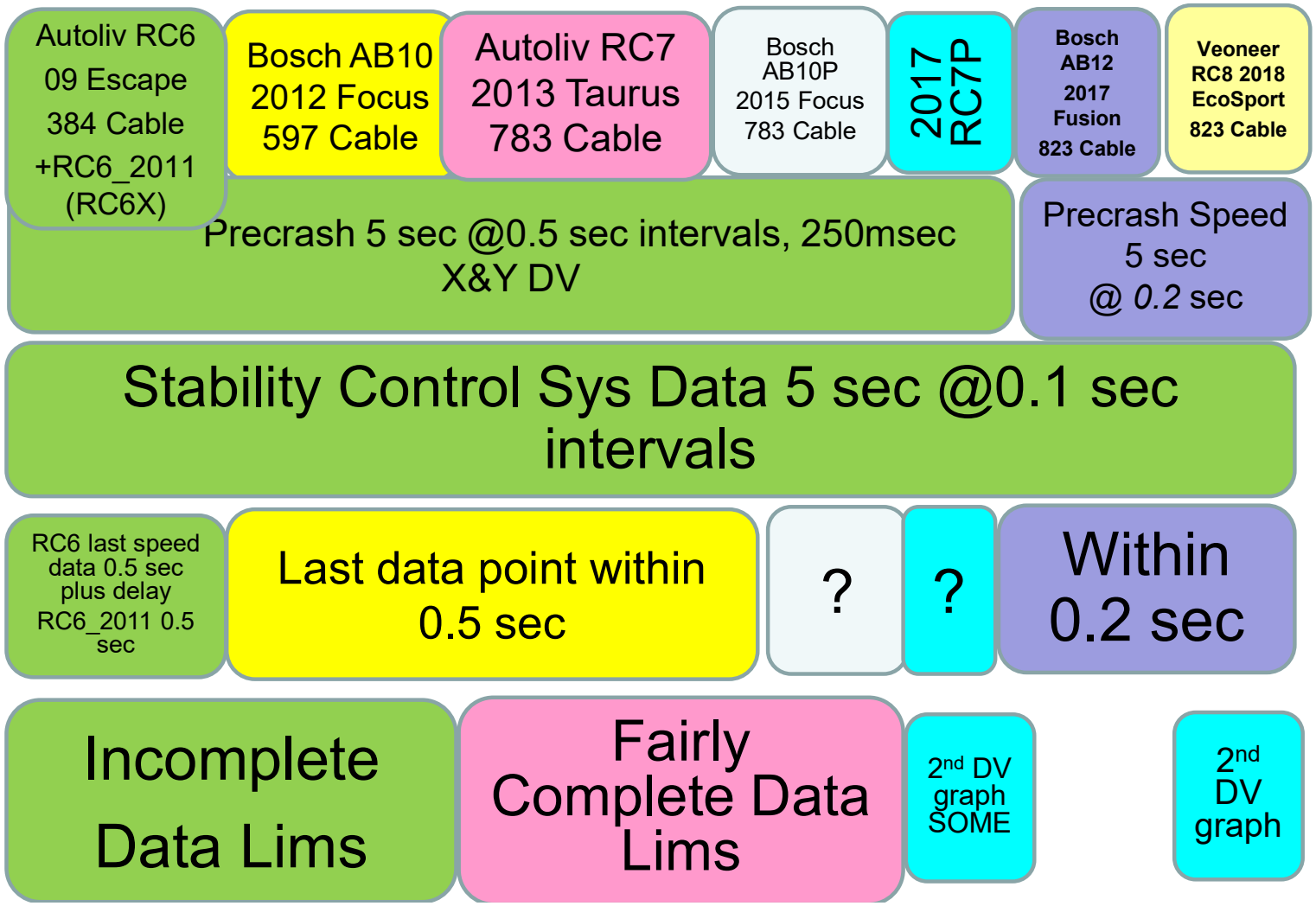
1. Odometer added
2. “Trip count” similar to key cycles but slightly higher,
3. Time Count (like Ford “Key on time”), Odometer
4. Pedestrian events if pedpro option bought

Ford 09+ CDR 23.2 Coverage 69%



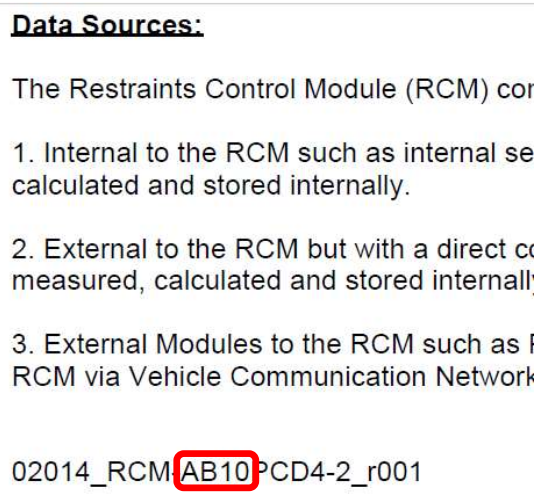
	Model Year	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
FORD/MERCURY (Merc Discontinued 2011)	500/Montego/Freestyle															
	Bronco/Bronco Sport													823		
	Cmax					783										
	Crown Vic/Grand Marquis															
	Econoline	384												823		
	Ecosport										823 RC8					
	Edge			384					783* AB10P				823			
	Escape/Mariner	384			384	783 AB10P		783*					823			
	Expedition	-	-	384				783 RC7			823					
	Explorer/Mountaineer	~~CDR~~/PCM	-	384		783			783*				823			
	F150	384						783* RC7		?						823
	F250-F550 Super Duty	-	-	384						783*	RC7P					
	F600-750 Medium Duty													783		
	Fiesta			384			783 RC7									
	Flex	~~CDR~~	384													
	Focus				597 AB10			783 AB10P								
	Fusion/Milan	-	384			783* AB10P				823* AB12				PHEV		
	GT										783					
	Mach E													823		
	Maverick													823		
	Mondeo													823		
	Mustang	-	384					783*	RC7P							
	Ranger	-	384									823*				
	Sport Trac (pickup bed)	PCM/~~RCM~~		~~CDR~~												
	Taurus/Sable	-	384			783 RC7										
	Transit T150-T350							783 RC7						823		
	Transit Connect		~~CDR~~	-	-	778*	783	RC7						823		
	Windstar/Freestar/Monterey															
		2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
LINCOLN	Aviator (Explorer)												823			
	Continental									823*				?		
	MKC/Corsair (Escape Platform)							783*	AB10P				823*			
	MKS (Taurus Platform)		384			783					see aviator					
	MKZ, Zephyr (Fusion platform)	-	384			783	783*	RC7		823* AB'						
	MKT '10 (Flex Platform)	~~CDR/PCM~~	384	384												
	MKX/Nautilus (Edge Platform)			384					783*	AB10P		823				
	Town Car (Cr Vic)															
	Navigator (Expedition)	-	-	384				783	RC7		823					

Ford Part 563 EDR Subfamilies



Identifying Ford EDR families

1. At the end of the data limitations section, just below the last paragraph, on the left, Bosch puts what version of data limitations is being printed. For Ford, most data limitations are by family.
2. No difference between DL
   a) 2009 RC6 and
   b) 2011 RC6X.
3. Open cdrx **copy** in notepad, see RC6_2011



RC8 Precrash Data Look

Pre-Crash Data -5 to 0 sec [5 samples/sec] (First Record)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Speed, Vehicle Indicated, Quality Factor	Accelerator Pedal, % Full	Accelerator Pedal, % Full, Quality Factor	Service Brake, On/Off	Service brake, on/off Quality Factor	Engine RPM
- 5.0	59.8 [96]	OK	18	OK	OFF(Driver_Not_Braking)	OK	2,174

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record) - Table 1 of 2

Time (sec)	ABS Activity (Engaged, Non-Engaged)	Stability Control Active	Traction Control via Engine	Traction Control via Brakes	Stability Control Lateral Acceleration (g)	Stability Control Longitudinal Acceleration (g)
- 5.0	Off/non-engaged	Off/non-engaged (No)	Off/non-engaged (Inactive)	Off/non-engaged (Inactive)	-0.01	0.02

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record) - Table 2 of 2

Time (sec)	Stability Control Yaw Rate (deg/sec)	Stability Control Roll Rate (deg/sec)	Steering Wheel Angle (Pinion) (deg)	Steering Wheel Angle (Pinion) Quality Factor
- 5.0	0.06	-1.53	0.0	OK
- 4.9	-0.13	-1.18	0.0	OK

Other RC8 features

- Acceleration data added
- Second Delta V graph centered around deployment +/- 125ms (also in SOME RC7P)
- Unique decoder for occupant size (see next slide)

Pre-Crash Data -5 to 0 sec [1 sample/sec] (First Record)

Time (sec)	Occupant Size Classification, Front Passenger (Child size Yes/No [Hex value])	Extended Power Status
- 5.0	No [$03]	No Fault
- 4.0	No [$03]	No Fault

Ford Occupant Size Classification

Pre-Crash Data -1 sec (First Record)

Ignition cycle, crash	359
Frontal air bag warning lamp, on/off	OFF
Occupant size classification, front passenger (Child size Yes/No [Hex value])	No [$08]
Safety belt status, driver	Driver Buckled

- The Hex Value provides additional information than just child yes/no
- The Decoder is different for each supplier and major module type
- Look at end of data limitations for module type – Autoliv=RC5, RC6, RC7; Bosch=AB9, AB10

RC6/7	AB10	AB12	RC8	Classification
$01	$00	$00	$00	Empty
$02	$01	$05	$02	Child or Occupied Disable
$04	$05	$04	$01	Indeterminate (gray zone between child and adult)
$08	$02	$01	$03	Occupied Enable Small Adult (typical 110 lb 5th female)
$10	$03	$02	$04	Occupied Enable Medium (typical 172 lb 50th male)
$20	$04	$03	$05	Occupied Enable Large Adult (typical 223 lb 95th male)

RC8 Lower Recording Threshold?

- Data Limitations State 5mph DV over 150ms threshold for recording
- But events with less than 5mph are seen

Deployment Data (First Record)

Maximum Delta-V, Longitudinal (MPH [km/h])	-2.02 [-3.25]
Time, Maximum Delta-V Longitudinal (msec)	300.0

- Remember, nothing deployed unless a device is listed with a deploy time
- This is a non-deployment

Unusual Ford Event Names

- Fuel Cutoff Level 1 or 2 (RC7P, RC8, AB12)
- Unlocked Frontal, Side, or Rollover Event (RC7, RC8)
- Locked Frontal, Side, or Rollover Event (RC7, RC8)
- Frontal Accel, Side Accel or Rollover Threshold Exceeded (AB12)
- Shutdown Request Threshold Exceeded Level 1,2 (AB12)
- Event Record 1 or 2 (AB10P)

Multiple “events” (front, side, rollover) are really algorithms enabled – could be only one combined event in memory. Events recovered may say frontal but if algorithm activated twice there may be two events.

Some events like Fuel Cutoff 1 may NOT store data.

2017 MKZ AB12

Pre-Crash Data -5 to 0 sec [5 samples/sec] (First Record) - Table 1 of 2

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Speed, Vehicle Indicated, Quality Factor	Accelerator Pedal, % Full	Accelerator Pedal, % Full, Quality Factor	Service Brake, On/Off	Service brake, on/off Quality Factor	Engine RPM	Cruise Control Driver Accelerator Pedal Override
- 5.0	99.2 [160]	OK	0	OK	Off	OK	2,816	Cruise_Reg_Not_Overridden
- 4.8	98.7 [159]	OK	0	OK	Off	OK	2,800	Cruise_Reg_Not_Overridden
- 4.6	98.3 [158]	OK	0	OK	Off	OK	2,786	Cruise_Reg_Not_Overridden
- 4.4	98.2 [158]		0	OK	On	OK	2,770	Cruise_Reg_Not_Overridden
- 4.2	.8 [157]	OK	0	OK	On	OK	2,758	Cruise_Reg_Not_Overridden

Every 0.2 Sec!

Pre-Crash Data -5 to 0 sec [5 samples/sec] (First Record) - Table 2

Time (sec)	Brake Pre-Charge Request	Brake Assist Sensitivity Level	Brake Deceleration Request (m/s^2)	Large Driver Steering or Accel Pedal Input	Collision Mitigation System Fault	Collis Mitiga Syste Enabled
- 5.0	N/A	N/A	236	N/A	N/A	N/A
- 4.8	N/A	N/A	236	N/A	N/A	N/A

Collision Avoidance

2017 MKZ AB12 add'l data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (First Record)

Time (sec)	Driver Gear Selection (Auto Trans)	Wheel Torque Requested (Nm)	Total Arbitrated Brake Torque (Nm)	Ignition Status	Speed Control Status
- 5.0	Drive	532	0	Run	Off
- 4.5	Drive	120	0	Run	Off
- 4.0	Drive	-20	0	Run	Off

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record) - Table 1 of 2

Time (sec)	ABS Activity (Engaged, Non-Engaged)	Brake Powertrain Torque Request 1	Brake Powertrain Torque Request 2	Traction Control via Brakes	Stability Control Lateral Acceleration (g)	Stability Control Longitudinal Acceleration (g)
- 5.0	Non-engaged	No	Inactive	No	-0.02	0.09

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record) - Table 2 of 2

Had this before

Time (sec)	Stability Control Yaw Rate (deg/sec)	Stability Control Roll Rate (deg/sec)	Steering Wheel Angle (deg)	Steering Wheel Angle Quality Factor
- 5.0	0.06	0.24	-0.9	OK
- 4.9	-0.08	0.00	-0.6	OK
- 4.8	0.45	0.86	0.0	OK

Ford AB12 Anomaly

- File found with “Event Record 2” with no “Event Record 1”. Reported to Bosch.
- Polarity in AB12 is correct. Ford Steering, Yaw, and Lat Accel 2009+ should all be the same sign – and opposite the right hand rule - negative CW right turn, Positive for CCW left turn.

- Steering Wheel Angle if Applicable: positive value indicates left turn, and negative value would indicate right turn.

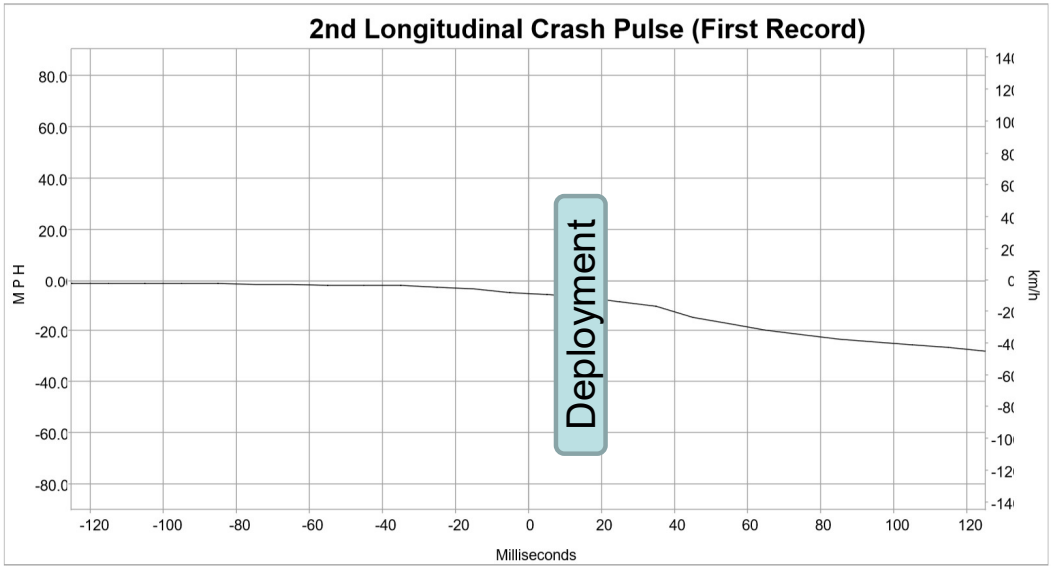
- Stability Control Lateral Acceleration if Applicable: Lateral Acceleration (Y-direction) is the acceleration along the lateral axis of the vehicle, reported as positive when accelerating to the left.

- Stability Control Longitudinal Acceleration if Applicable: Longitudinal Acceleration (X-direction) is the acceleration along the longitudinal axis of the vehicle, reported as positive when accelerating in a forward direction.

- Stability Control YawRate if Applicable: The YawAxis is the vertical axis of the vehicle, generally perpendicular to the plane of the road. A positive YawRate is counter-clockwise when observing the vehicle from above.

New RC7P Feature- 2017 F250

- Ford has added a second graph to capture the deployment decision +/-125ms. This is important in hit-curb-then-hit-tree cases where the deployment was previously off the end of the mandated 250ms after wakeup graph



Ford Global Real Timer

Added for AB12/ RC7P/RC8

Timer starts when car is built, is hooked to battery, runs continuously

Month of build is on door jamb sticker (exact day/time is not)

Can be used to tie your recording to your crash

Great for knowing a more precise time between events when not within 5 seconds

Pre-Crash Data -1 sec (First Record)

Ignition cycle, Crash	1,782
Frontal Air Bag Warning Lamp, On/Off	Off
Safety Belt Status, Driver	Belted
Seat Track Position Switch, Foremost, Status, Driver	Rearward
Seat Track Position Switch, Foremost, Status, Front Passenger	Rearward
Safety Belt Status, Front Passenger	Unbelted
Brake Telltale	Off
ABS Telltale	Off
ESC/TC Telltale	Off
ESC/TC Off Telltale	Default Mode
Powertrain Wrench Telltale	Off
Powertrain Malfunction Indicator Lamp (MIL) Telltale	Unchanged Off
Global Real Time (seconds)	47,951,087

09+ Ford Stability Control Data

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record)

Times (sec)	Stability Control Lateral Acceleration (g)	Stability Control Longitudinal Acceleration (g)	Stability Control Yaw Rate (deg/sec)
- 1.4	0.086	-0.045	0.12
- 1.3		-0.031	1.5
- 1.2		-0.647	-1.5
- 1.1		-0.75	-2.5
- 1.0		-0.709	-1.5
- 0.9		-0.763	-1.5
- 0.8		-0.826	-1.5
- 0.7		-0.792	-0.75
- 0.6		-0.895	1.25
- 0.5	0.014	-0.83	1.87
- 0.4	0.03	-0.835	0.75
- 0.3	0.022	-0.607	0.37
- 0.2	0.107	-0.653	0.75
- 0.1	0.05	-0.823	-3.37
0.0	-0.004	-0.811	-3.75

Reaction

Integrate lat accel - get velocity – integrate again – get displacement

Drag Factor!

Integrate yaw – get **approach angle change**!! Take tangent of cumulative angle, multiply by speed, get **sideways movement** – how far out of lane vehicle went!!

Bosch AB10P Unique Info

1. Longitudinal Accel Stability Control Info is positive for slowing down, negative for speeding up – *opposite of what data limitations say*.
2. Lateral Accel data is positive right, negative left – different than the other Ford modules.
3. Steering and Yaw remain positive left and are identified as such in data limitations.

RC6+ Steering Angle Glitch

1. Ford stores one value for steering relative to original zero position set at factory
2. If vehicle steering is serviced, original zero may be lost and system learns a new one from average of all new driving.
3. EDR may write down ORIGINAL position relative to zero – so a FEW models serviced may appear to have WRONG value for zero.
4. No warning in DL's about this yet

2020/21 F250+ no EDR events

- Ford accidentally turned off the EDR in 20/21 F Super Duty (F250, 350, 450, and 550) ACM's built from 2/18/2019 thru 7/10/2021
- 12/3/2021 Ford issued a Customer Satisfaction Program 21P19 advising of this. Vehicle brought to dealers for other service will reflash the ACM to turn the EDR back on.
- You can ask dealer to run VIN and tell you if 21P19 has been performed before you go to download it so ask not to waste your time.

Reading Police Explorers

- Some upfitters rewire the DLC and steal the low/medium speed CAN bus pins that connect to ACM for other police circuitry.
- “Original” DLC plug may be shoved up under dash
- If ACM will not communicate via upfitted DLC plug, look for the original DLC plug under dash and use that with 784 adapter.

Backpowering New Fords

1. Airbag system will stays on until car slows to a near stop. Switched battery feed replaces ignition switch feed.
2. Airbag fuse (interior fuse box) Micro 3

19	7.5A [7]	Extended power module.

3. 2020 Explorer moves to engine compt

146	15A	Battery electronic control module. ACM Power

4. ALSO REQUIRES GATEWAY (INTERIOR FUSE BOX)

13	7.5A	Steering column control module. Smart datalink connector. GATEWAY Instrument cluster.
7	10A	Smart data link connector power. DLC

For Ford EDR **not supported** by Bosch CDR
See “Unsupported Fords” on my website

Most requests can go directly to Ford’s suppliers. A few must go through Ford legal:

Email to EDRFORD@FORD.COM

(this is an address for a Bowman and Brooke outside counsel that handles these requests for Ford)

Contact Rick Ruth for additional information prior to making request

As of this writing, Veoneer (former Autoliv) charges $2000 to all, Bosch and Conti are no charge to law enforcement but $1500 to private parties.

Veoneer and Conti now say they cannot download modules >15 years old

Available on my website

Getting EDR Data from unsupported Fords

Richard R. Ruth, P.E.

(this copy has been updated 12 22 2022, after the original presentation at the Jan 2016 CDR Summit)

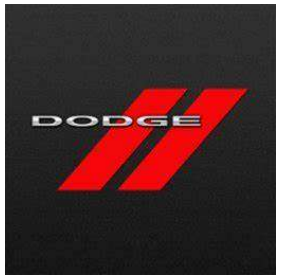


Stellantis (formerly FCA)

Includes Chrysler, Jeep, Dodge, Ram, Fiat, Lancia, Renault, Citroen and Peugot Data Analysis













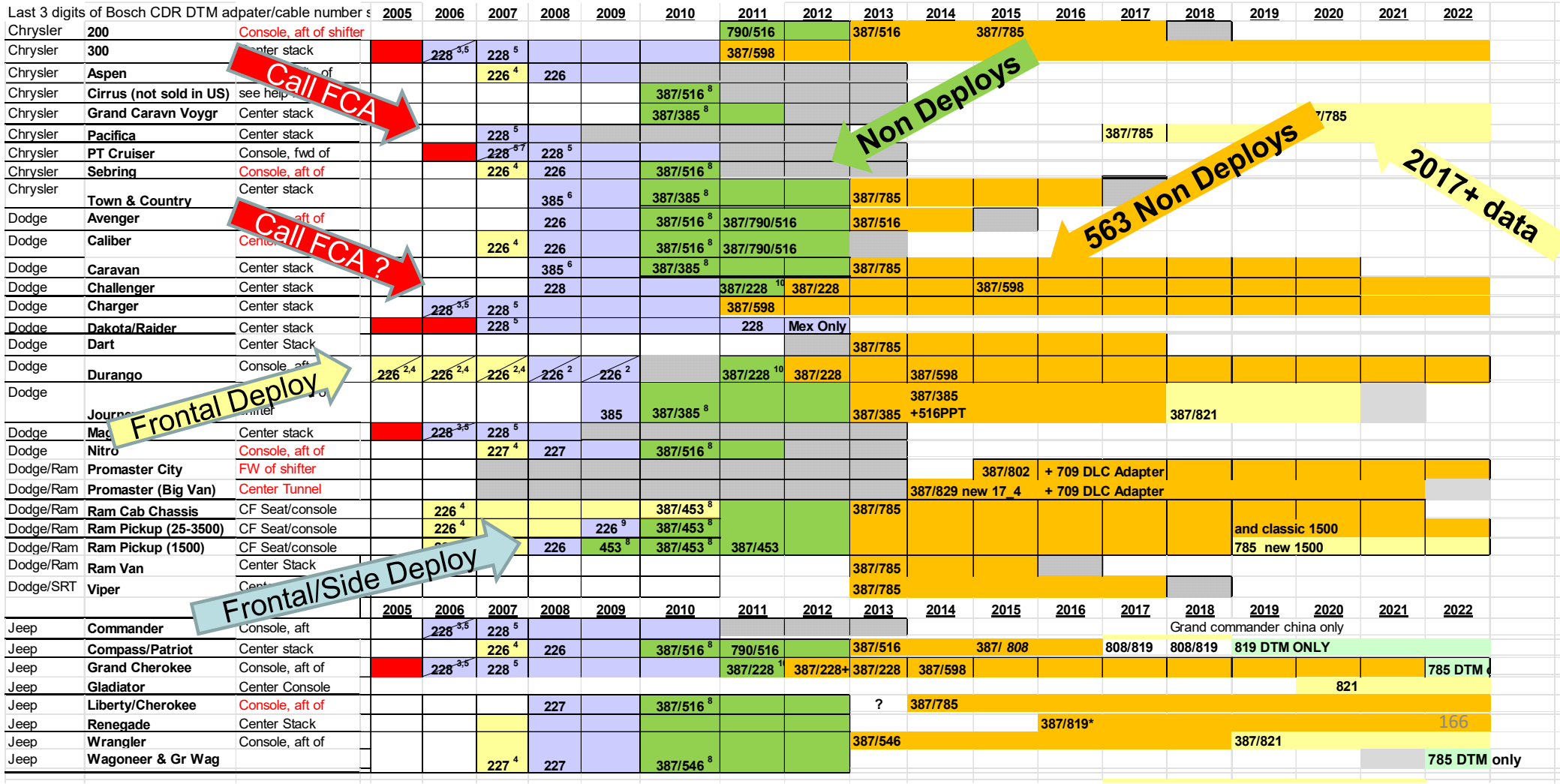








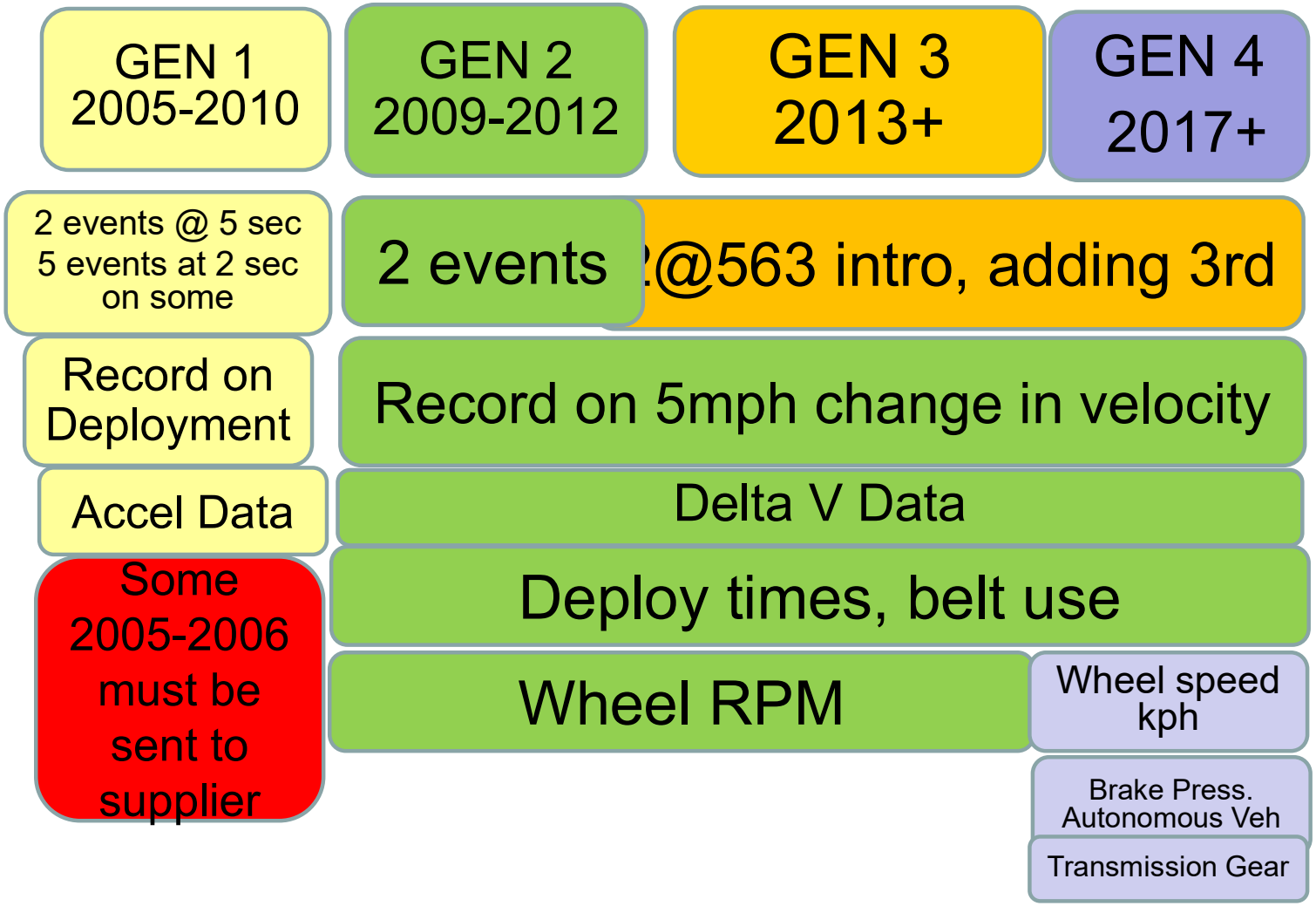
Stellantis 21.5 CDR Coverage 64%



Last 3 digits of Bosch CDR DTM adpater/cable number s			2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Chrysler	200	Console, aft of shifter							790/516		387/516		387/785							
Chrysler	300	nter stack		228 3,5	228 5				387/598											
Chrysler	Aspen	of			226 4	226														
Chrysler	Cirrus (not sold in US)	see help						387/516 8												
Chrysler	Grand Caravn Voygr	Center stack						387/385 8												
Chrysler	Pacifica	Center stack			228 5										387/785					
Chrysler	PT Cruiser	Console, fwd of			228 5,7	228 5														
Chrysler	Sebring	Console, aft of			226 4	226		387/516 8												
Chrysler	Town & Country	Center stack				385 6		387/385 8			387/785									
Dodge	Avenger	aft of				226		387/516 8	387/790/516		387/516									
Dodge	Caliber	Cent			226 4	226		387/516 8	387/790/516											
Dodge	Caravan	Center stack				385 6		387/385 8			387/785									
Dodge	Challenger	Center stack				228			387/228 10	387/228			387/598							
Dodge	Charger	Center stack		228 3,5	228 5				387/598											
Dodge	Dakota/Raider	Center stack			228 5				228	Mex Only										
Dodge	Dart	Center Stack									387/785									
Dodge	Durango	Console, aft	226 2,4	226 2,4	226 2,4	226 2	226 2		387/228 10	387/228		387/598								
Dodge	Journ	hifter					385	387/385 8			387/385	387/385 +516PPT				387/821				
Dodge	Mag	Center stack		228 3,5	228 5															
Dodge	Nitro	Console, aft of			227 4	227		387/516 8												
Dodge/Ram	Promaster City	FW of shifter											387/802	+ 709 DLC Adapter						
Dodge/Ram	Promaster (Big Van)	Center Tunnel										387/829 new 17_4		+ 709 DLC Adapter						
Dodge/Ram	Ram Cab Chassis	CF Seat/console		226 4				387/453 8			387/785									
Dodge/Ram	Ram Pickup (25-3500)	CF Seat/console		226 4			226 9	387/453 8									and classic 1500			
Dodge/Ram	Ram Pickup (1500)	CF Seat/console				226	453 8	387/453 8	387/453								785 new 1500			
Dodge/Ram	Ram Van	Center Stack									387/785									
Dodge/SRT	Viper	Cent									387/785									

			2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Jeep	Commander	Console, aft		228 3,5	228 5											Grand commander china only				
Jeep	Compass/Patriot	Center stack			226 4	226		387/516 8	790/516		387/516		387/ *808*		808/819	808/819	819 DTM ONLY			
Jeep	Grand Cherokee	Console, aft of		228 3,5	228 5				387/228 1	387/228+	387/228	387/598								785 DTM
Jeep	Gladiator	Center Console																821		
Jeep	Liberty/Cherokee	Console, aft of				227		387/516 8			?	387/785								
Jeep	Renegade	Center Stack												387/819*						166
Jeep	Wrangler	Console, aft of									387/546						387/821			
Jeep	Wagoneer & Gr Wag				227 4	227		387/546 8												785 DTM only

FCA EDR Evolution (under dev't)



2018+ Output New Look –

Removed multiple brake elements, added ABS/SC, moved strg to P1, hi res
Two tables at 0.1 sec interval, others tables @ 0.25 sec intervals

Pre-Crash Data [10 samples/sec] (Most Recent Event - table 1 of 2)
(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Pre-Crash Recorder Status	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full	Service Brake	Engine RPM	ABS Activity	Stability Control	Steering Input (deg)
-5.0	Complete	96 [155]	0.00	Off	3,878	No	On	1
-4.9	Complete	96 [154]	0.00	Off	3,835	No	On	-1
-4.8	Complete	96 [154]	0.00	Off	3,726	No	On	-5
-4.7	Complete	96 [154]	0.00	Off	3,621	No	On	-10
-4.6	Complete	96 [154]	0.00	Off	3,489	No	On	-18
-4.5	Complete	95 [154]	0.00	Off	3,344	No	On	-21
-4.4	Complete	95 [153]	0.00	Off	3,210	No	On	-20
-4.3	Complete	95 [153]	0.00	Off	3,109	No	On	-19
-4.2	Complete	95 [153]	0.00	Off	3,068	No	On	-19
-4.1	Complete	95 [152]	0.00	Off	3,108	No	On	-18
-4.0	Complete	95 [152]	0.00	Off	3,079	No	On	-18

Wheel RPM replaced by speed

See driven vs non driven, left vs right, ABS working…

Pre-Crash Data [10 samples/sec] (Most Recent Event - table 2 of 2)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	ABS MIL	Yaw Rate (deg/sec)	Wheel Speed LF (km/h)	Wheel Speed RF (km/h)	Wheel Speed LR (km/h)	Wheel Speed RR (km/h)
-5.0	Off	0.40	154.37	154.52	154.55	154.52
-4.9	Off	-0.48	154.28	154.16	154.25	154.25
-4.8	Off	-1.76	154.40	154.09	153.88	153.77
-4.7	Off	-2.80	154.28	153.95	153.88	153.67
-4.6	Off	-4.24	154.01	153.64	153.85	153.55
-4.5	Off	-6.80	153.98	153.16	153.80	153.19
-4.4	Off	-7.12	153.80	152.88	153.70	152.80
-4.3	Off	-5.84	153.49	152.67	153.37	152.62
-4.2	Off	-5.12	153.13	152.19	153.25	152.38
-4.1	Off	-4.64	152.80	152.04	152.91	151.91
-4.0	Off	-4.64	152.49	151.73	152.46	151.65
-3.9	Off	-4.00	152.13	151.55	152.13	151.31
-3.8	Off	-3.68	151.83	151.25	151.95	151.13

Current Gear & More @ 0.25 sec

Pre-Crash Data [4 samples/sec] (Most Recent Event - table 1 of 2)

(the most recent sampled values are recorded prior to the event)

NEW

Changes at END of shift

Time Stamp (sec)	Pre-Crash Recorder Status	Raw Manifold Pressure (kPa)	PCM MIL	ETC Lamp	Current Gear	Reverse Gear Status
-5.00	Complete	20.00	Off	Off	D5	Not_Reverse
-4.75	Complete	20.00	Off	Off	D5	Not_Reverse
-4.50	Complete	21.60	Off	Off	D5	Not_Reverse
-4.25	Complete	23.20	Off	Off	D5	Not_Reverse
-4.00	Complete	23.20	Off	Off	D6	Not_Reverse
-3.75	Complete	23.20	Off	Off	D6	Not_Reverse
-3.50	Complete	23.20	Off	Off	D6	Not_Reverse
-3.25	Complete	23.20	Off	Off	D6	Not_Reverse
-3.00	Complete	22.40	Off	Off	D6	Not_Reverse
-2.75	Complete	21.60	Off	Off	D6	Not_Reverse
-2.50	Complete	21.60	Off	Off	D7	Not_Reverse
-2.25	Complete	21.60	Off	Off	D7	Not_Reverse
-2.00	Complete	21.60	Off	Off	D7	Not_Reverse
-1.75	Complete	21.60	Off	Off	D7	Not_Reverse

‹ 10.63 in ‹

Tire pressure @ 0.25 sec intervals

Pre-Crash Data [4 samples/sec] (Most Recent Event - table 2 of 2)

(the most recent sampled values are recorded prior to the event)

NEW

Time Stamp (sec)	Tire Pressure Position	Tire Pressure Status	Tire Pressure Value (PSI)	Cruise Control Lamp Status	Cruise Control Engaged Status	ACC On/Off	ACC Set Speed (MPH [km/h])
-5.00	LHF	Normal	34	Off	Not_Engaged	Off	0 [0.0]
-4.75	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-4.50	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-4.25	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-4.00	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-3.75	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-3.50	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-3.25	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-3.00	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-2.75	RHF	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-2.50	RHF	Normal	33	Off	Not_Engaged	Off	0 [0.0]

Integrating FCA Yaw Rate

From 08 charger case

- 3 deg/sec * 0.1 sec =	-0.3 degrees
-11 deg/sec * 0.1 sec =	-1.1 degrees
-18 deg/sec * 0.1 sec =	-1.8 degrees
-15 deg/sec * 0.1 sec =	-1.5 degrees
- 9 deg/sec * 0.1 sec =	-0.9 degrees
- 4 deg/sec * 0.1 sec =	-0.4 degrees
- 1 deg/sec * 0.1 sec =	-0.1 degrees
Total Swerve Right	6.1 degrees

Note 1: this is how far nose of car is pointing right – *CG* motion may *lag*

Note 2: Polarity opposite RH rule this model

New Chrysler Info

1. High G Frontal sensors not likely to clip.
2. “Sensor Design Range Exceeded” line on graph means clipping above that level
3. Buckle switch NOT normally used to determine if pre-tensioner fires (Exceptions on some non US built vehicles)
4. Fiat Spider = Mazda Miata, threshold <5mph
5. Master Cylinder Pressure on 2018 Charger!
6. 2019+ not reading in 21.4, revert to 19.5 to read
7. 2019+ Compass must be read DTM – see help file
8. Odometer in EDR may vary from display by <10km

New Issues

- 2023 Grand Cherokee not communicating with CDR

Hyundai/Kia EDR Tools

- Starting Jan 2022, you may purchase U.S. DLC only kits from Crash Data Group **$5150 per brand** including software 1st yr.
- 1ea of DTM cables 1-24 + CFCI adapter in a case $8500 from CDG
- Full Hyundai kit from GIT comes with 13 DTM cables thru #24, Kia with 15 DTM cables thru #23. 10 Cables are in both. New cables cost $300 each. Cables #25-30 not included in full kits as of last check.
- Hyundai full kit: Part G0Z**H**DMN001 last known price ***$6450,*** Kia full kit: Part GIT0Z**K**DMN001 ***$6950 + tax+ship (price may have gone up – lots of others did),***
- Software updates **$825/yr/** brand from GIT Tool Company or Crash Data Group. Current V58 Hyundai/Genesis, V54 Kia.
- CAN-FD (Flexible Data Rate) DTM adapter needed for 2020 HV80 & other new models cost approx. $1,080
- Contacts: Patrick Dadson pdadson@gitamerica.com (888)549-4977
- Scott Baker **crash@crashdatagroup.com** (800) 280-7940

Additional Info

1. K/H Interface modules are identical, but you must buy both kits to get software for both. Software updates won't download unless the **correct brand interface module** is attached to your laptop and the ID & password keyed to that interface are used. US Kit update request must come from within US borders (VPN not honored).
2. I read K/H modules for others for $300. More people are buying Hyundai kits now that we have 10+ model years on the road and you can buy DLC only kits.

Officially Supported US 2013+ Models V58H/54K

Model	Model Years															
KIA	2010	2011	2012	2013*	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	
CADENZA (Azera sister)					19(VG)	19	19	13(YG)	13	13	13	13				
EV6													29+CFCI (CV)			
FORTE	10	10	10	10 (TD)	6 (YD)	6	6	12(YD,YDM)	12	21 or 28 (BDm)				21		
K5												26(D3LA)				
K900						11 (KH)	11	11	11	21 or 28 (RJ)						
NIRO (DE EV, DE HEV, DE PHEV)								21(DE)	21	21	21	21	21	30+CFCI (SG)		
OPTIMA	No	4	4	4 (TF/QF)	4	4	14 (JF,JFA)*	14	14	14	14					
OPTIMA HYBRID	No	4	4	4 (TF HEV)	4	4	4	23(JF HEV)	23	23	23					
RIO	No	No	8	8(UB)	8	8	8	8	21 (SC)	21	21	21	21	21		
RONDO (Canada only 2011 MY+)	No	No	9	(not officially supported since non-US, read as a Kia Rio)												
SEDONA	No	No	No	No	10(VQ)	12(YP)	12	12	12	12	12	12	29+CFCI (KA4)			
SELTOS												28(SP2)	28	28		
SORENTO	No	3	3	3(XM)	3,7(XM)	3,7	13 (UMA)	13	13	13	13	29+CFCI (MQ4)		29		
SOUL	1 see note 1	1 see note 1	1	1(AM)	12(PS)	12(PSEV)	12	12	12	12,26(SK3 EV	26 (SK3)	26	26	26		
SPORTAGE (SL)	No	Yes	2	2(SL)	2	2	2	13(QL)	13	13	13	13	30+CFCI (SG)			
STINGER (CK)									21 (CK)	21	21	21	21	21		
TELLURIDE (ON)											25(ON)	25	25	25		
HYUNDAI	2010	2011	2012	2013*	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	
ACCENT (RB)	No 10	No?	5 see note 2	5(RB)	5	5	5	5	21(HC)	21	21	21	21	21		
AZERA (HG)		likely	Yes	7(HG)	7	7	7	7								
ELANTRA Coupe(JK), Sedan(UD/MD	No	Yes	8	8(JK/UD/MD)	8	8	8	21(AD, ADA)	21	21	21	28 (CN7)				
ELANTRA GT (GD) (aka "i30")	old model		new model	6/12(GD)	6,12	6,12	6,12	6,12	21(PD)	21	21	21				
EQUUS (VI)(stretch genesis)	?	yes?	yes?	9,11(VI)	9,11(VI)	9,11(VI)	9,11(VI)									
GENESIS (Coupe BK, Sedan BH/DH)	9 shave keyway	9 shave keyway	?	9(BK, BH)	9	13(DH)	13		2018	2019	2020	2021	2022	2023		
GENESIS GV60													29+CFCI (JW1)			
GENESIS G70	YELLOW = COULD BE EITHER OF 2 CABLES, SEE WHICH FITS									21,25(IK)	21,25	21,25	21,25	21,25		
GENESIS GV70													29+CFCI (JK1)			
GENESIS G80								13(DH)	13	13	13	29+CFCI (RG3)				
GENESIS GV80											29+CFCI (JX1)					
GENESIS G90								13(HI)	13	13	13	13	13	30+CFCI (RS4)		
IONIQ EV and HEV								21(AE)	21	21	21	21	21			
IONIQ 5													29+CFCI (NE EV)			
KONA (i20, Rio sister)small SUV)									24 (OS)	24	24	24	24	24		
KONA EV										27 (OS EV)	27	27	27	27		
NEXO										26 (FE)	26	26	26	26		
PALISADE											25(LX2)	25	25	25		
SANTA CRUZ				2013*	2014	2015	2016	2017	2018	2019	2020		30+CFCI (NX4A OB)			
SANTE FE	9 shave keyway note 3	9 shave keyway	9 shave keyway	7(NC,DMA)	7	7	7	7	7	25 (TMA)	25	25	25	25		
SONATA		4	?	(YFA, YF HEV	4	14 (LF) 4 (HEV)	14 (LF HEV)	14	14	14	25(DN8A)	25	25	25		
TUCSON	Yes	Yes	2	2(LM)	2	2	13(TL)	13	13	13	13	13	30+CFCI			
VELOSTER coupe	out of productio		?	5(FS)	5	5	5	5	5	14 (JS)	14					
VELOSTER N										28(JSN)	25	25	25	25		
VENUE											30(QX)	30				
Gold = 2013 Part 563 Compliant		K/H Common Cables		2013*	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	

New Data -Smart Cruise Control info 21 Palisades

Pre-Crash Information 2 (-5 ~ 0 sec)

Time (sec)	Gear shift selector display	SCC information (Main switch status)	SCC information (Set Speed)	SCC information (SPEED UNIT)	SCC information (Mode status)	SCC information (Fail info)
-5.0	D	ON	85	MPH	Engaged	System without Error
-4.5	D	ON	85	MPH	Engaged	System without Error

Auto Braking info 21 Palisades

Pre-Crash Information 3 (-5 ~ 0 sec)

Forward Collision Alert

Time (sec)	FCA (Function status)	FCA (Warning Level)	FCA (Fail info)
-1.0	ON	No Warning / Invalid or Not supported	Normal
-0.9	ON	No Warning / Invalid or Not supported	Normal
-0.8	ON	Level 1	Normal
-0.7	ON	Level 2	Normal
-0.6	ON	Level 2	Normal

Audible

0.2g Autobrake

Level 3 =Max autobrake

Backpowering Kia/Hyundai Vehicles

- The K-H Interface normally gets its power thru the DLC plug. It does not have a separate power plug.
- Back powering required using a second fuse block adapter to power the DLC plug.
- Kent Boots has procured the mating connector to a diagnostic port on the interface that has power input pins.
- He sells a cable for $75 plus $10 shipping that mates the interface to Kia/Hyundai and Bosch 2.5mm round pin power supplies
- Email Kent kent@getedrdata.com

Pre-2013 Kia Hyundai may have data

Model	Vehicle Size, Descrip	Model Years 2010	2011	2012	2013*
KIA		Cable #	Cable #	Cable #	Cable #
CADENZA (VG, 2014) (Azera sister)	112.0" WB	YG requires cable 13 per software			
FORTE (TD 2013, YD 2014))	104.3"WB, compact	10	10	10	10 (TD)
K900 (KH 2015)	119.9"WB				
NIRO					
OPTIMA (2013 TF QF, 2016 JF, JFA)	110.0" WB sedan	No	4	4	4 (TF/QF)
OPTIMA HYBRID	110.0" WB sedan	No	4	4	(TF HEV)
RIO (UB)	94.5"WB, compact	No	No	8	8(UB)
RONDO (Canada only 2011 MY+)		No	out of production		
SEDONA (VQ '14, YP '15)	113.8/118.9" WB miniv	No	No	No	No
SORENTO (XM to 2015)(UMA 2016)	106.3" WB, crossover	No	3	3	3(XM)
SOUL (AM 2013, PS 2014+)	100.4" WB	1 see note 1	1 see note 1	1	1(AM)
SPORTAGE (SL)	103.9" WB, crossover	No	Yes	2	2(SL)
HYUNDAI					
ACCENT (RB)	94.5"WB, compact	No 10	No?	5 see note 2	5(RB)
AZERA (HG)	109.4"WB, sedan		likely	Yes	7(HG)
ELANTRA Coupe(JK), Sedan(UD/MD)	104.3"WB, compact	No	Yes	8	8(JK/UD/MD)
ELANTRA GT (GD) (aka "i30")	former Elantra Touring	old model		new model	6(GD)
EQUUS (VI)(stretch genesis)	119.9" WB, luxury seda	?	yes?	yes?	9(VI)
GENESIS (Coupe BK, Sedan BH/DH))	115.6" WB, luxury seda	9 shave keyway	9 shave keyway	?	9(BK, BH)
GENESIS G90					
SANTE FE (Sport DMA, NC)	106.3" WB, crossover	9 shave keyway note 3	9 shave keyway	9 shave keyway	7(NC,DMA)
SONATA (YF sedan, YF HEV)	110.0" WB sedan		4	?	4(YF)
TUCSON (LM to 2015)(TL in 2016)	103.9" WB, crossover	Yes	Yes	2	2(LM)
VELOSTER coupe (FS)	104.3"WB, compact	out of productior		?	5(FS)

2010-2012 EDR Access

1. The tool requires VIN input and only allows read of 2013+ VINS.
2. Input surrogate 2013 VIN to communicate with 2010-2012 models. Report will list surrogate VIN used.
3. Decoder changed slightly for 2013. Not all data decodes properly.
4. 2010-12 file admissibility must be presented skillfully (get help).
5. **Data successfully admitted in Virginia, Florida, and California.**

K/H 2010-2012 data reliability?

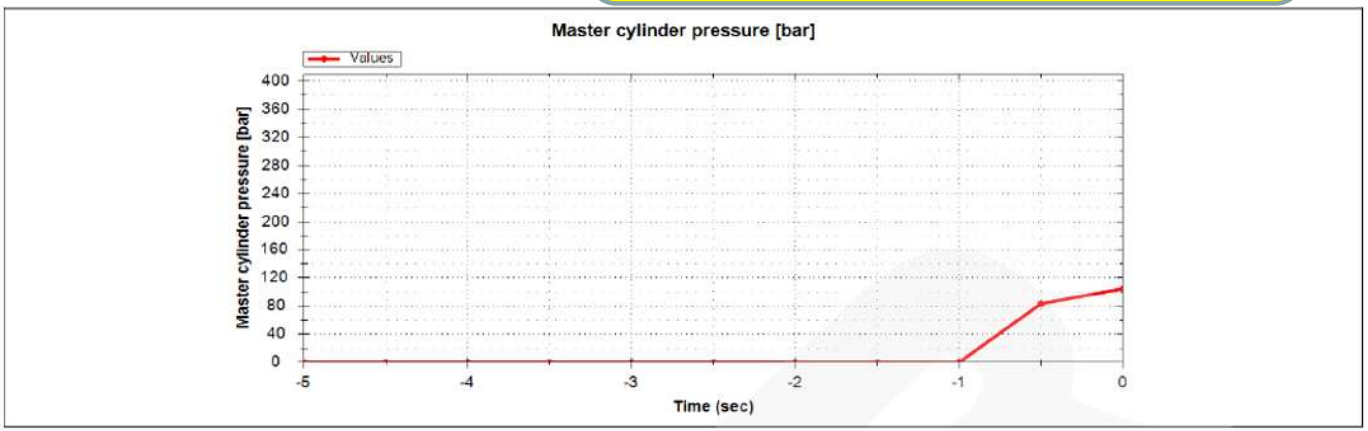
1. In 50+ NHTSA crashes, speed, delta V and belt use appear OK (one exception)
2. Steering angle, and accel data appear implausible on some
3. Key cycles at download not in all models
4. SAE 2014-01-0502 has details.
5. Additional testing on 2012 Soul reported in SAE 2015-01-1445.
6. 2021-01-0905 **2 crash tests of 2012 Hyundai Genesis Coupe EDR.**
7. Admitted to court in VA, CA, one other

Brake Pres. added to 2019 Forte

< Event 1 >

Master cylinder pressure

Units in Bars. 1 Mpa=10 bars
Toyota max scale is 12 Mpa=120 bar



Num	Time (sec)	Master cylinder pressure [bar]
1	-5.0	0.2
2	-4.5	0.2
3	-4.0	0.2
4	-3.5	0.2
5	-3.0	0.2
6	-2.5	0.2
7	-2.0	0.2
8	-1.5	0.2
9	-1.0	0.2
10	-0.5	83.7
11	0.0	104.6

Some 2012 Honda not supported

- For 2012 Pilot and Odyessy help file Note 1 indicates due to production variations some vehicles may not be supported by CDR, and will repond “no communication with module”.
- Contact ahmmediation@ahm.honda.com for assistance in reading those ACM’s. (ahm stands for American Honda Motor). (reported on CDR Tool group Oct 2023)

Honda 23.0 CDR Coverage

Model	Location	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
ACURA		Last 3 digits of Bosch CDR cable number					**Gen 2**		**Gen 3**			**Gen 4**		
RL	Under dashboard cente		**547**											
TL	Under dashboard cente		**547**											
TSX	Under dashboard cente		**547**											
ZDX	Under dashboard cente		**547**											
MDX	Under dashboard cente		**547**		**387/789**			**387/810**				**?**	**846**	
NSX	Under dashboard cente							**387/789**						
RDX	Under dashboard cente			**547**			**387/789**		**387/825**					
ILX	Under dashboard cente			**547**			**387/789**							
RLX	Under dashboard cente				**387/789**									
TLX	Under dashboard cente					**387/789**				**387/810**		**846**		
INTEGRA														**846**
HONDA			**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	
ACCORD	Under dashbord center			**387/789**			**387/810**		**387/825**					
CIVIC 2 DR	Under dashbord center		**547**				**387/810**						**846**	
CIVIC 4 DR	Under dashbord center		**547**	**387/789**			**387/810**						**846**	
CLARITY												**387/810**		
CROSSTOUR	Under dashbord center		**547**											
CR-V	Under dashbord center		**547**				**387/789**	**387/810**					**846**	
CR-Z	Under dashbord center		**547**											
FCX Clarity	Under dashbord center							**387/810**						
FIT	Under dashbord center	EV	**547**			**387/789**		**387/810**					**846**	
HR-V							**387/789**		**387/810**				**846**	
INSIGHT	Under dashbord center		**547**							**387/810**				
ODYSSEY	Under dashbord center		**547**		**387/789**			**387/810**						
PASSPORT	Under dashbord center									**387/810**				
PILOT	Under dashbord center		**547**				**387/789**	**387/810**						**846**
RIDGELINE	Under dashbord center		**547**					**387/810**						

2011- NO time series EDR But may have Delta V single value

Honda only assists law enforce -ment voluntar ily

Honda EDR Evolution

- 2012-2015 Gen 1 – One set DL's
  - Roll rate polarity varies by supplier
    (polarity added under graph)
  - Accel data instantaneous (not useful)
- 2016-17 Gen 2 adds AEB, Lane Departure data
- 2018 – Gen 3 adds "IG SW status"
  - Airbag deploys even if ign off until car stops
  - Markets outside US/Japan do not get precrash data
- 2022 – Gen 4 – time -0.5 to TRG, Rear Belt Status, Lifetime event counter

Gen 2 New Data Elements

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 2 of 3

Time Stamp (sec)	PCM Derived Accelerator Pedal Position, % full	Forward Collision Warning (Not Warning/ Warning)	Collision Mitigation Braking System (Not Engaged/ Engaged)	Collision Mitigation Braking System, Forward Collision Warning (On/Off)	Lane Departure Warning (Not Warning/ Warning)	Road Departure Mitigation (Not Engaged/ Engaged)	Road Departure Mitigation, Lane Departure Warning (On/Off)
-5.0	13	Not warning	Not engaged	On	Not warning	Not engaged	Off
-4.5	13	Not warning	Not engaged	On	Not warning	Not engaged	Off

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 3 of 3

Time Stamp (sec)	Adaptive Cruise Control (Not Engaged/ Engaged)	Adaptive Cruise Control (On/Off)	Lane Keeping Assist (Not Engaged/ Engaged)	Lane Keeping Assist (On/Off)	Cruise Control (Not Engaged/ Engaged)	Cruise Control (On/Off)
-5.0	Not engaged	Off	Not engaged	Off	Not Engaged	Off
-4.5	Not engaged	Off	Not engaged	Off	Not Engaged	Off

GEN 4 New Data Elements

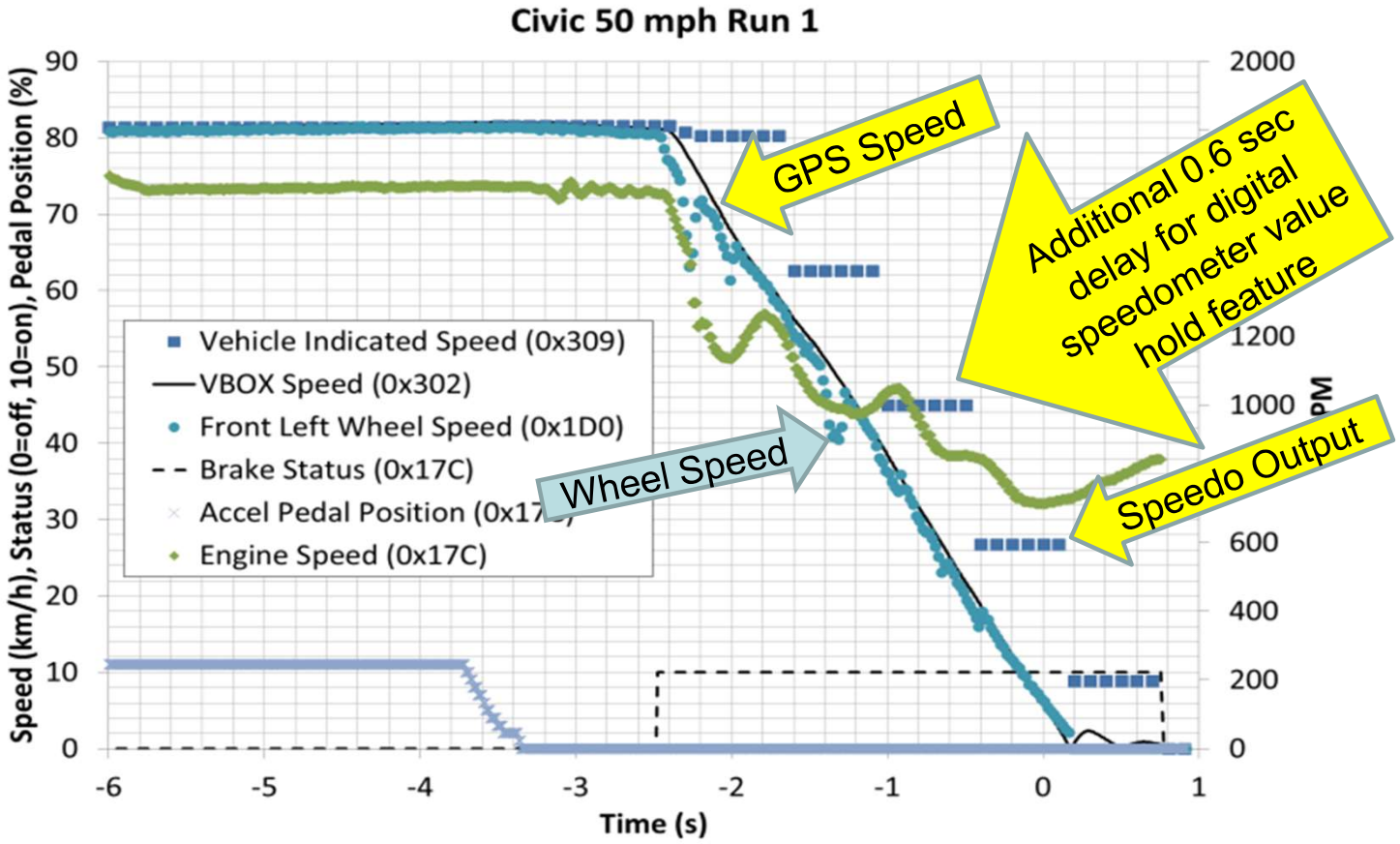
System Status at Event (Impact Event Record 1)

Seat track position switch, foremost, status, driver	No
Safety belt status, left second row passenger	Off
Safety belt status, middle second row passenger	Off
Safety belt status, right second row passenger	Off
Ignition cycle, crash	1,011
ECU life timer, crash	225h:58m:50s
Time from Time zero to End of Event (msec)	150
Time between T-0.5 and Time zero (msec)	215
Lifetime event number	1
Multi-Event Number (Event 1/2)	1

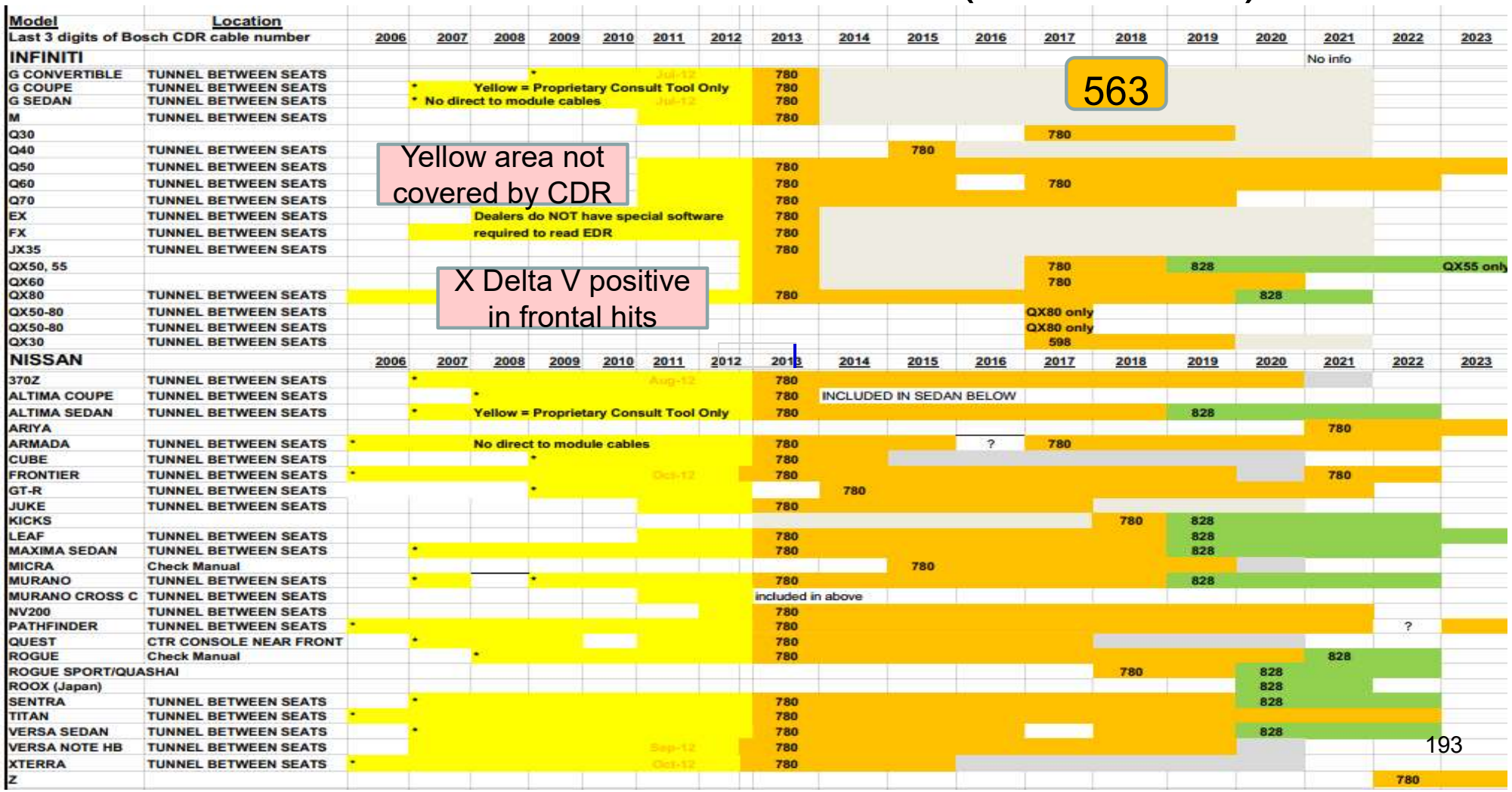
Reminders: UNIQUE TO HONDA

1. Most manufactures record the speed calculated in the powertrain control module as it is being sent **TO** the speedometer on the CAN bus.
2. Honda records the speed being displayed **ON** the speedometer, after the PCM calculated speed has been dampened or “smoothed”.
3. The last data point labeled 0.0 is sampled at AE. -0.5 is -0.01 to -0.50. While sampled at AE, speedometer processing delays what is being reported, so it lags the true speed at impact during hard braking.

2012 Civic Speed Reporting Delay



NISSAN EDR's 23.0 (CDR 51%)



Nissan Pre-Crash Data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % full	Engine RPM	Motor RPM	Service Brake (On, Off)	Steering Input (deg)
-5.0	24 [39]	16	2650	0	Off (Brake Not Activated)	-32
-4.5	23 [37]	11.5	1750	0	Off (Brake Not Activated)	-2
-4.0	21 [34]	9	1200	0	On (Brake Activated)	28
-3.5	15 [24]	9	1150	0	On (Brake Activated)	14
-3.0	13 [21]	4	950	0	On (Brake Activated)	4
-2.5	11 [17]	4	950	0	On (Brake Activated)	4
-2.0	5 [8]	4	900	0	On (Brake Activated)	-48
-1.5	3 [5]	4	850	0	On (Brake Activated)	-98
-1.0	1 [2]	4	850	0	On (Brake Activated)	-160
-0.5	3 [5]	10.5	1100	0	Off (Brake Not Activated)	-194
0.0	6 [9]	16	1150	0	Off (Brake Not Activated)	-218

Hybrid/Electric or Trans Input

Trans input tells you convertere slip

Table 2 – not rq'd

At AE

Nissan Long ΔV Polarity Error in SOME

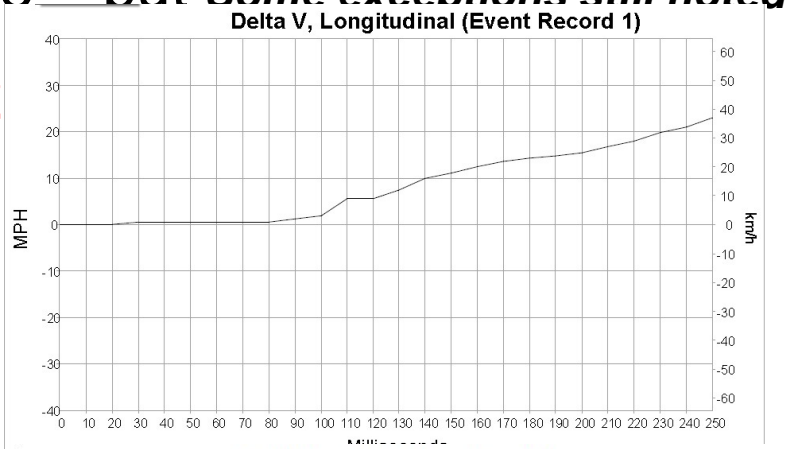
Altima fixed - Front hit *was* positive – but ***Some exceptions still noted!***

2013 Altima in CDR 12.2

System Status at Event (Event Record 1)

Frontal Air Bag Suppression Switch Status	S airbag inhibit)
Maximum Delta-V, Longitudinal (MPH [km/h])	29 [47]
Time, Maximum Delta-V, Longitudinal (msec)	300
Maximum Delta-V, Lateral (MPH [km/h])	5 [8]
Time, Maximum Delta-V, Lateral (msec)	210
Maximum Acceleration, Longitudinal (g)	32.5
Time, Maximum Acceleration, Longitudinal (msec)	102.5
Maximum Acceleration, Lateral (g)	36
Time, Maximum Acceleration, Lateral (msec)	112.5

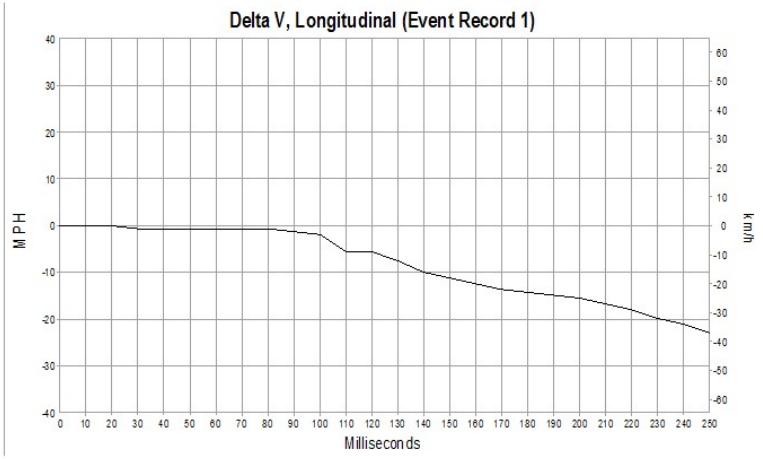
+29 at 300ms



2013 Altima in CDR 16.5

S airbag inhibit)	
Maximum Delta-V, Longitudinal (MPH [km/h])	-29 [-47]
Time, Maximum Delta-V, Longitudinal (msec)	300
Maximum Delta-V, Lateral (MPH [km/h])	5 [8]
Time, Maximum Delta-V, Lateral (msec)	210
Maximum Acceleration, Longitudinal (g)	-32.5
Time, Maximum Acceleration, Longitudinal (msec)	102.5
Maximum Acceleration, Lateral (g)	36
Time, Maximum Acceleration, Lateral (msec)	112.5

-29 at 300ms



Nissan Recording Threshold

1. Several files have been read with events under 5mph Delta V over 150ms.
2. From 2013-2019 we believed all Nissans had a wakeup threshold of AE
3. At 2020 EDR User Summit, it was reported that tests were done on a 2014 Pathfinder at low speeds. No events were created below 5mph Delta V, events were created over 5mph.
4. Threshold could be model dependent!

VW/Audi Brands EDR

1. EDR in 2014 EOS, most 2015+ VW/Audi.
2. EDR was installed in anticipation of the US NPRM requiring EDR's by 9/1/2014, which was never implemented.
3. **Bentley** has EDR 2016+.
4. **Lamborghini** has EDR in 2015+ Adventadore/Hurracan, 2019+ URUS
5. **Porsche no EDR** *until 2019 Taycan, + Most 2021's*.
6. US only 2015-17, Worldwide 2018+ in transverse engine platforms

VW/Audi 21.3 Oct 2021 34+% (NC in 23.0)

Model	Location	2009-13	2014	2015	2016	2017	2018	2019	2020	2021	2022
Last 3 digits of Bosch CDR adapter/cable number				See help file for Rolls Royce							
All VW/Audi Require the 387 Adapter to read DTM			Gold = US only (Canada not covered)				Yellow=Worldwide				
VW											
Arteon	UNDER CENTER STACK						805				
Atlas	UNDER CENTER STACK						805				
Beetle	UNDER CENTER STACK			804			Canada added				
CC	UNDER CENTER STACK			804							
Crafter	UNDER CENTER STACK						805				
EOS	UNDER CENTER STACK		806								
Golf, eGolf	UNDER CENTER STACK			805			Worldwide				
Jetta	UNDER CENTER STACK			804			805				
Neo											
Passat	UNDER CENTER STACK			804			805				
Polo	UNDER CENTER STACK						805				
Routan	UNDER CENTER STACK										
Sportwagen + Alltrack				Golf platform, not called out separtely in help file							
T-Roc	UNDER CENTER STACK						805	805			
T-Cross	UNDER CENTER STACK										
Tiguan	UNDER CENTER STACK			804		805					
Touareg	UNDER GEAR SELECTOR			804				500/826			
Touran	UNDER CENTER STACK						805				
AUDI				2015	2016	2017	2018	2019	2020	2021	2022
A1	Under Center Stack							805			
A3,S3, RS3	Under Center Stack			805					842		
A4, S4, A4 allroa	Tunnel between 1st/2nd row seats			804		500/813					
A5, R5, S5, RS5	Tunnel between 1st/2nd row seats			804			500/813				
A6, A6L, S6, Ava	Tunnel between 1st/2nd row seats			804			500/826				
A7, S7, RS7	Tunnel between 1st/2nd row seats			804			500/826				
A8, A8L, S8	Tunnel between 1st/2nd row seats			804			500/826				
Q2	Under Center Stack						805				
Q3	Under Center Stack			804				805			
Q4											842
Q5, SQ5	Tunnel between 1st/2nd row seats			804			500/813				
Q7	Under Center Stack					500/813			500/826		
Q8	Under Center Stack						500/826				
R8	Under Center Stack					804					
TT, TTS				not covered at this time							
eTron	Under Center Stack							500/826			
The source data for this chart is the Bosch CDR help file for Version 19.6.		Always consult the latest help file for the most accurate coverage									
Copyright Ruth Consulting LLC 2021- Authorized for distribution by UNF-IPTM & SAE											

Chrysler Minivan Sister

No “back model” coverage

2023 and 2024 now covered in 23.1

VW/Audi Events

- Recorder holds up to 6 events in Chron order.
- Event types are Front, Rear, Side, Rollover, Pedestrian (if equipped), External Trigger, and HV (High Voltage disconnect). External triggers are Automatic Driver Assistance System (ADAS) events and are not crashes. If the driver is not braking hard enough, ADAS will override and brake harder.
- DL's say crash event threshold is 8 kph (5mph) lat or long Delta V over 150ms for ND's, D's will record at any magnitude. External Triggers may record with 0 ΔV.
- A small # of ND events have <5mph, this is the exception not the rule.

VW/Audi Details

1. Same data set for all event types
2. Time from last precrash to AE noted
3. Some 2017+ modules use Flex Ray/500
4. Date/Time of Crash from clock (GPS if clock not set) in 2015+ Audi models, phased into VW models
5. Newer models require hood to be open to read through DLC, have gateway
6. EDR Speed is as reported on the speedometer

VW/Audi Speed Data +5%?

- A 2021 Audi A6 was tested in Europe by setting "External Trigger" (Auto Emergency Braking) events and measure reported speed vs GPS speed. Authors found the reported values to be 5% higher than actual speed.[1]
- Two 2021 VW ID4 IIHS tests run at 64.4 kph reported 64 kph in EDR.[2] Further testing needed?

1 "Event data recordings triggered by automated emergency braking", European Association For Accident Research (EVU 30th annual conf July 2023) by Spec and Van Essen (PDF) Event data recordings triggered by automated emergency braking (researchgate.net)

2 "Late Model EDR Accuracy" WREX 2023 by Ruth

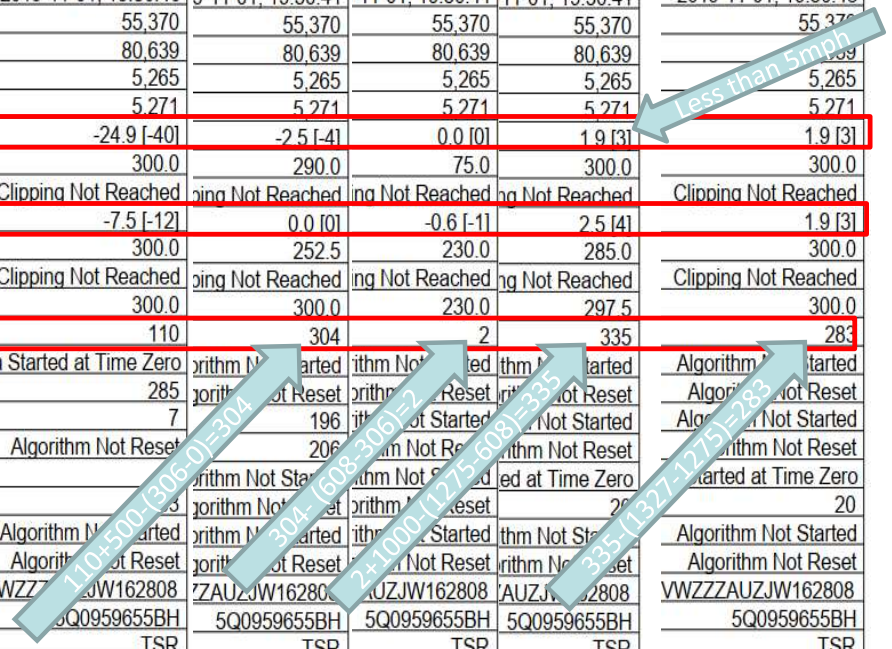
VW/Audi Auto Emergency Braking

- IPTM is not aware of any pictures stored or any way to access pictures taken by the AEB system.
- Sources within VW advise if you put a good scan tool on a vehicle after a crash, there is a speed stored in the camera module calculated from images and not from the vehicle speed sensor.
- If tires are locked or otherwise not rolling at vehicle speed, the camera source may be a more accurate speed.

Threshold & Time between events (Five Overlaid)

System Status at Event (Record 5)

Event Counter at Event	1	2	3	4	5
Event Type	Frontal	Frontal	Frontal	ta Not Available	Data Not Available
Multi-Event, Number of Events	1, Event	2, Event	3, Event	4, Event	5, Event
Time from Initial Event to Current Event (msec)	0.0	306.0	608.0	1,275.0	1,327.0
Vehicle Clock, Date and Time at Event (YYYY-MM-DD, HH:M	2019-11-01, 19:50:40	9-11-01, 19:50:41	-11-01, 19:50:41	11-01, 19:50:41	2019-11-01, 19:50:43
Vehicle Mileage (km)	55,370	55,370	55,370	55,370	55,37
Operating Time (min)	80,639	80,639	80,639	80,639	
Ignition Cycle at Event (Cycles)	5,265	5,265	5,265	5,265	5,265
Ignition Cycle at Download (Cycles)	5,271	5,271	5,271	5,271	5,271
Maximum Delta-V, Longitudinal (MPH [km/h])	-24.9 [-40]	-2.5 [-4]	0.0 [0]	1.9 [3]	1.9 [3]
Time, Maximum Delta-V, Longitudinal (msec)	300.0	290.0	75.0	300.0	300.0
Clipping Time, Longitudinal Acceleration Sensor (msec)	Clipping Not Reached	ping Not Reached	ing Not Reached	ng Not Reached	Clipping Not Reached
Maximum Delta-V, Lateral (MPH [km/h])	-7.5 [-12]	0.0 [0]	-0.6 [-1]	2.5 [4]	1.9 [3]
Time, Maximum Delta-V, Lateral (msec)	300.0	252.5	230.0	285.0	300.0
Clipping Time, Lateral Acceleration Sensor (msec)	Clipping Not Reached	ping Not Reached	ing Not Reached	ng Not Reached	Clipping Not Reached
Time, Maximum Delta-V, Resultant (msec)	300.0	300.0	230.0	297.5	300.0
Time from Last Speed Data Sample (Precrash) to Time Zero	110	304	2	335	283
Time from Time Zero to Algorithm Start (Front) (msec)	ithm Started at Time Zero	orithm	ithm	thm	Algorithm
Time from Time Zero to Algorithm Reset (Front) (msec)	285	orit	orithm	rit	Algori
Time from Time Zero to Algorithm Start (Side) (msec)	7	196	it	Not Started	Alg
Time from Time Zero to Algorithm Reset (Side) (msec)	Algorithm Not Reset	206	m Not Re	ithm Not Reset	
Time from Time Zero to Algorithm Start (Rear) (msec)		orithm Not Sta	hm Not S	ed at Time Zero	
Time from Time Zero to Algorithm Reset (Rear) (msec)		gorithm Not	orithm	20	20
Time from Time Zero to Deployment (Rollover) (msec)	Algorithm N	orithm N	ith	thm Not Sta	Algorithm Not Started
Time from Time Zero to Algorithm Reset (Rollover) (msec)	Algorit	orit	n Not Reset	rithm No	Algorithm Not Reset
Vehicle Identification Number (VIN)	WVWZZ	ZAUZJW16280	UZJW162808	AUZJ	VWZZZAUZJW162808
Part Number, ACM	5Q0959655BH	5Q0959655BH	5Q0959655BH	5Q0959655BH	5Q0959655BH
Supplier ID, ACM	TSR	TSR	TSR	TSR	TSR
Production Date, ACM	171117	171117	171117	171117	171117
Supply Voltage (Before Event) (V)	14.7	12.6	12.6	12.6	12.6
Complete File Recorded	Completed Successfully	eted Successfully	ted Successfully	ed Successfully	Completed Successfully



Subaru CDR EDR Coverage 23.0

Model	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Bosch CDR900 Interface Required for true Subaru models													
Cables 614 and 616 were originally released for Toyota models but also service some Subaru's													
SUBARU US Models													
Ascent	Not in production								835			?Ne	835
BRZ	n/a	n/a	Read as Toyota brand 86 model with Cable 614 (CAN+ or CDR900 interface)									844	ase
Crosstrek	n/a	n/a	616										
Forester	No	No	616*	*					835				
Impreza	No	614*	*	614*/616							Japan		
Legacy	No	No	614		836					844			
Outback	No	No	614		836					844			
WRX (inlcudes STI)	No	No	No	No	614		616	614/616				844	844
Tribeca	No	No	No	No	Discontinued								
NON-US Models													
Exiga (Australia, Japan)			614										
XV (Europe)		614					614/616			616			
Levorg (Japan)					614		614/616				844	844	

NOTE: Bosch CDR coverage phased in from software versions 19.1 to 19.4 to 19.6 to 21.0, rechecked with 21.5

Prior to this the Subaru dealer ship tester, the Denso DST-I with Subaru Select Monitor (SSM3 and SSM4) software had to be used.

* if equipped with Gen 3 ACM there may not be an EDR

The source of this information is the Bosch CDR help file. Always consult the help file for the latest information.

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Subaru EDR Tools

1. Bosch CDR 900 interface and 21.3+ software reads all model and model years except for 2013-2020 BRZ, which you read as a Toyota brand Model 86. The 21.4+ software is rejecting the BRZ, reload 19.6 software to read it until Bosch fixes that.
2. Subaru records both a front and side event in anglular crashes. Use the first in time for Delta V.
3. Recent attempts to read a 2023 Crosstrek listed as not covered were successful using a 2021 VIN.

CDR 2015+ Outback Precrash Data

Pre-Crash Data -5 to 0 Sec (Record 1)

Time (sec)	-5.0	-4.5	-4.0	-3.5	-3.0	-2.5	-2.0	-1.5	-1.0	-0.5	0.0
Speed, Vehicle Indicated (MPH [km/h])	21.1 [34]	22.4 [36]	23.6 [38]	24.9 [40]	24.9 [40]	23.0 [37]	19.3 [31]	16.2 [26]	10.6 [17]	7.5 [12]	6.8 [11]
Accelerator Pedal, % Full (%)	23	28	24	0	0	0	0	0	0	0	0
Service Brake, On/Off	OFF	OFF	OFF	OFF	ON	ON	ON	ON	ON	ON	ON
Engine RPM (RPM)	1,900	2,100	2,300	1,800	1,300	1,100	1,200	1,100	1,000	1,000	1,000
ABS Activity	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Stability Control (On, Off, Engaged)	ON	ON	ON	ON	ON	ON	ON	ON	ON	ON	ON
Steering Input (deg)	0.0	0.0	0.0	0.0	0.0	-2.5	-2.5	-5.0	-5.0	-2.5	-2.5

TESLA 3.5% of new sales!



1. Remember the definition of an EDR is something that records a time series of data **as a result of a crash**.
2. Tesla may record information as a result of turning the key on, but that is not an event data recorder. More on this later.
3. Tesla did not have any EDR in the ACM of its original 2008 Roadster models.
4. Tesla has installed EDR in The 2012+ Model S, the 2015+ Model X, and the 2017 Model 3.
5. Effective March 2018, Tesla announced they will sell a publicly available tool to download the Tesla EDR

Tesla EDR

- Kit includes:
  - TIV-144: In vehicle cable for Tesla Model S and X
  - TIV-145: In vehicle cable for Tesla legacy Model S
  - TIV-996: In vehicle cable for Tesla Model 3
  - TD2M-601: Direct to Module cable for Tesla Model X and 3
  - TD2M-602: Direct to Module cable for Tesla Model S
  - TD2M-139: Direct to Module cable for Tesla Model S (NEW CABLE)
  - AC Power Adapter
  - PCAN-USB adapter
  - Hard Shell Case



COST: $1400.00 as of 4/15/2023

Getting the Data

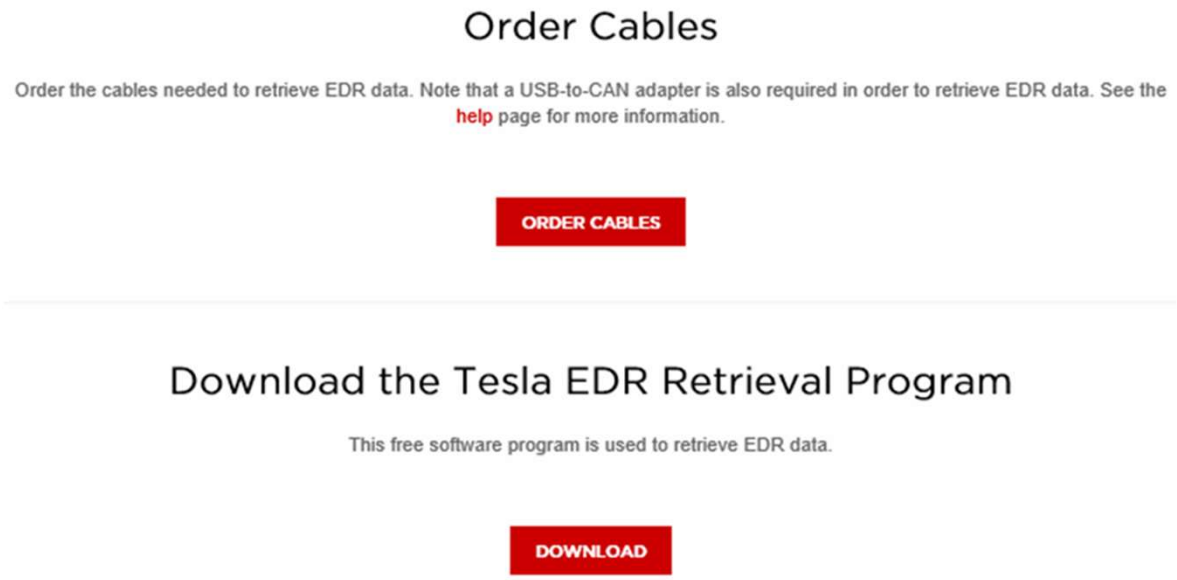
Coverage



Model	2008-11	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
ROADSTER	NO EDR												
Model S	Not in Prod								2/26/2019 GEN 2			6/10/2021 GEN 3	
Model X	Not in Production										2021+ GEN 2		
Model 3	Not in Production										10/5/2020 GEN 2		
Model Y	Not in Production												

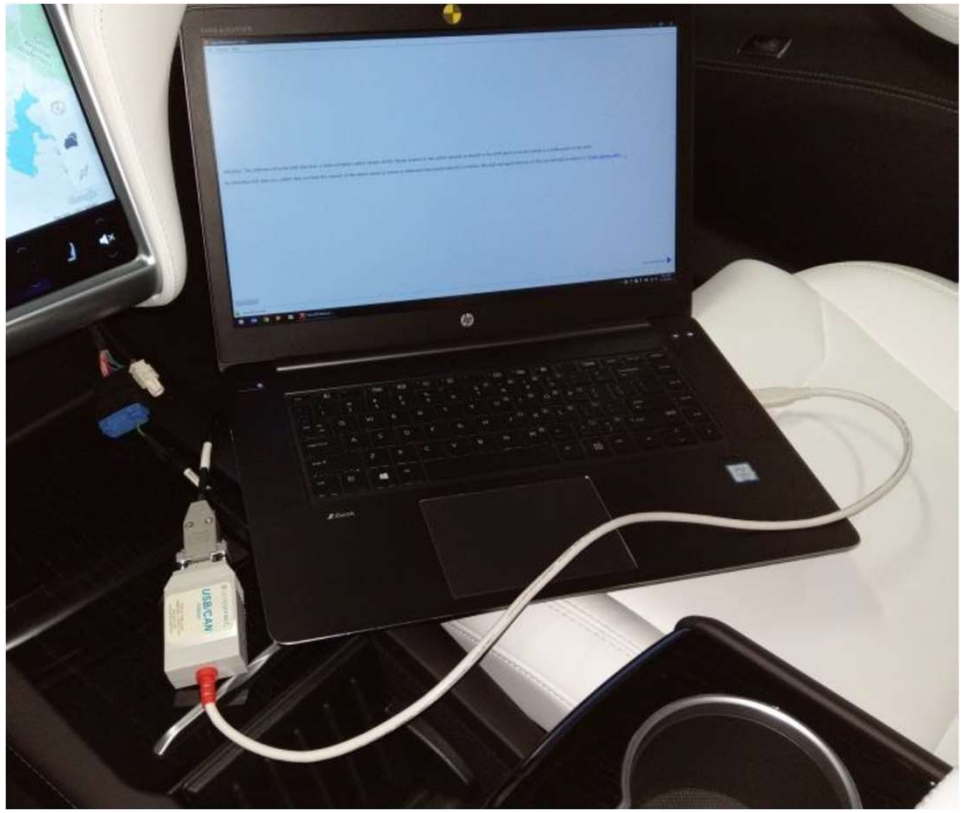
TESLA

- More information is available on their website https://edr.tesla.com
- See “Other sources” later



The Tesla EDR Tool

1. Tesla provides the software free of charge on their website that installs on your laptop. No special computer is needed.
2. The hardware kit sells for a list price of $1200.00.
3. The equivalent of the Bosch CDR interface manual is the “PCAN-USB” adapter. It is very small, and reminds you more of a USB to serial adapter, just a slight bit larger.

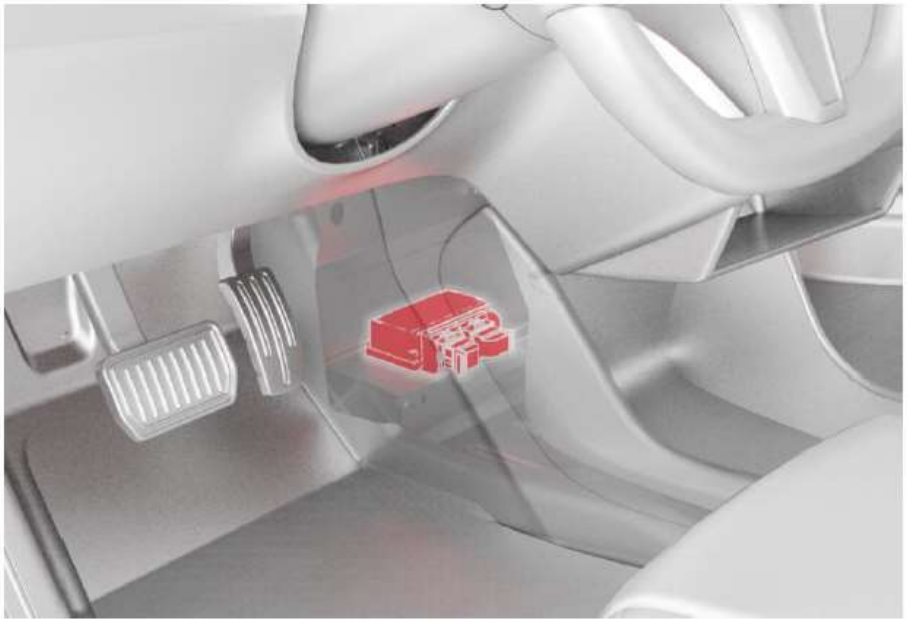


Plug in the PCAN adapter to the Body Harness. Model X and Model 3 are in other places & use a different connector described in Tesla's EDR guides by model at **edr.tesla.com/help**.



TESLA also has 3 direct to module connectors

If the In-Vehicle Connection procedure is unsuccessful, you might be able to connect to the RCM directly, and establish a direct-to-module connection. The RCM is located forward of the center console, below the instrument panel and center screen.



Some 2018 Model 3 Cables must be replaced

External Power must be used for DTM readouts

·. Connect 12V power to the Tesla Model S Direct-To-Module EDR Retrieval Harness.

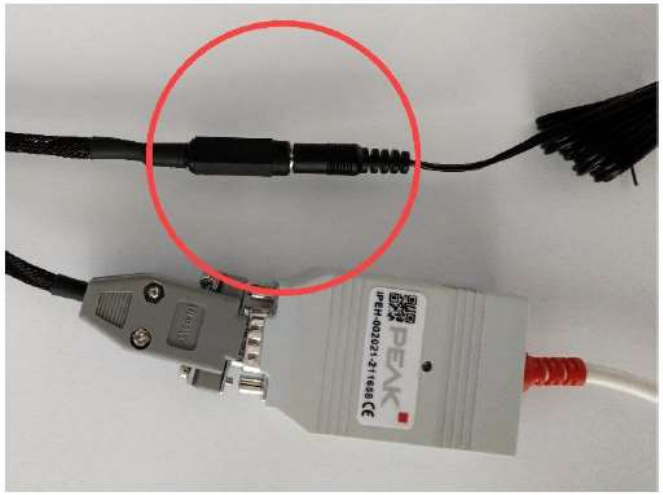


Figure 20. Connecting 12V power to the EDR retrieval harness

·. Proceed to “Retrieve Data” on page 26 to complete the data retrieval procedure.

The Tesla Guide (one per model)

- The Guide is VERY thorough and helpful 20+ pages
- It gives step by step instructions for the hardware assembly and to navigate the EDR program
- There are photographs of connector and module locations
- There are instructions for disassembly to get to the connectors and modules

Tesla EDR

- Data varies by model

	Model S	Model 3	Model X
PreCrash Data			
Vehicle Speed	2Hz	5Hz	2Hz
Accelerator Pedal	2Hz	5Hz	2Hz
Rear Motor Speed	2Hz	5Hz	2Hz
Service Brake	2Hz	2Hz	2Hz
Stavility Control	2Hz	2Hz	2Hz
ABS	2Hz	2Hz	2Hz
Steering Wheel Angle	2Hz	10 Hz	10 Hz
Lateral Acceleration	Not Reported	10 Hz	10 Hz
Longitudinal Acceleration	Not Reported	10 Hz	10 Hz
Roll Rate	Not Reported	10 Hz	10 Hz
Yaw Rate	Not Reported	10 Hz	10 Hz

Tesla EDR

- Data varies by model

	Model S	Model 3	Model X
Crash Data			
Longitudinal Δv	300ms	300ms	250ms
Lateral Δv	300ms	300ms	250ms
Longitudinal Acceleration	500Hz (300ms)	2000Hz (60ms)	2000Hz (60ms)
Lateral Acceleration	500Hz (300ms)	2000Hz (60ms)	2000Hz (60ms)
Normal Acceleration	Not Reported	Not Reported	50Hz (1.4s)
Roll Angle	100Hz (6s)	Not Reported	Not Reported

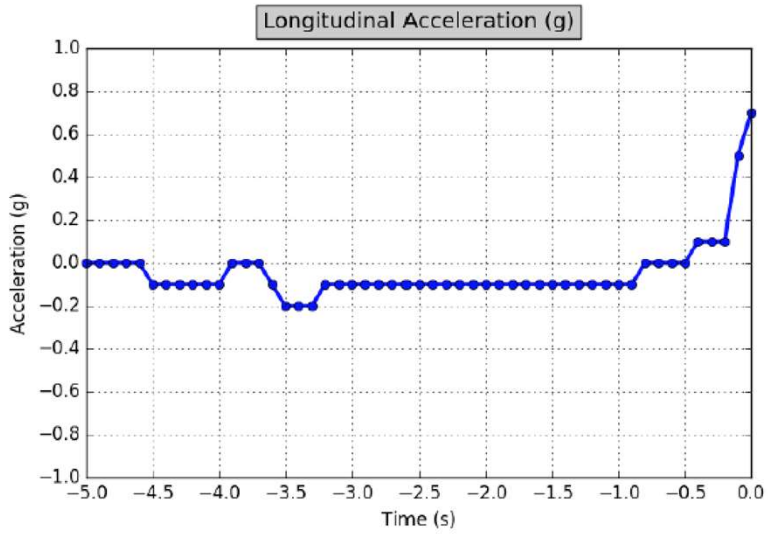
Tesla EDR Sample File

- Model 3 Data

Time (sec)	Service Brake	Stability Control	ABS Activity
-5.0	Off	Off	Off
-4.5	Off	Off	Off
-4.0	Off	Off	Off
-3.5	Off	Off	Off
-3.0	Off	Off	Off
-2.5	Off	Off	Off
-2.0	Off	Off	Off
-1.5	Off	Off	Off
-1.0	Off	Off	Off
-0.5	Off	Off	Off
0.0	On	Off	Off
-2.6	6.0	10.8	714
-2.4	6.0	13.2	746
-2.2	6.0	14.8	784
-2.0	7.0	12.4	807
-1.8	7.0	12.4	845
-1.6	7.0	12.4	883
-1.4	7.0	11.2	905
-1.2	8.0	12.0	922
-1.0	8.0	10.4	941
-0.8	8.0	4.8	932
-0.6	8.0	0.0	916
-0.4	8.0	0.0	888
-0.2	8.0	0.0	843
0.0	7.0	0.0	749

Tesla EDR Sample File

- Model 3 Data



Time (s)	Acceleration (g)
-5.0	0.0
-4.9	0.0
-4.8	0.0
-4.7	0.0
-4.6	0.0
-4.5	-0.1
-4.4	-0.1
-4.3	-0.1
-4.2	-0.1
-4.1	-0.1
-4.0	-0.1
-3.9	0.0
-3.8	0.0
-3.7	0.0
-3.6	-0.1
-3.5	-0.2
-3.4	-0.2
-3.3	-0.2

Time (s)	Acceleration (g)
-3.2	-0.1
-3.1	-0.1
-3.0	-0.1
-2.9	-0.1
-2.8	-0.1
-2.7	-0.1
-2.6	-0.1
-2.5	-0.1
-2.4	-0.1
-2.3	-0.1
-2.2	-0.1
-2.1	-0.1
-2.0	-0.1
-1.9	-0.1
-1.8	-0.1
-1.7	-0.1
-1.6	-0.1
-1.5	-0.1

Time (s)	Acceleration (g)
-1.4	-0.1
-1.3	-0.1
-1.2	-0.1
-1.1	-0.1
-1.0	-0.1
-0.9	-0.1
-0.8	0.0
-0.7	0.0
-0.6	0.0
-0.5	0.0
-0.4	0.1
-0.3	0.1
-0.2	0.1
-0.1	0.5
0.0	0.7

Troubleshooting

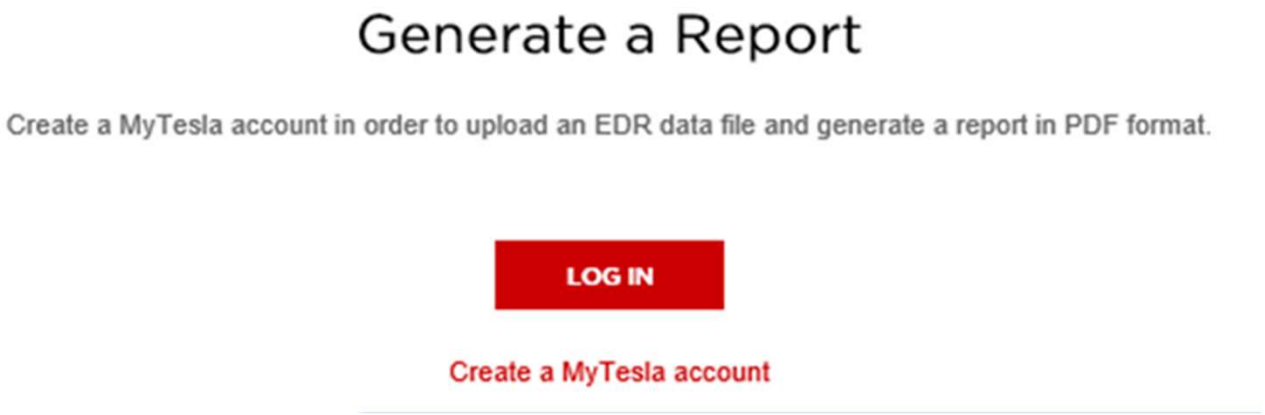
1. Some users report a “time out” error when trying to read data. The vehicle is going back to sleep. **Putting your foot on the brake pedal** and keeping it there for the whole readout keeps the vehicle alive.
2. Low battery power accentuates the time out problem. Tesla NEMA 5-15 adapter?



3. Or see the Help Guide for information how to backpower through the First-Responder loop.

Converting Raw Data to a Report

The Tesla software extracts a raw data file. To convert the raw data to a report, you must upload the raw data to the Tesla Server. A report is immediately returned in the latest version of software.



Worried about report conversion being biased by Tesla?

1. Data is Encrypted.
2. The report is returned IMMEDIATELY. There is no time for Tesla to alter any data to Tesla's benefit.
3. However, some clients insist that no one has a copy of the data except them.
4. There is a "Save without VIN sequence" option to de-identify data
5. There is no alternative.

What about OTHER data Tesla Data Records?

1. Most of the time you will NOT NEED to access the other data Tesla records, the EDR should be sufficient. But if you need a longer time history, Tesla records MANY data parameters every one second plus, when anything electronically monitored happens. It gets uploaded to the cloud.
2. It is NOT an "event" data recorder, it is a continuous recorder (does **NOT** meet Part 563 definition of a data recorder) (so **NO** PUBLICLY AVAILABLE tool to read it)
3. Law Enforcement contact Tesla and demonstrate proper legal authority
4. Tell them what you are interested in (they have so much you have to narrow it down)

What about OTHER data Tesla Data Records?

1. Modern Tesla have recorded video that can be uploaded to Tesla
2. This data can be retrieved from Tesla
3. Contact legal@tesla.com



BMW CDR US 21.5 Coverage (no change to 23.0)

See help file for Rolls Royce

Model	Location	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Last 3 digits of Bosch CDR adapter/cable number shown												
BMW	**All BMW require 387 Adapter for DTM**											
1 Series			Not sold in US						500/822			
2 Series	Driver side, behind IP, rt of strg			796								796 OR 5(
2 Series xdrive	Center console between seats										500/822	
3 Series	Passenger side, behind the glove box		796								500/822	
Active Hybrid 3	Driver side, behind IP, rt of strg			796								
4 Series	Passenger side, behind the glove box		796					500/822				
5 Series	Passenger side, behind the glove box		796								500/822	
6 Series	See Service Manual								500/822			
6 Series GT	Passenger side, behind the glove box		796			500/796	500/822					
7 Series	Passenger side, behind the glove box								500/822			
8 Series	Passenger side, behind the glove box		796								500/822	
X3 xDrive	Center Tunnel, Between Seats			796					500/822			
X5	Center Tunnel, Between Seats				798	500/798	500/807		also X2			
X1, X2	See Help File - Various				796			500/822			500/807	
X3, X4	Center Tunnel, Between Seats				796						500/822	
X6	See Service Manual								500/822			
X7	Center Tunnel, Under Dash				798				500/822			
Z4	See Service Manual				798				500/822			
iX												500/822
i3, i8				500/807								

No “back model” coverage

500=flex ray

2023 now covered in 23.1

The source data for this chart is the Bosch CDR help file for Version 21.5

Always consult the latest help file for the most accurate coverage

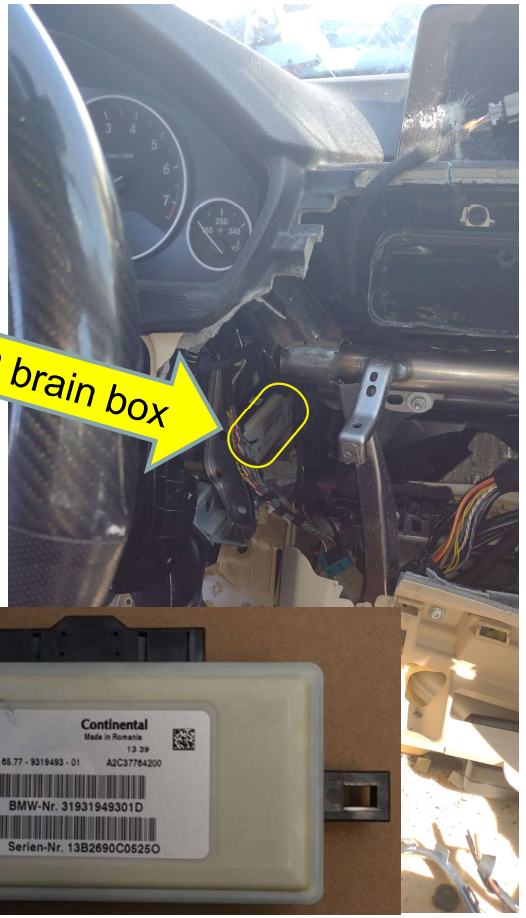
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SOME BMW have 2 “Airbag Modules”. Data in ACSM in the instrument panel



<- Sensor only module on center tunnel
No ACSM nomenclature on it

ACSM EDR brain box



Advanced Crash Safety Module (ACSM) (brain box) has the data. Other module with the arrow contains sensors, is hard mounted to body.

ACSM EDR brain box



More data in 2020 X5 “ACSM5”

Coverage for ACSM6 added in 23.0

- Recent files show day/date/time of crash and second/third row seat belt use, more

Multi-Event, Number of Events	1. Event
Time From Previous Event to Current Event (msec)	Data Not Available
Time From Last Speed Data Sample (Precrash) to Time Zero (msec)	57
Time From Initial Event to Current Event (msec)	Data Not Available
Complete File Recorded, Public, Prio 1 Data	Completed Successfully
Ignition Cycle, Crash (Cycle)	1,315
Ignition Cycle, Download (Cycle)	1,316
Vehicle Clock, Date and Time at Event (YYYY-MM-DD, HH:MM:SS)	2020-12-18, 09:03:54
Vehicle Mileage (km)	13,285
Operating Time (min)	20,794
Safety Belt Status, 2nd Row, Left Side	Not Belted
Safety Belt Status, 2nd Row, Center	Not Belted
Safety Belt Status, 2nd Row, Right Side	Not Belted
Safety Belt Status, 3rd Row, Left Side	Data Not Available
Safety Belt Status, 3rd Row, Center	Data Not Available
Safety Belt Status, 3rd Row, Right Side	Data Not Available

Mercedes CDR Coverage 21.2 36%

Model	Start of Sales	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Last 3 digits of Bosch CDR DTM cable number											
Mercedes	EDR Start	All Mercedes use 387 adapter for direct to module downloads									
A-Class Limo								832			
AMG-GT					800						
B-Class EV	after 4/1/2014		598								
B-Class	after 6/2/2014			598					832 Limo		
C Class Sedan	after 9/1/2014			801							
C Class Coupe	tart of 2015 M.Y.			800	801						
CLA Class	after 6/2/2014			598					832		
CLS-Class				800					801		
E Class (includes coupe)	after 4/1/2013		800			801 w steering data					
EQA/B/C									801 w strg		
G Class	after 11/1/2014			598				801			
GL Class	after 7/11/2014			598	?	?					
GLA Class	after 6/2/2014			598						832	
GLC Class					598		801			832	
GLE Class					598				832		
GLK	after 12/2013		800		?	?					
GLS Class					598				832		
GT								801			
G wagon									801		
Maybach S600					801						
ML-Class	after 7/11/2014			598	?						
S-class Sedan	after 9/2013		801							500/841	
S-Class Coupe	after 9/1/2014			801							
SL	after 12/2013		800								
SLC							800				
SLK	after12/2013		800								
Sprinter								832			
Vito					598					832	
Smart & 2016 with "E"	after 9/1/2014			809							
Smart with VIN 4th "F"					820						
Smart for two, coupe, cabrio							820				
Metris							598				

No "back model" coverage

2023 Only released in China in 23.0

Mercedes Details

1. Mercedes phased in EDR in anticipation of the NPRM requiring EDR's for Sept. 2014
2. Precrash speed, brake, throttle (563 Table 1) & lat/longitudinal ΔV.
3. Can hold 5 events, D's lock, ND's FIFO
4. Recording threshold is 5 mph over 150ms
5. Speed is avg. of 4 wheel speed sensors
6. The 2017 E Class has Steering, ABS, Stab Cont'l. This marks a departure from their philosophy to record only the minimum required by law.

Mercedes Key Cycles

- Some Mercedes have unusually high key cycle counts for their age, and since event.
- Other brands have 2500 key cycles per yr or 20,000km.
- Working theory is that counter increments whenever key fob leaves range and comes back into range.

Ignition Cycle, Crash (cycle)	11,639
Ignition Cycle, Download (cycle)	12,113
Vehicle Mileage (km)	20,950
Operating Time (min)	29,712
Vehicle Identification Number	4JGDA5JB2HA******
Event Counter (counts)	1

Mercedes Operating Timer

- In one model, the mileage divided by the operating timer implied the vehicle traveled an average speed of 500 mph
- Obviously this can’t be true, issue is under investigation

New Mercedes ADS Module

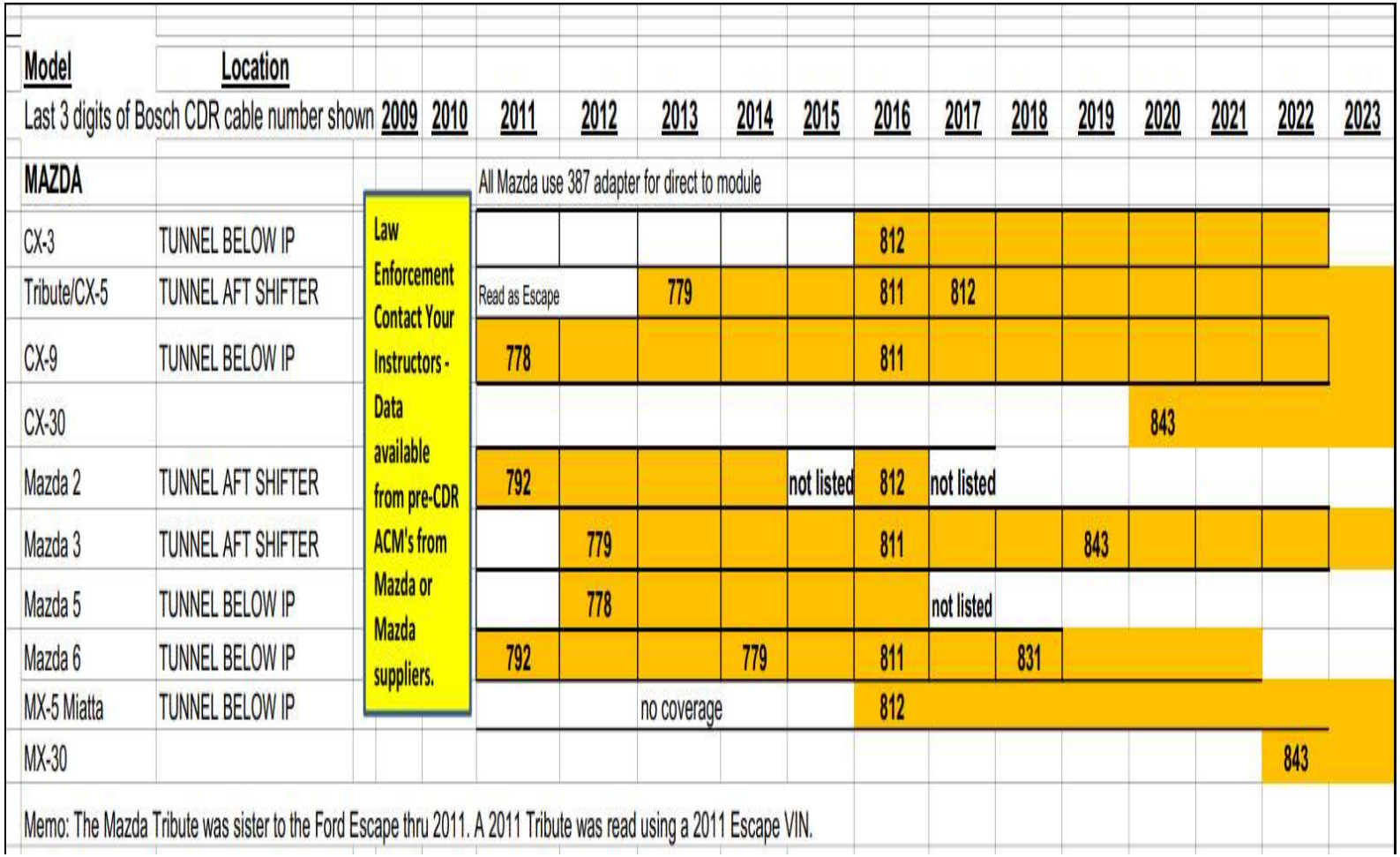
2023	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024

Mkt	Year/Make	Model	Module	New CDR2		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
2 California and Nevada	2024 Mercedes-Benz	EQS (sedan)	ADS	CDR 2	Data	CDR 900	OBD Only CDR 900 - click here	does not support direct to module downloads	N/A

New Mercedes ADS Module

- Stores sensor data relevant to autonomous driving
- Five Event Capability
- Stage 1 “Unusual longitudinal driving dynamics” (ND)
- Stage 2 “Detected Collision
- -30 seconds thru +5 at 5Hz
- Store in Chron order, most recent first
- Requires CDR2 program to read the data, DLC only

Mazda CDR 23.0 Coverage 44%



Model	Location	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Last 3 digits of Bosch CDR cable number shown																
MAZDA				All Mazda use 387 adapter for direct to module												
CX-3	TUNNEL BELOW IP								812							
Tribute/CX-5	TUNNEL AFT SHIFTER			Read as Escape		779			811	812						
CX-9	TUNNEL BELOW IP			778					811							
CX-30													843			
Mazda 2	TUNNEL AFT SHIFTER			792				not listed	812	not listed						
Mazda 3	TUNNEL AFT SHIFTER				779				811			843				
Mazda 5	TUNNEL BELOW IP				778					not listed						
Mazda 6	TUNNEL BELOW IP			792			779		811		831					
MX-5 Miatta	TUNNEL BELOW IP					no coverage			812							
MX-30															843	

Law Enforcement Contact Your Instructors - Data available from pre-CDR ACM's from Mazda or Mazda suppliers.

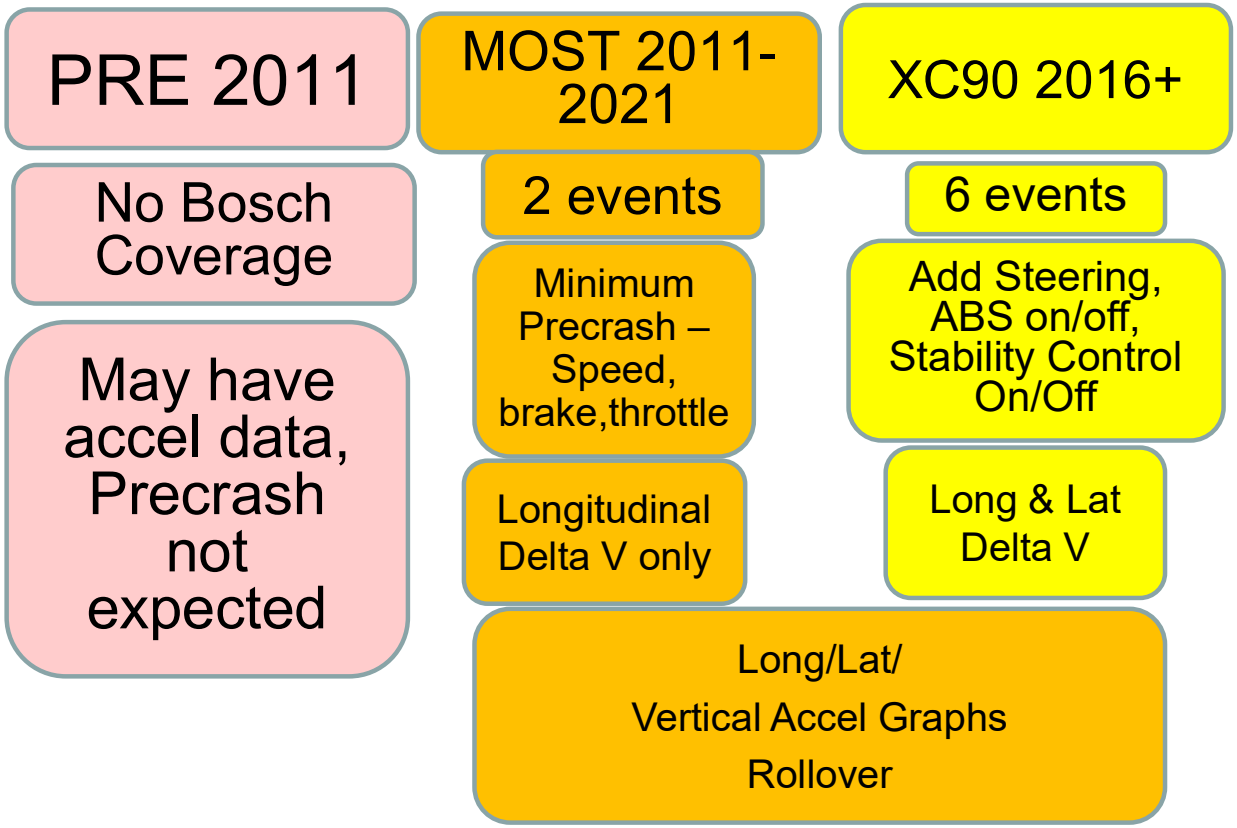
New CX-50 support added 23.1

Memo: The Mazda Tribute was sister to the Ford Escape thru 2011. A 2011 Tribute was read using a 2011 Escape VIN.

Volvo CDR 23.0.2 Coverage 32%+

Model	Location	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Last 3 digits of Bosch CDR cable number sh															
VOLVO		All Volvo use 387 Adapter for DTM images													
C40	CTR TUNNEL UNDER CONSOLE													500/830	
S60	CTR TUNNEL UNDER CONSOLE		799							500/830					
S80, S80L	CTR TUNNEL UNDER CONSOLE														
S90	CTR TUNNEL UNDER CONSOLE								816						
V40	CTR TUNNEL UNDER CONSOLE								816 Euro						
V60	CTR TUNNEL UNDER CONSOLE		799							500/830					
V70	CTR TUNNEL UNDER CONSOLE				799										
V90	CTR TUNNEL UNDER CONSOLE	Old Volvo DV Law Enf only call Volvo							816						
XC40	CTR TUNNEL UNDER CONSOLE									500/830					
XC60	CTR TUNNEL UNDER CONSOLE				799					816					
XC70	CTR TUNNEL UNDER CONSOLE				799										
XC90	CTR TUNNEL UNDER CONSOLE							816							

Volvo EDR Evolution



Volvo will only read in run mode, not accessory mode

Volvo anomalies observed

- Side curtain airbags deploy when frontal airbags deploy
- In 2-event crashes, time between events is always 0, data element appears to be not functioning
- Many times precrash is the same for both events, so we assume crashes are <0.5 sec apart
- (May be in all models, but the above was noticed in 2013-2016 60 series files)

2016+ XC90 Sample Volvo File

(vs. 2013 Volvo with minimum data)

Pre-Crash Data -1 Sec (Event Record 1)

Ignition Cycle, Crash	2,764
Safety Belt Status, Driver	On, Belted
Safety Belt Status, Passenger	On, Belted
Frontal Airbag Warning Lamp	Off
Frontal Airbag Suppression Switch Status, Front Passenger	On
Seat Track Position Switch, Foremost, Status, Driver	Not Equipped
Seat Track Position Switch, Foremost, Status, Front Passenger	Not Equipped
Occupant Size Right Front Passenger Child	Not Equipped

2020 still not added to other models

Added Data for XC90

Pre-Crash -5 to 0 sec (Event Record 1)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full (%)	Service Brake (On, Off)	Steering input (%)	ABS Activity	Stability control status
-5.0	43.5 [70.0]	100.0	Off	0.0	Off	On
-4.5	47.8 [77.0]	100.0	Off	0.0	Off	On
-4.0	52.2 [84.0]	100.0	Off	2.0	Off	On
-3.5	55.9 [90.0]	100.0	Off	3.0	Off	On
-3.0	58.4 [94.0]	100.0	Off	3.0	Off	On
-2.5	61.5 [99.0]	100.0	Off	11.0	Off	Engaged
-2.0	62.8 [101.0]	100.0	Off	6.0	Off	Engaged
-1.5	65.2 [105.0]	100.0	Off	-9.0	Off	Engaged
-1.0	67.1 [108.0]	100.0	Off	-8.0	Off	Engaged
-0.5	69.6 [112.0]	100.0	Off	-9.0	Off	Engaged
0.0	70.2 [113.0]	100.0	Off	-13.0	Off	Engaged

2016+ XC90 Sample Volvo File

2021 still not added to other models

System Status at Retrieval

Vehicle Identification Number	YV1LF
Application Diagnostic Database Part Number	31387255 AL
Ignition Cycle, Download	2,765
Lifetime Operating Timer (sec)	2,154,311

System Status at Event (Event Record 1)

Data Area Status, Event Record 1	Locked, Data Stored
Data Area Read Status, Event Record 1	Data Not Read
Complete File Recorded (Yes/No)	Yes
Multi-Event, Number of Events (1,2)	Event Number 1
Time from Preceding Event (sec)	Written but No Data Available
Maximum Delta-V, Longitudinal (MPH [km/h])	< -62 [-100]
Time, Maximum Delta-V, Longitudinal (msec)	100
Maximum Delta-V, Lateral (MPH [km/h])	-19.9 [-32.0]
Time, Maximum Delta-V, Lateral (msec)	>300

Clipped Value (at Max)

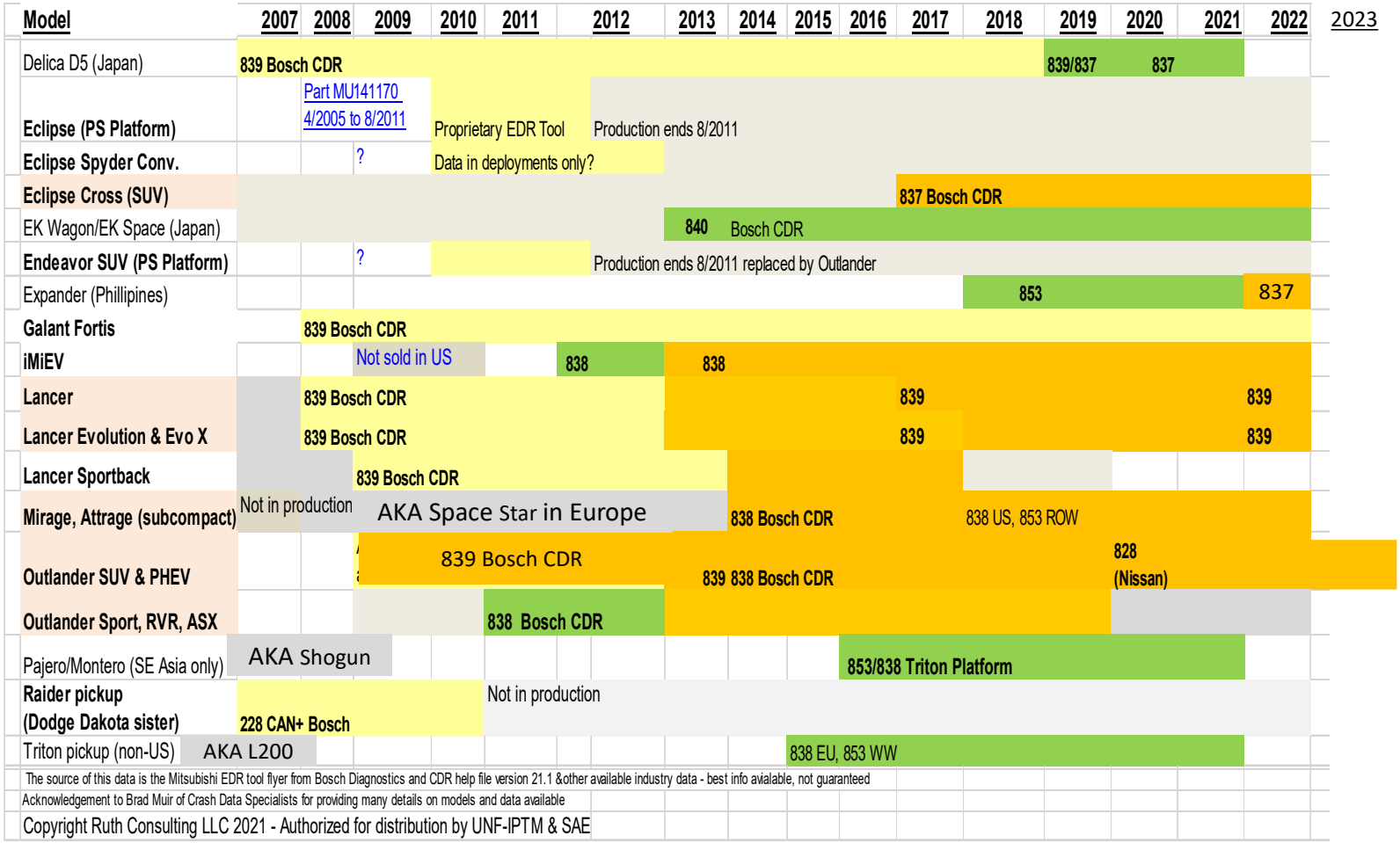
Not present in 2013, Added for XC90

[redacted]



1. Bosch CDR software 21.1 includes significantly expanded Mitsubishi coverage. Mi
2. The only models not officially covered are 2021+ Outlander Sport and pre-2013 that may be spoofable as Dodge Calibers .
3. The original Mitsubishi proprietary tool ($7,000) is not needed for any vehicles. Software has not been updated for years.
4. Early Mitsubishi readouts had some incomplete files retrieved (no useful data retrieved). CDR coverage is not expected to affect that issue, it will still exist.
5. Mitsu sales volume is less than 1% (about 0.6%)

US Mitsubishi EDR's (and some worldwide) Most now supported by CDR, some not



Help file cable links suggest some Mitsubishi listed in help file as supported are not (yet) – 23.0 release notes add part numbers to 23.0 but notes are still on cable links

MITSUBISHI ACM CABLE - P/N: 1699200783 / CABLE ID# 837

This CDR cable is required for direct-to-module imaging of 2018 and newer Eclipse Cross vehicles sold into the various markets. This cable will be available end of June 2019.

MITSUBISHI ACM CABLE - P/N: 1699200784 / CABLE ID# 838

This CDR cable is required for direct-to-module imaging of 2011 and newer Mitsubishi vehicles (see below) sold into the various markets. This cable will be available end of June 201

2011 - 2019 ASX, Outlander Sport & RVR

2014 - 2019 Outlander

2018 - 2019 Outlander PHEV

*2014 and newer Mirage

*2017 and newer Attrage

*2012 and newer iMiEV

* not currently supported in CDR software, support planned for a future release

MITSUBISHI ACM CABLE - P/N: 1699200785 / CABLE ID# 839

This CDR cable is required for direct-to-module imaging of 2007 and newer Mitsubishi vehicles (see below) sold into the various markets. This cable will be available end of June 201

*2008 and newer Galant Fortis

*2007 and newer Delica-D5

*2009 and newer Lancer Evolution

*2007 - 2013 Outlander

* not currently supported in CDR software, support planned for a future release

Jaguar/Land Rover

- ACM's must be sent to the supplier Veoneer in France or Bosch in Budapest, Hungary.
- If ACM is not in hand yet, contact Jag/LR to learn who the supplier is. They will give you contact info for supplier.
- Euro suppliers charge higher prices – Bosch as high as $4500
- Some Bosch ACM's have little useful data – ask what you will get before committing to pay!

Newly published research

1. Every year at the SAE World Congress in April new research is presented.
2. A very brief summary follows, I encourage you to read these papers.
3. Other good sources of EDR info include Collision Magazine

https://www.collisionmagazine.com/article-index/

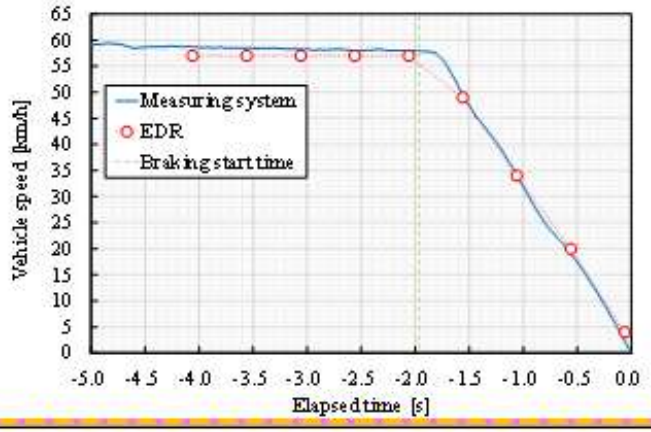
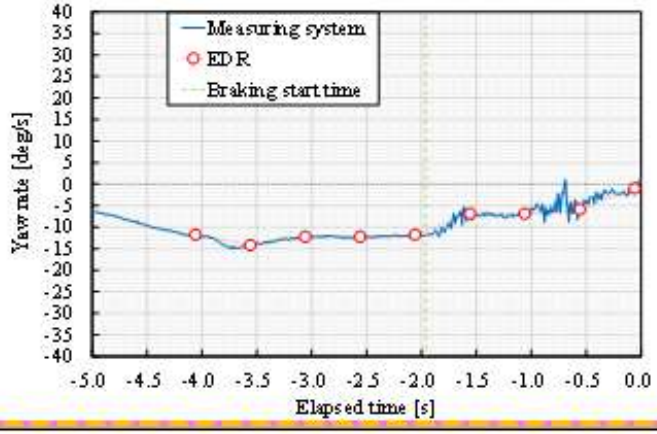
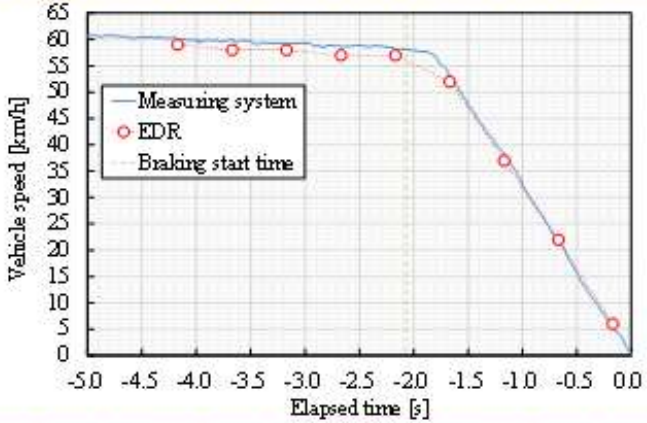
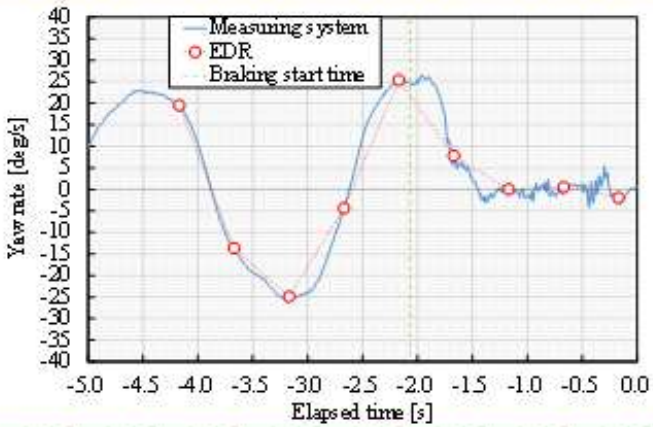
& Accident Reconstruction Journal.

EDR @ SAE 2022 #1

2022-01-0837 Study on estimation of traveling trajectory using the recording data in the event data recorder by Hideki Matsumura, *Japan* *Institute for Traffic Accident Research and Data Analysis (ITARDA)*

Tested a 2019 RAV4 with a 17 EDR recording speed and yaw at 0.5 second intervals and GPS instrumentation. When tested in a single curve starting from a known straight ahead path the yaw rate simulated the vehicle path very well. When tested in a slalom it was necessary to adjust the data to the known end point and rotate the time series to best fit the known start and intermediate points. If the sampling rate was higher the fit would be better.

SAE 2022-01-0837 data

	Vehicle Speed	Yaw rate
Curve		
Slalom		

EDR @ SAE 2022 #2

2022-01-0821 **Validation of Frozen Speedometer Readings Against EDR Data**

By Christopher H. Goddard & Steve Anderson of Collision Science & Govt of Ontario Canada

With the assistance of the collision reconstruction community, a large dataset of 236 vehicles with frozen speedometer readings were compared with EDR and other corroborating methods. This paper reviews the current state of knowledge, compares the instrument readings of each of the 236 vehicles against the available EDR data or other corroborating method. It then assesses each case against the criteria proposed by Goddard and Price [3, 4] to assess if the cases that are being filtered out for selection, are accurate readings. It was found the existing criteria was an effective filter in removing the majority of cases capable of producing erroneous readings. However, it was found that some erroneous readings were present in low speed impacts. With the addition of a minimum speed criteria, the likelihood of an erroneous reading was greatly reduced. At recorded speedometer readings over 80 Km/h, the corroboration with the EDR speed had a standard deviation of 4%.

SAE WCX 2023 EDR Session

- **2023-01-0628 Data Collection for Incident Response for Vehicles with Autonomous Systems**

HVEDR Papers

- 2023-01-0009 **Examination of Detroit Assurance® 4.0 Video Radar Decision Unit (VRDU) Records for Use in Crash Analysis**
- **2023-01-0010 Testing of Heavy Truck Advanced Driver Assistance Systems and Crash Mitigation Systems**
- **2023-01-0011 An Evaluation of the Performance of the WABCO OnGuard ACTIVE Collision Mitigation System in a 2019 Freightliner M2 106**
- **2023-01-0012 Examination of Bendix® Data Recording (BDR) Records for Use in Crash Analysis**

WREX 2023

- April 2023 Conference had many worthwhile presentations, and many instrumented crash tests
- 2 Gigs of data can be downloaded by attendees
- Find someone who went and get a copy

Appeal for Files

- Manufacturers keep coming out with new generations of EDR's
- The way I keep up is asking my network of former students and conference attendees to send me files from new vehicles
- PLEASE SEND ME FILES FROM 2022/2023 MY VEHICLES
- If you don't de-identify them, I will
- Helpful if I know which way they turned to check polarity of steering, yaw, lat accel

Event Data Recorder Update

QUESTIONS????

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